

Strategic Deindustrialization and Industrial Policy Games*

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Abstract

This paper asks whether a country can use trade to deindustrialize a geopolitical rival. In our model, imports reduce domestic employment and learning by doing. If they remain high long enough, industrial knowledge can fall below a critical threshold, leaving the importer in a low-productivity state even after imports recede. A geopolitical rival may deliberately cause this outcome for security reasons. A large exporter with market power may also do so for commercial gain, because a less productive customer pays more for future imports. We then study policy responses. A sufficiently high tariff deters the exporter, while an employment subsidy can preserve domestic industrial knowledge without restricting trade. Under conditions we characterize, the subsidy threat alone deters deindustrialization.

Keywords: learning by doing; multiple steady states; hysteresis; deindustrialization; trade and national security; industrial policy; tariffs; sectoral export surplus.

JEL codes: F12, F13, F52, L52, O14.

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1 Introduction

This paper studies whether one country can use trade to deindustrialize another. We ask when an import surge can cause a lasting loss of productive capacity, when an exporter would choose to cause that loss, and how the importing country can prevent it.

These questions matter when production sustains knowledge that is lost as production contracts. An importing country gains access to cheaper goods, but the resulting decline in domestic employment can reduce its future productivity. In sectors such as semiconductors and shipbuilding, industrial capacity also supports military production. A geopolitical rival may therefore value the loss of that capacity for security reasons. Recent policy reflects both concerns: the U.S. CHIPS and Science Act, the European Chips Act, and the export-control measures of 2024–26 treat the capacity to manufacture advanced goods as a matter of national security rather than ordinary commerce. These episodes motivate our analysis, but by themselves establish neither that trade caused the loss of industrial capacity nor that exporters intended it.

We develop a dynamic model of two countries, East and West, in which the East exports capital goods to the West in exchange for consumption goods, either currently or through claims on future consumption. Productivity in each country depends on industrial knowledge shared across domestic producers. Knowledge accumulates through learning by doing, but individual households do not internalize the contribution of their work to future productivity. The West is competitive and, initially, its government does not intervene. The East’s government chooses exports, taking into account both their commercial return and the security benefit of reducing Western industrial capability. The weight it places on that benefit, its security weight, is the model’s central parameter. The surplus we study is a sectoral surplus in capital goods, not necessarily a current-account surplus.

Our first result is that a temporary import surge can cause a permanent decline in industrial capacity. Productivity is S-shaped in knowledge. When the feedback between productivity and employment is sufficiently strong, the economy has two stable steady states separated by an unstable knowledge threshold. Imports displace domestic production and reduce employment. Lower employment slows learning, and knowledge falls. A sufficiently large and persistent surge pushes the West below the threshold. Once there, it converges to the low-industrialization steady state even after imports return to their original level.

Our second result characterizes the exporter’s incentive to cause this decline. When commercial incentives alone are not strong enough to justify deindustrializing the West, there is a strictly positive critical security weight above which the East optimally does so. The export path has two phases. The East first ships enough capital goods to push Western knowledge down to the critical threshold. Once the West is trapped, the East cuts exports back to the commercial level appropriate for a low-productivity customer. The West’s own employment and knowledge dynamics then sustain the decline. We prove this result, and the policy results below, in a benchmark in which productivity takes a high value above the knowledge threshold and a low value below it.

Security is not necessary for this outcome. East and West have the same preferences and technology, but they play different roles in the trading game. The West is competitive,

while the East is a large exporter that recognizes how its exports affect the price of capital goods. By reducing Western productive capacity, the East can raise the price the West pays for future imports. This creates a purely commercial motive for deindustrialization. The strategy requires a temporary sacrifice, since the East must first ship enough capital goods to reduce Western employment and knowledge. When future gains matter enough, that sacrifice is worthwhile; otherwise, and absent a security motive, the equilibrium is autarky.

Our third result concerns the West's response. A sufficiently high tariff deters deindustrialization. Below the deterrent level, the East absorbs the tariff and continues to export enough to push Western knowledge across the threshold; above it, the East abandons the attempt. The deterrent tariff rises with the East's security motive. But the West need not observe that motive if it is willing to prohibit trade altogether. In the benchmark, a prohibitive tariff protects against every security motive and carries no static cost, because two industrialized countries would not trade absent a strategic motive. The informational problem matters only if the West wants to deter the East without shutting down trade.

An employment subsidy addresses the domestic learning externality directly. The policy supports employment when imports threaten to push Western knowledge below the critical threshold. We give conditions under which the West is willing to provide enough support to prevent that crossing. With lump-sum financing and a subsidy rate that can respond to the size of the East's exports, the policy deters deindustrialization from the high steady state at any security weight. The East anticipates that additional exports cannot push the West into the low-productivity state and therefore does not initiate the surge. Along the equilibrium path, the subsidy is not paid. Unlike a prohibitive tariff, the policy does not restrict imports. This distinction has no welfare value in our analytical benchmark because two industrialized countries do not trade in the absence of the strategic motive.

The analytical deterrence result relies on a sharp threshold in productivity and security losses. With smooth schedules, numerical solutions continue to exhibit deterrence for security weights well above the benchmark's critical security weight; at still higher weights, we do not find an equilibrium (Appendix F). In these computations, the West pays a positive subsidy along the equilibrium path, equal to the subsidy that corrects its domestic learning externality.

The analysis makes the importer's domestic knowledge stock central to industrial policy. The West does not need to know why the East exports. It needs to know how much domestic employment is required to preserve productive capacity and how imports affect that employment. Vulnerability is therefore sectoral rather than aggregate. The relevant objects are the import flow, domestic employment, and the knowledge sustained by that employment. The aggregate trade balance says little about how close an industry is to losing the capacity to sustain itself.

Related literature. The paper relates to three strands of research. The first studies learning by doing, industrial development, and persistent trade effects. [Bahk and Gort \(1993\)](#), [Benkard \(2000\)](#), and [Irwin and Klenow \(1994\)](#) provide evidence on the accumulation, depreciation, and diffusion of production knowledge. [Baldwin \(1969\)](#) and [Dasgupta and Stiglitz \(1988\)](#) study the implications of learning for infant-industry protection. Our multiplicity

of steady states descends from the literature on increasing returns and development traps (Murphy et al., 1989; Matsuyama, 1991); here an international flow of capital goods, rather than domestic spillovers, moves the economy between them. Krugman (1987) shows how learning can make specialization persistent, while Baldwin and Krugman (1989) show how sunk costs can give temporary shocks lasting trade effects. We study a knowledge threshold below which an established industry cannot recover on its own and a foreign government’s incentive to push it across that threshold.

The second strand studies strategic trade policy. Brander and Spencer (1985) analyze export subsidies that shift profits toward domestic producers. Costinot et al. (2014) study capital controls as a means of manipulating intertemporal terms of trade. Recent work studies trade policy that responds to security risk (Traiberman and Rotemberg, 2023) and how geopolitical conflict reshapes trade (Adolph et al., 2025). Our exporter can improve its future terms of trade by reducing its customer’s productive capacity. A security motive provides an additional return from the same policy. We characterize how the importing country’s response changes the incentive to pursue it.

The third strand connects international imbalances, industrial policy, and geopolitical competition. Pflueger and Yared (2024) study the relationship between financing advantages and hegemony. The closest contemporaneous paper is Cesa-Bianchi et al. (2026), which shows how industrial policies combined with weak domestic demand generate surpluses that reduce innovation and productivity abroad. Our contribution is to characterize a threshold beyond which the importer’s loss persists after the surplus ends, an exporter’s choice to cause that loss, and an equilibrium response that can deter it. The mechanism operates through a sectoral flow of goods and does not require a particular current-account balance.

Section 2 presents the motivating episodes. Section 3 develops the learning mechanism and the conditions for lasting deindustrialization. Section 4 studies the exporter’s choices, and Section 5 analyzes the importer’s policies. Section 6 provides the numerical illustration. Sections 7 and 8 discuss implications and conclude. The appendices contain proofs, extensions, and the numerical analysis of the smooth economy.

2 Motivating Facts

This section describes three episodes that motivate the analysis: the concentration of solar manufacturing in China, the concentration of commercial shipbuilding in East Asia, and the declining U.S. share of semiconductor fabrication. Each raises a question about the relationship between access to imported goods and the capacity to produce them domestically. We focus on the location of production, changes in industrial leadership, and efforts to expand domestic capacity.

In solar photovoltaics, manufacturing became concentrated across the production process. The International Energy Agency places China’s share above 80 percent at each of the five principal stages: polysilicon, ingots, wafers, cells, and modules (International Energy Agency, 2022). China’s manufacturing share was more than twice its share of global demand. Over the preceding decade, manufacturing had shifted toward China from Europe, Japan, and the

United States. This expansion also lowered production costs and made solar power more affordable. The same development therefore increased access to inexpensive equipment and concentrated the capacity to supply it. The IEA identifies cost competitiveness, with China the lowest-cost location for every stage, as a key challenge to diversifying supply.

Commercial shipbuilding illustrates concentration among several exporting countries. In 2023, China delivered more than half of the world’s new ship tonnage. China, South Korea, and Japan together accounted for about 95 percent, leaving the rest of the world, including the United States and Europe, about one twentieth of the total ([United Nations Conference on Trade and Development, 2024](#)). The episode also shows why the model’s two-country structure is an abstraction: the concentration of production can involve several competing exporters.

Semiconductors illustrate a decline in relative manufacturing capacity alongside efforts to expand domestic production. The U.S. share of global fabrication capacity fell from 37 percent in 1990 to 10 percent in 2022 ([Semiconductor Industry Association and Boston Consulting Group, 2020, 2024](#)). Mainland China accounted for 24 percent in 2022, and for a third of logic capacity at 28 nanometers and older, while China, Taiwan, South Korea, and Japan together held about three quarters. These shares describe the location of fabrication rather than the entire semiconductor industry. The U.S. decline is relative, not absolute: U.S. fabrication capacity grew by 11 percent between 2012 and 2022. The outcome of the policy response is not settled. The industry’s 2024 projections have the U.S. share rising to 14 percent by 2032 with the investments supported by the CHIPS and Science Act, and falling to 8 percent without them ([Semiconductor Industry Association and Boston Consulting Group, 2024](#)). These are conditional projections.

These episodes motivate a theory of how trade changes the capacity to produce. They do not establish that imports caused a permanent loss through learning by doing, or that an exporting government intended such a loss. Production shares alone cannot identify either claim. The model supplies the missing economic structure. Employment sustains industrial knowledge, and a contraction in production can reduce the knowledge available to future producers. We ask when this feedback gives a temporary import surge lasting effects, when an exporter benefits from causing those effects, and when domestic policy prevents them. Section 7 returns to historical precedents, including the decline of India’s cotton-textile industry.

3 A Baseline Model

This section studies one economy, the West, that receives an exogenous flow of capital goods. We describe its households and its learning-by-doing technology, reduce the equilibrium to a law of motion for industrial knowledge, and characterize the steady states. When the feedback between productivity and employment is strong enough, the economy has a low and a high steady state separated by an unstable one, and a large enough flow of capital goods eliminates the high steady state. We then show that a temporary flow can have a permanent effect, using a resource windfall as the example, and derive the subsidy that

corrects the learning externality.

Time, denoted t , is discrete, and the horizon is infinite: $t = 0, 1, 2, \dots$. There are two countries: East and West. We ignore the East for the moment. The baseline is intended to isolate the basic economic mechanisms that carry over into the more complicated setting in which the East behaves strategically.

A representative household living in the West has preferences defined over streams of consumer goods (c_t) and labor (n_t) given by:

$$\max_{\{c_t, n_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t [c_t - v(n_t)] \quad (1)$$

Consumer goods are produced with a stock of capital goods (k_t) according to:

$$c_t = f(k_t) \quad (2)$$

Assumption 1 (Preferences and technology). *The labor-disutility function $v : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is twice continuously differentiable, satisfies $v(0) = v'(0) = 0$ and $v''(n) > 0$ for all $n \geq 0$, and has $\lim_{n \rightarrow \infty} v'(n) = \infty$. The production function $f : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is continuous on $\mathbb{R}_+$, twice continuously differentiable on $\mathbb{R}_{++}$, satisfies $f(0) = 0$, $f'(k) > 0$, and $f''(k) < 0$ for all $k > 0$, and has $\lim_{k \downarrow 0} f'(k) = \infty$ and $-kf''(k)/f'(k) < 1$ for all $k > 0$.*

New capital goods are produced by individual households using their labor (n_t) and an accumulation of *collective* domestic knowledge (h_t) that determines domestic labor productivity $\alpha(h_t)$. This latter technology is given by:

$$\alpha(h_t) = \alpha_L + \frac{\alpha_H - \alpha_L}{1 + \exp(-\gamma(h_t - \chi))} \quad (3)$$

where $\alpha_L > 0$ is baseline productivity, $\alpha_H > \alpha_L$ is the productivity frontier, χ is the inflection point, and $\gamma > 0$ governs the steepness of learning externalities.

There is also an exogenous flow of capital goods (m_t) that can be interpreted as arising from a natural resource or another external transfer. In the sequel, the East is modeled explicitly, and the same flow is interpreted as imports of capital goods exchanged for a claim on the West. For now, we refer to m_t as imports of capital goods. Together, the sum of domestic production and imports of capital goods is given by:

$$k_t = \alpha(h_t)n_t + m_t \quad (4)$$

Finally, the body of domestic collective knowledge evolves according to

$$h_{t+1} = (1 - \delta)h_t + \lambda \mathbf{n}_t \quad (5)$$

where $\mathbf{n}_t$ denotes aggregate (or per capita) employment, $\delta \in (0, 1)$ captures depreciation (organizational forgetting), and $\lambda > 0$ governs the intensity of learning by doing.

3.1 Equilibrium Conditions

An individual household maximizes utility (1) subject to (2) and (4) taking the evolution h_t, m_t as given. Because an individual household has a negligible impact on the evolution of collective knowledge, each household's choice problem is static in nature. Substituting (2) into the objective (1), a household's desired labor supply at each date t is characterized by:

$$v'(n_t) = \alpha(h_t) f'(\alpha(h_t)n_t + m_t) \quad (6)$$

In equilibrium, $n_t = \mathbf{n}_t$ for all t . With this latter condition imposed, equations (5) and (6) characterize the model's equilibrium dynamics for a given initial condition, h_0 and an exogenous sequence $\{m_t\}$.

In a steady state with $m_t = m$, employment and the stock of knowledge satisfy:

$$v'(n^*) = \alpha(h^*) f'(\alpha(h^*)n^* + m) \quad (7)$$

Setting $h_{t+1} = h_t = h^*$, $n_t = \mathbf{n}_t = n^*$ and using (5), we have:

$$h^* = (\lambda/\delta) n^* \quad (8)$$

Substituting (8) into (7) reduces the steady state to a single equation in n^* ,

$$v'(n^*) = \alpha\left(\frac{\lambda}{\delta} n^*\right) f'\left(\alpha\left(\frac{\lambda}{\delta} n^*\right) n^* + m\right) \quad (9)$$

Multiple steady states are possible here because of the S-shaped property of $\alpha(\cdot)$. The remainder of this section makes that statement precise: we reduce the equilibrium to a scalar dynamical system, characterize its steady states and their stability, and show how a permanent change in the exogenous supply of new capital goods m affects the steady state.

3.2 Equilibrium Dynamics

Equation (6) determines the equilibrium level of employment as a function of the levels of domestic knowledge and imports. Write the solution of (6) as the *employment function*

$$n_t = \phi(h_t; m) \quad (10)$$

Lemma 1 (Employment function). *Under Assumption 1, for every $(h, m) \in \mathbb{R}_+^2$, equation (6) has a unique solution $\phi(h; m) > 0$. The function ϕ is continuously differentiable and satisfies $\phi_h(h; m) > 0$ and $\phi_m(h; m) < 0$. Effective capital $k(h; m) \equiv \alpha(h)\phi(h; m) + m$ is strictly increasing in both arguments, with*

$$k_m = 1 + \alpha\phi_m = \frac{v''}{v'' - \alpha^2 f''} \in (0, 1), \quad k_h = \alpha' \phi + \alpha\phi_h = \frac{\alpha' [\phi v'' + \alpha f']}{v'' - \alpha^2 f''} > 0.$$

Proof. Fix (h, m) and let $a = \alpha(h) > 0$. The left-hand side of equation (6), $v'(n)$, is continuous and strictly increasing from zero to infinity. The right-hand side, $af'(an + m)$, is

strictly positive and strictly decreasing in n , with a possibly infinite limiting value as $n \downarrow 0$. Hence, equation (6) has a unique solution $\phi(h; m) > 0$. At the solution, let $k = \alpha\phi + m$. The comparative statics are

$$\phi_h = \frac{\alpha' [f' + \alpha\phi f'']}{v'' - \alpha^2 f''} \quad \text{and} \quad \phi_m = \frac{\alpha f''}{v'' - \alpha^2 f''}$$

Since $f'' < 0$, it follows immediately that $\phi_m < 0$. For ϕ_h , observe that

$$f' + \alpha\phi f'' = f' \left[1 - \left(-\frac{k f''}{f'} \right) \frac{\alpha\phi}{k} \right]$$

Since $\alpha\phi/k \leq 1$ and $-k f''/f' < 1$, the bracketed term is strictly positive. Because $\alpha' > 0$, it follows that $\phi_h > 0$. For the last claim, substitute the expressions for ϕ_m and ϕ_h : $1 + \alpha\phi_m = (v'' - \alpha^2 f'' + \alpha^2 f'')/(v'' - \alpha^2 f'')$, and $\alpha'\phi + \alpha\phi_h = \alpha'[\phi(v'' - \alpha^2 f'') + \alpha f' + \alpha^2 \phi f'']/(v'' - \alpha^2 f'')$. Both denominators are positive and both numerators are positive. An inflow therefore raises effective capital, but by less than one for one: part of it crowds out domestic production. $\square$

The economics underlying Lemma 1 are straightforward. Condition (6) equates the marginal cost of employment, $v'(n_t)$, with its marginal benefit, the product $\alpha(h_t)f'(k_t)$ of productivity and the marginal product of effective capital. For any given level of domestic production $\alpha(h_t)n_t$, an exogenous increase in imported capital goods m_t lowers the marginal product of capital and, hence, lowers the marginal benefit of employment. As the return to domestic employment declines, the resources devoted to this activity are optimally curtailed.

Substituting the employment function into the law of motion for collective knowledge gives

$$h_{t+1} = G(h_t; m) \equiv (1 - \delta)h_t + \lambda\phi(h_t; m) \quad (11)$$

A steady state is a fixed point $G(h^*; m) = h^*$. Define the steady-state residual $D(h; m) \equiv G(h; m) - h = \lambda\phi(h; m) - \delta h$. Thus, h^* is a steady state if and only if $D(h^*; m) = 0$. Moreover, $G_h(h; m) = 1 - \delta + \lambda\phi_h(h; m) > 0$ and $D_h(h; m) = G_h(h; m) - 1$. Therefore, a steady state is locally asymptotically stable if $D_h(h^*; m) < 0$ and unstable if $D_h(h^*; m) > 0$.

The multiplicity mechanism requires the equilibrium law of motion $G(\cdot; m)$ to be S-shaped relative to the 45-degree line. This shape is natural in the present environment. When the stock of knowledge is either low or high, the productivity schedule $\alpha(h)$ is relatively flat, so an increase in knowledge produces only a modest employment response. Over the intermediate range, however, $\alpha(h)$ is steep, and the resulting increase in employment reinforces knowledge accumulation through learning by doing. If this feedback is sufficiently strong, the slope of $G(\cdot; m)$ rises above one over an intermediate interval before falling below one again.

This geometry is familiar from models of development traps, coordination failures, technology adoption, and hysteresis under learning by doing. It produces two stable fixed points separated by an unstable threshold. Because the employment response $\phi(h; m)$ is endogenous, however, the S-shape of α alone does not guarantee the required shape of G . We therefore impose the following restriction directly on the induced steady-state residual $D(h; m) = G(h; m) - h$.

Assumption 2 (Shape restriction). *There exists $m_1 > 0$ such that D is twice continuously differentiable on $\mathbb{R}_{++} \times [0, m_1]$. For every $m \in [0, m_1]$, $D_h(\cdot; m)$ has exactly two interior zeros $h_1(m) < h_2(m)$, with*

- $D_h(h; m) < 0$ for $h < h_1(m)$,
- $D_h(h; m) > 0$ for $h_1(m) < h < h_2(m)$,
- $D_h(h; m) < 0$ for $h > h_2(m)$.

Moreover,

$$D_{hh}(h_1(m); m) > 0 \quad \text{and} \quad D_{hh}(h_2(m); m) < 0,$$

and the functions h_1 and h_2 are continuously differentiable on $[0, m_1]$.

Proposition 1 (Steady-state multiplicity and the import fold). *Suppose Assumption 2 holds and*

$$D(h_1(0); 0) < 0 < D(h_2(0); 0), \quad D(h_2(m_1); m_1) < 0.$$

Then there exists a unique fold $\bar{m} \in (0, m_1)$ such that:

- (i) *For $0 \leq m < \bar{m}$, the economy has exactly three distinct steady states,*

$$h_L^*(m) < h_M^*(m) < h_H^*(m).$$

- (ii) *At $m = \bar{m}$, the economy has exactly two distinct steady states. The middle and high steady states coalesce at $h_2(\bar{m})$ in a saddle-node bifurcation, characterized by the tangency*

$$D(h_2(\bar{m}); \bar{m}) = 0 \quad \text{and} \quad D_h(h_2(\bar{m}); \bar{m}) = 0, \tag{12}$$

equivalently $G(h_2(\bar{m}); \bar{m}) = h_2(\bar{m})$ and $G_h(h_2(\bar{m}); \bar{m}) = 1$.

- (iii) *For $\bar{m} < m \leq m_1$, the economy has a unique steady state, namely the low-level steady state $h_L^*(m)$.*

- (iv) *On the three-steady-state region $0 \leq m < \bar{m}$, the two outer steady states are strictly decreasing in imports while the middle steady state is strictly increasing:*

$$\frac{dh_L^*}{dm} < 0, \quad \frac{dh_M^*}{dm} > 0, \quad \frac{dh_H^*}{dm} < 0.$$

The surviving low-level steady state is likewise strictly decreasing in m for $\bar{m} < m \leq m_1$.

The proof is in [Appendix A](#).

For $m < \bar{m}$, the low and high steady states are locally asymptotically stable, while the middle steady state is unstable. For $\bar{m} < m \leq m_1$, the surviving low steady state is locally asymptotically stable. Both statements follow from the sign of the residual $D(\cdot; m)$ between its zeros, because $G(\cdot; m)$ is increasing: D is positive just below the low and high steady

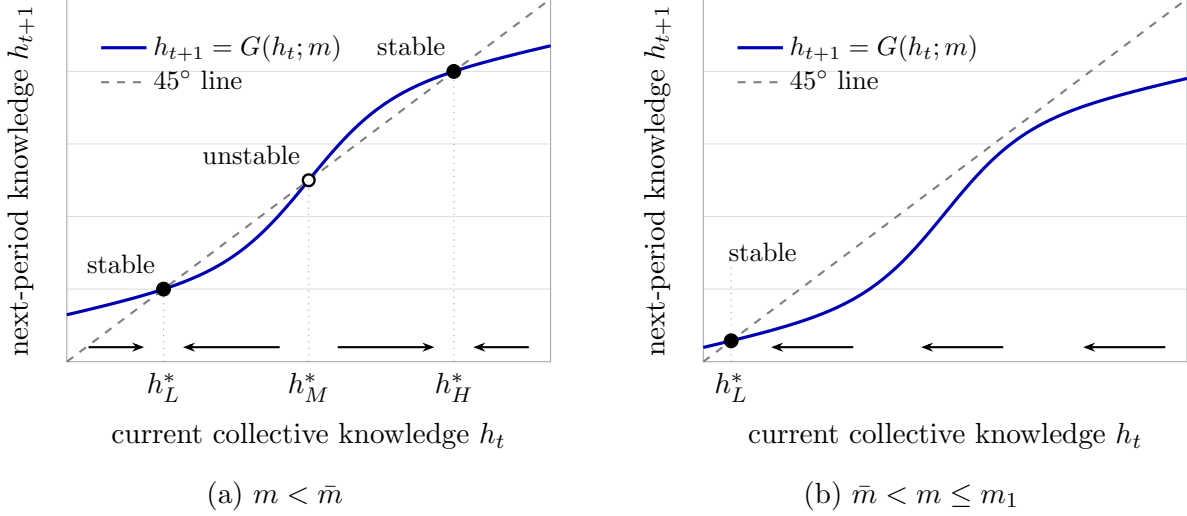


Figure 1: The law of motion $G(\cdot; m)$ against the 45° line. Panel (a): for $m < \bar{m}$ the S-shaped map crosses the line three times; the low and high steady states are stable and the middle one is unstable, and the arrows show the direction of motion. Panel (b): for $\bar{m} < m \leq m_1$ the map crosses once, the surviving low steady state is stable, and knowledge converges to it from every starting point. Both panels are drawn for a stylized map.

states and negative just above them, so nearby knowledge moves monotonically toward each of them, and the pattern is reversed around the middle steady state.

When three steady states coexist, the unstable one divides the state space in two: since $G(\cdot; m)$ is increasing, an economy starting above $h_M^*(m)$ converges to the high steady state and one starting below it converges to the low steady state. We therefore call $h_M^*(m)$ the *separatrix*, and the two intervals it divides the *basins of attraction* of $h_H^*(m)$ and $h_L^*(m)$. We call the low steady state the *trap*, and say that an economy is *trapped* when its knowledge lies in the basin of attraction of the low steady state.

Figure 2 collects Proposition 1 in the space of steady states. As m rises, the two stable branches fall and the separatrix rises (part (iv)), so the high steady state and the separatrix travel toward each other. At $\bar{m}$ they collide and annihilate. Beyond it only the low branch remains. The mechanism that drives the rest of the paper operates entirely along this diagram: whatever raises m moves the economy to the right, and a large enough increase leaves it with no high-industrialization steady state to return to.

3.3 A Resource Windfall: Dutch Disease with Hysteresis

Nothing in the analysis so far requires the flow m to come from an adversary, or indeed from abroad. As noted earlier, m_t is simply an exogenous flow of capital goods, and a natural resource is one reading of it. Suppose the economy sits at the high steady state $h_H^*(m_0)$ and a resource is discovered: a Lucas tree appears whose dividend is paid in capital goods, which enter effective capital through (4) exactly as any other source of m does. If the dividend is large enough that $m' > \bar{m}$, Proposition 1(iii) applies and the high steady state ceases to exist.

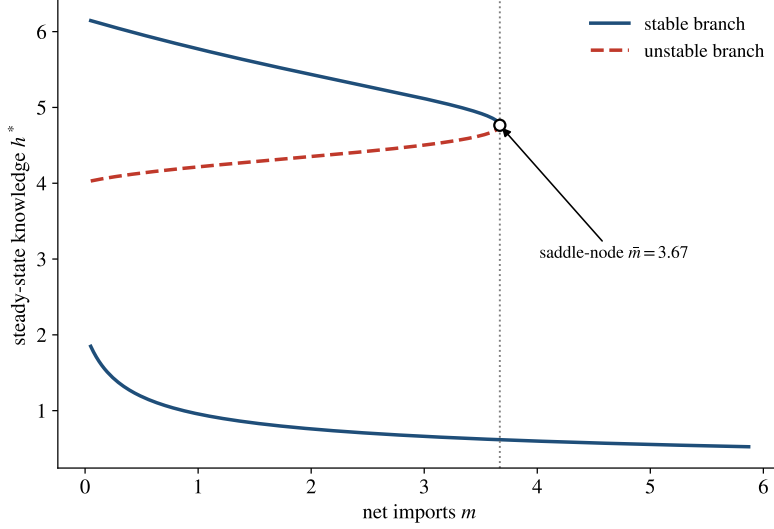


Figure 2: Steady-state knowledge h^* as a function of net imports m . The two stable branches (h_L^* falling, h_H^* falling) and the unstable branch (h_M^* rising) meet at the saddle-node $\bar{m}$, beyond which only the low branch remains.

The map falls to the configuration of panel (b) of Figure 1: the only remaining steady state is the low-industrialization state, and knowledge declines toward it. A windfall thus contracts tradable manufacturing, which is the familiar Dutch-disease outcome, though it arrives here through the return to industrial employment rather than through a real appreciation.¹

What the S-shaped map adds is that the damage can outlast its cause. A resource boom is typically temporary: the tree eventually evaporates and the flow reverts to m_0 . Whether the economy recovers depends only on which side of the separatrix $h_M^*(m_0)$ it has reached by the time the dividend stops. In stating this we write $G^T(h; m)$ for the T -fold composition of the map $G(\cdot; m)$ with itself, that is, the knowledge stock reached after T periods starting from h when the flow is held fixed at m , with the convention $G^0(h; m) = h$.

Proposition 2 (Hysteresis: temporary booms, permanent losses). *Let $m_0 < \bar{m}$, so that three steady states coexist, and let the economy begin at $h_0 = h_H^*(m_0)$. Suppose $m_t = m' > \bar{m}$ for $t = 0, \dots, T-1$ and $m_t = m_0$ for all $t \geq T$, so that $h_T = G^T(h_0; m')$. Define the critical duration*

$$\bar{T}(m') \equiv \min \{T \geq 1 : G^T(h_H^*(m_0); m') < h_M^*(m_0)\}. \quad (13)$$

Then

- (i) h_t is strictly decreasing for $t = 0, \dots, T$. If the boom never ended, that is, if $m_t = m'$

¹In the canonical account (Corden and Neary, 1982) a windfall raises domestic demand and the real exchange rate, draws labor out of tradable manufacturing, and reverses when the windfall ends. Here there is no real appreciation and no reallocation across sectors: the additional capital goods lower the return to industrial employment, and because the knowledge dynamics have an unstable threshold, a boom that outlasts the critical duration of Proposition 2 leaves a permanent loss. Krugman (1987) embeds learning by doing in a Dutch-disease setting. We add the multiplicity and the hysteresis.

at every date, then h_t would decrease monotonically and converge to $h_L^*(m')$;

(ii) $\bar{T}(m')$ is finite. If $T \geq \bar{T}(m')$ the economy converges to $h_L^*(m_0)$. If $T < \bar{T}(m')$ it converges to $h_H^*(m_0)$, with one exception: if $T = \bar{T}(m') - 1$ and $h_T = h_M^*(m_0)$ exactly, the economy remains at the separatrix. The exception arises for at most countably many boom sizes m' ;

(iii) $\bar{T}(m')$ is nonincreasing in m' .

The proof is in [Appendix A](#).

The path in [Proposition 2](#) does not depend on what agents expect. Preferences are quasilinear in consumption and the household holds no assets, so its labor choice solves the static condition (6). The learning externality is external: no household takes account of the fact that its own employment raises tomorrow's productivity. Equilibrium employment is therefore the function $\phi(h_t; m_t)$ of the current state alone, and the transition path is identical whether the duration T of the boom is perfectly foreseen at date 0 or arrives as a complete surprise. This is a property of the competitive equilibrium, not of learning by doing in general. It is what makes the trap a pure externality problem rather than a problem of coordinating expectations, and it is why the corrective policy below is a wedge rather than a device for selecting among self-fulfilling paths.

That policy is easy to characterize here, where there is one country and no strategic interaction. A planner who values the same household preferences but internalizes (5) solves

$$\max_{\{n_t\}} \sum_{t=0}^{\infty} \beta^t [f(\alpha(h_t)n_t + m_t) - v(n_t)] \quad \text{subject to} \quad h_{t+1} = (1 - \delta)h_t + \lambda n_t. \quad (14)$$

Let $\beta^t \nu_t$ be the Lagrange multiplier on the law of motion, with $\nu_t \geq 0$ the social value of the knowledge created at date t . The planner's first-order condition for employment is

$$v'(n_t) = \alpha(h_t) f'(k_t) + \lambda \nu_t, \quad (15)$$

which differs from the competitive condition (6) by exactly the term $\lambda \nu_t$, the social value of the knowledge created by an extra unit of work. The competitive economy therefore under-employs at every date, weakly, and strictly at every date at which $\nu_t > 0$. Too little employment means too little learning, so the whole law of motion sits below its efficient counterpart, and the separatrix is harder to stay above than it should be.

Proposition 3 (The corrective subsidy). *Let $\{n_t^P, h_t^P\}$ be a solution to (14), and let $\nu_t \geq 0$ be the multiplier on the knowledge law along it, so that (15) and (17) hold. Suppose labor income is subsidized at rate τ_t and the cost is rebated lump sum, so the household's condition becomes $v'(n_t) = (1 + \tau_t) \alpha(h_t) f'(k_t)$. The competitive allocation from the same h_0 coincides with $\{n_t^P, h_t^P\}$ if and only if*

$$\tau_t = \frac{\lambda \nu_t}{\alpha(h_t^P) f'(k_t^P)} \quad \text{for every } t. \quad (16)$$

The subsidy is strictly positive whenever $\nu_t > 0$, and at given (h_t, ν_t) it is increasing in the flow m_t .

Proof. If the two allocations coincide, the household's condition and (15) both hold at (h_t^P, n_t^P) , and subtracting gives $\tau_t \alpha(h_t^P) f'(k_t^P) = \lambda \nu_t$, which is (16). Conversely, suppose the rates are set by (16) and argue by induction on t . Suppose $h_t = h_t^P$, which holds at $t = 0$. At the stated rate the household's condition reads $v'(n) = \alpha(h_t^P) f'(\alpha(h_t^P)n + m_t) + \lambda \nu_t$, whose left side is strictly increasing in n and right side strictly decreasing, so it has a unique solution; n_t^P solves it by (15). Hence $n_t = n_t^P$ and $h_{t+1} = h_{t+1}^P$. Positivity is immediate. For monotonicity, at given (h_t, ν_t) employment solves (15), and differentiating gives $dk_t/dm_t = v''/(v'' - \alpha^2 f'') > 0$ as in Lemma 1, so k_t rises with m_t , $f'(k_t)$ falls, and the ratio in (16) rises. $\square$

The proposition takes a solution of the planner's problem as given. The problem is not concave, so its first-order conditions (15) and (17) are necessary at an interior optimum but do not identify one, and the solution need not be unique; Appendix C shows that at low discount factors the problem has two basins of attraction. What the proposition delivers is the form of the wedge that decentralizes whichever solution the planner selects, as a sequence of rates set along that solution.²

The subsidy (16) is forward-looking: it depends on the multiplier ν_t , which must be computed along the planner's path. At a steady state it does not have to be. The first-order condition of problem (14) for h_{t+1} is

$$\nu_t = \beta \left[\alpha'(h_{t+1}) n_{t+1} f'(k_{t+1}) + (1 - \delta) \nu_{t+1} \right], \quad (17)$$

which says that the value of the knowledge created today is the extra output it produces tomorrow, plus its undepreciated value thereafter, discounted. At a steady state $\nu_t = \nu_{t+1}$, so the multiplier is $\beta \alpha'(h) n f'(k) / [1 - \beta(1 - \delta)]$ and the subsidy of Proposition 3 takes the closed form $\beta \lambda \alpha'(h) n / (\alpha(h) [1 - \beta(1 - \delta)])$.

The instrument is thus indifferent to where the flow comes from. It responds to the depressive effect of m_t on employment, not to its origin, so the same wedge answers a resource dividend and a rival's export surplus; only the path of m_t differs. Two further implications follow from Proposition 2. Since only $m' > \bar{m}$ destroys the high steady state, a policy that keeps the dividend out of effective capital (saving it abroad rather than converting it into capital goods) preserves multiplicity however large the discovery. And since the loss becomes permanent only after $\bar{T}(m')$ periods, the policy problem carries a deadline as well as a target: keeping the dividend out of effective capital works before the separatrix is crossed and not after.

Appendix B asks how much of the multiplicity survives when policy pays for the knowledge. It indexes the correction by the share ζ of the steady-state wedge that the subsidy pays, so that $\zeta = 0$ is laissez-faire and $\zeta = 1$ reproduces the steady states of the planner's first-order conditions, and follows the steady-state set as ζ rises. Enough policy removes the

²A solution exists: flows are bounded above and v is unbounded, so employment can be confined to a compact interval without loss, and the discounted objective is continuous on the resulting product space. Condition (17) is the first-order condition for a perturbation of h_{t+1} alone, with n_t and n_{t+1} adjusting, and is therefore necessary at any interior optimum; its bounded solution, the discounted sum of future marginal products of knowledge, is nonnegative because $\alpha' > 0$.

low steady state through a saddle-node, so the trap is a property of the allocation mechanism, not of the technology. Internalization alone need not do it: the intensity at which the low steady state disappears can be many times the Pigouvian wedge, as it is at the parameters of Section 6.³ The same argument, run in the flow rather than in the policy, gives an export fold $\underline{x}(\zeta)$: a country that sustains a large enough surplus of capital goods loses its low steady state too, and the fold falls with ζ . Section 6 reports both folds.

4 Trade Between Two Countries

Section 3 treated the flow of capital goods as exogenous. This section makes it a choice. A second economy, the East, produces the capital goods, sells them to the West at a market price, and chooses how much to ship. The model is stated once, for a general security weight ψ : the East's welfare falls with the West's industrial capability at a rate governed by ψ , and $\psi = 0$ is ordinary commerce. We derive the equilibrium conditions for general ψ and solve the problem in a benchmark where exact results are available. There is a critical security weight. Above it the East runs a surplus commerce would not run, and the West is trapped. Below it commerce between the two industrialized countries is autarky, given that the benchmark assumes no static gain from trade between them and that a condition on the primitives keeps the critical security weight above zero. The paper's measure of predation is the export in excess of its commercial level.

Throughout, the two countries share the same technology and the same quasilinear preferences over consumption and leisure.

4.1 Environment

Each country $j \in \{E, W\}$ has the technology of Section 3: output $c_t^j = f(k_t^j)$, effective capital $k_t^j = \alpha(h_t^j)n_t^j + m_t^j$ with m_t^j the country's net imports of the capital good, productivity (3), and knowledge that evolves by (5).

West. The West is a decentralized market economy. Consumption is the numeraire, and the imported capital good trades at the relative price q_t (in consumption units). Western households maximize $\sum_{t=0}^{\infty} \beta^t [c_t^W - v(n_t^W)]$ by choosing employment n_t^W , the quantity of imports $x_t \geq 0$, and one-period real bond issuance b_t^W , subject to the budget

$$c_t^W + R_{t-1} b_{t-1}^W = f(k_t^W) - q_t x_t + b_t^W, \quad k_t^W = \alpha(h_t^W) n_t^W + x_t, \quad (18)$$

taking (q_t, R_t) as given. Optimality gives two conditions each period, a labor margin and an import margin:

$$v'(n_t^W) = \alpha(h_t^W) f'(\alpha(h_t^W) n_t^W + x_t), \quad f'(\alpha(h_t^W) n_t^W + x_t) = q_t, \quad (19)$$

³Whether the low steady state can be part of the *efficient* allocation is a separate question. Appendix C shows that it can, but only under heavy discounting: the planner's problem then has two basins of attraction, the history dependence of Skiba (1978) and Dechert and Nishimura (1983), and at the discount factor used here its low basin is empty.

Because utility is linear in consumption, the bond condition pins the safe rate at the rate of time preference:

$$1 = \beta R_t \implies R_t = 1/\beta. \quad (20)$$

The import margin says the West buys capital goods until their marginal product equals the price. Combined with the labor margin, it reproduces the employment function of Section 3, $n_t^W = \phi(h_t^W; x_t)$, with the flow now equal to the traded quantity. The only change is what stands behind the flow: a market price $q_t = f'(k_t^W)$ rather than an unexplained transfer. Everything Section 3 established about ϕ applies unchanged. A larger inflow depresses Western employment (Lemma 1), and Western knowledge follows the law of motion (11) at $m = x_t$.

East. The East is governed by a social planner. It produces capital goods, ships $x_t \geq 0$ of them to the West, and is paid the world price q_t per unit, which it holds as a claim on the West, its reserves B_t^E . Shipping x_t reduces the East's own effective capital one-for-one, so domestic output is $f(\alpha(h_t^E)n_t^E - x_t)$ and the flow budget is

$$c_t^E + (B_t^E - R_{t-1}B_{t-1}^E) = f(\alpha(h_t^E)n_t^E - x_t) + q_t x_t, \quad (21)$$

with $m_t^E = -x_t$ and $m_t^W = x_t$: consumption plus net reserve accumulation equals domestic output plus export earnings. Feasibility requires $0 \leq x_t \leq \alpha(h_t^E)n_t^E$.

The East's objective. The Eastern planner's per-period welfare adds one term to the common consumption-leisure payoff:

$$W_t^E = c_t^E - v(n_t^E) - \psi g(h_t^W), \quad (22)$$

with the security-loss function

$$g(h^W) \equiv \mathcal{D}(\alpha(h^W)), \quad (23)$$

where $\mathcal{D}$ is strictly increasing. Here $\psi \geq 0$ is the weight on the security term. One reading is that ψ is the per-period probability of a geopolitical contingency, a war, a sanctions episode, or a technological embargo, and $\mathcal{D}$ is the damage the West could inflict in that contingency with the industrial capability $\alpha(h^W)$. The damage depends on how much of that capability converts to military use, most of it for semiconductors, machine tools, or shipyards and little of it for food, textiles, or services; that conversion rate, and a fixed mobilizable labor stock, are scales inside $\mathcal{D}$. By (23), the West's knowledge enters the East's payoff only through $\alpha(h^W)$, so g is increasing in h^W and inherits the S-shape of α . Consumption enters linearly, as in the Western household problem (1): the two countries share the same preferences over consumption and leisure and differ only in this term, which the West does not carry. At $\psi = 0$ the term is absent.

The reserve asset and market clearing. The West pays for its imports with consumption goods, now or later. The asset market clears at $B_t^E = b_t^W$, and the capital-good market

clears at the traded quantity x_t . The asset is real, denominated in consumption, and safe; by (20) it earns $R_t = 1/\beta$. The pattern of trade is therefore simple. The East exports capital goods and imports consumption goods, and the bond moves only the timing of the second flow. In a steady state with a constant shipment x , price q , and bond position $\bar{B}$, the West ships $qx + (R - 1)\bar{B}$ units of consumption to the East each period, the import bill plus the interest on its debt, and the East's trade balance is a deficit of $(R - 1)\bar{B}$ matched by interest income of the same size. The surplus this paper studies is a surplus in the capital good, not a current-account surplus. Along a path the split between consumption goods now and claims is indeterminate: utility is linear in consumption on both sides and the return equals the rate of time preference, so the West is indifferent between paying and borrowing, and the East between consuming and lending. What is pinned down is the present value. We impose the no-Ponzi condition $\lim_{t \rightarrow \infty} \beta^t b_t^W = 0$ on the West's debt, under which the present value of the West's consumption exports equals the present value of its capital-good imports plus its initial debt. Nothing below turns on the size of the East's claim.

4.2 The East's Problem

The East is a Stackelberg leader. At date 0 it commits to a policy path, anticipating how the West will respond; the West is competitive and re-optimizes each period. The East cannot choose Western employment directly. Its only instrument is the export flow x_t , and Western employment then follows from the West's own employment function (10). This is what makes the problem a Ramsey problem: the East optimizes over allocations that the Western economy will support as a competitive equilibrium.

The East faces two prices and treats them differently. In the bond market it is a price-taker: the safe return is $R_t = 1/\beta$, pinned by the West's bond condition (20). In the goods market it is large. The West's employment function (10) gives Western employment, $n_t^W = \phi(h_t^W; x_t)$, and the West's import margin (19) ties the price the West pays to the quantity it receives. The East chooses its price and quantity subject to that demand schedule.

The East chooses sequences of its consumption, domestic employment, exports, reserves, the price, and the two knowledge stocks,

$$\{ c_t^E, n_t^E, x_t, B_t^E, q_t, h_{t+1}^E, h_{t+1}^W \}_{t=0}^\infty,$$

to solve

$$V^E(h_0^E, h_0^W, B_{-1}^E) = \max \sum_{t=0}^{\infty} \beta^t \left[c_t^E - v(n_t^E) - \psi g(h_t^W) \right] \quad (24)$$

subject to, for all $t \geq 0$, the East's budget, the two knowledge laws, the West's import

demand, and feasibility:

$$c_t^E + B_t^E - R_{t-1}B_{t-1}^E = f(\alpha(h_t^E)n_t^E - x_t) + q_t x_t, \quad (25)$$

$$h_{t+1}^E = (1 - \delta)h_t^E + \lambda n_t^E, \quad (26)$$

$$h_{t+1}^W = (1 - \delta)h_t^W + \lambda \phi(h_t^W; x_t), \quad (27)$$

$$q_t = f'(\alpha(h_t^W)\phi(h_t^W; x_t) + x_t), \quad (28)$$

$$x_t \geq 0, \quad B_t^E \geq 0, \quad x_t \leq \alpha(h_t^E)n_t^E, \quad (29)$$

with initial conditions (h_0^E, h_0^W, B_{-1}^E) given and the safe return $R_t = 1/\beta$ from (20).

Constraints (27) and (28) are the implementability constraints of the Ramsey program: the West's knowledge law with Western employment replaced by the employment function, $n_t^W = \phi(h_t^W; x_t)$, and the West's import margin at that employment. They are the only channels through which the East's choices reach the Western economy. The budget (25) determines c_t^E .⁴

The first-order conditions are recorded here for interpretation; the analysis of Section 4.3 argues from plans. Let $\beta^t \mu_t^E$ and $\beta^t \mu_t^W$ be the Lagrange multipliers on the two laws of motion (26) and (27), and let $\beta^t \vartheta_t$ be the multiplier on the demand constraint (28). Because the objective is linear in consumption, the multiplier on the budget (25) is β^t . We sign μ_t^W so that raising h_{t+1}^W is costly: $\mu_t^W \geq 0$ is the value to the East of a one-unit reduction in West knowledge. The first-order condition for q_t gives $\vartheta_t = x_t$: relaxing the demand constraint by one unit of price is worth the x_t units sold at that price. At an interior optimum, the conditions for n_t^E and x_t are

$$v'(n_t^E) = \alpha(h_t^E) f'(\alpha(h_t^E)n_t^E - x_t) + \lambda \mu_t^E, \quad (30)$$

$$f'(\alpha(h_t^E)n_t^E - x_t) = \underbrace{q_t + x_t \frac{\partial q_t}{\partial x_t}}_{\text{marginal export revenue}} + \underbrace{\lambda \mu_t^W \left(-\frac{\partial \phi(h_t^W; x_t)}{\partial x_t} \right)}_{\text{marginal deindustrialization benefit}}, \quad (31)$$

and the conditions for h_{t+1}^E and h_{t+1}^W are

$$\mu_t^E = \beta \left[\alpha'(h_{t+1}^E) n_{t+1}^E f'(\alpha(h_{t+1}^E)n_{t+1}^E - x_{t+1}) + (1 - \delta) \mu_{t+1}^E \right], \quad (32)$$

$$\mu_t^W = \beta \left[\psi g'(h_{t+1}^W) + x_{t+1} \left(-\frac{\partial q_{t+1}}{\partial h_{t+1}^W} \right) + \mu_{t+1}^W \left((1 - \delta) + \lambda \frac{\partial \phi(h_{t+1}^W; x_{t+1})}{\partial h_{t+1}^W} \right) \right], \quad (33)$$

where $\partial q_t / \partial x_t < 0$ and $\partial q_t / \partial h_t^W < 0$ by (28) and Lemma 1. At a corner with $x_t = 0$, (31) holds with $\leq$ replacing $=$.

Condition (33) lists the two reasons the East values a reduction in h_{t+1}^W . The first term is the security loss $\psi g'$. The second is a terms-of-trade gain: $\partial q / \partial h^W < 0$, so a less industrialized West pays a higher price, on each of the x_{t+1} units it buys. The security weight enters

⁴The West's budget is not a separate constraint on the East's problem. West consumption does not enter the East's objective, and by Walras's law the East's budget, asset-market clearing $B_t^E = b_t^W$, and the West's budget together imply the world resource constraint $c_t^E + c_t^W = f(k_t^E) + f(k_t^W)$, so imposing any two delivers the third. The West's budget only pins down c_t^W , which we assume stays nonnegative.

the system only through the first term. Conditions (30) and (32) do not involve ψ : the security motive enters the East's employment optimality condition only through the shipment x_t .

The export condition (31) prices one shipped unit. The left-hand side is its cost: one unit less of effective capital at home, worth $f'(\alpha(h_t^E)n_t^E - x_t)$ in consumption. The first term on the right is marginal export revenue: the price q_t , net of the terms-of-trade loss $x_t \partial q_t / \partial x_t < 0$ on the units already sold. The second term is the deindustrialization benefit: shipping more depresses Western employment ($\partial \phi / \partial x < 0$, Lemma 1), which lowers h_{t+1}^W , valued by the East at μ_t^W .

The East's financial claim is payoff-irrelevant. With the safe rate pinned at $R_t = 1/\beta$ by (20), the East's intertemporal margin holds with equality for any path of B_t^E , so its level is indeterminate. The East effectively consumes its export earnings, and the single payoff-relevant state it steers is the West's knowledge h_t^W .

4.3 Predation and Welfare

This subsection contains the main result of the section: there is a critical security weight above which the East's optimal plan drives the West across its threshold. The plan is a finite *flood*, a run of shipments that carries the West's knowledge down to the threshold, followed by a return to ordinary commerce with a trapped customer. The East's problem is not concave, so we do not argue from first-order conditions. Instead we let the steepness γ of (3) grow without bound, so that the productivity schedule becomes a step, α_L at and below χ and α_H above it. This *sharp-threshold benchmark* keeps every mechanism of the model and removes one thing, the curvature of α between its two plateaus. In it we prove the results analytically. The subsection first reduces the East's problem to a problem with a single state, the West's knowledge. It then derives the critical security weight, first in an economy in which the East receives nothing for its shipments and then with paid shipments; asks whether commerce alone, with no security motive, can produce predation; characterizes the optimal export path above the critical security weight; and compares the East's gain from predation with the West's loss.

Write x_t^c for the East's optimal export at $\psi = 0$, the *commercial export*. The part of a shipment above its commercial level is the part that only the security motive explains, and it is the paper's measure of predation. In the benchmark below the commercial export between two industrialized countries is zero, so every unit shipped to an industrialized West is predation.

In the benchmark a country's productivity depends only on which side of χ its knowledge lies; call the side its *regime*, $i \in \{H, L\}$. Write $\varphi_i(x)$ for the West's employment when its regime is i and it imports x , the solution of $v'(n) = \alpha_i f'(\alpha_i n + x)$, and $q_i(x) = f'(\alpha_i \varphi_i(x) + x)$ for the price it pays. Write $n^E(x)$ for the solution of $v'(n) = \alpha_H f'(\alpha_H n - x)$, and define

$$\Pi_i(x) \equiv f(\alpha_H n^E(x) - x) - v(n^E(x)) + q_i(x) x. \quad (34)$$

Assumption 3 (Sharp-threshold benchmark). *The productivity schedule is the step*

$$\alpha(h) = \begin{cases} \alpha_L & h \leq \chi, \\ \alpha_H & h > \chi, \end{cases} \quad 0 < \alpha_L < \alpha_H, \quad (35)$$

and:

- (i) $\lambda\varphi_L(0) < \delta\chi < \lambda\varphi_H(0)$: in autarky the benchmark has a low steady state below χ and a high steady state $h_H = \lambda\varphi_H(0)/\delta$ above it.
- (ii) Both countries begin at h_H .
- (iii) Π_H attains its maximum at $x = 0$, and the ratio $-\Pi'_H(x)/(-\lambda\varphi'_H(x))$ is strictly increasing on $(0, \infty)$.
- (iv) There is a feasible shipment x with $\lambda\varphi_H(x) < \delta\chi$. If the East ships this x at every date, the West's knowledge falls below χ in finite time, so the East can move the West from regime H to regime L .

The four parts do distinct work. Part (i) is a condition on the decentralized West: at low productivity its employment is too low to sustain a knowledge stock of χ , and at high productivity it sustains more than χ . The decentralized West therefore has an autarky steady state in each regime. Part (ii) starts both countries industrialized, and this is the interesting configuration. A West that starts at or below χ is already trapped, so there is nothing to destroy. An East that starts at or below χ cannot trap the West and remain pre-industrial: a flood requires labor, and at low productivity that labor industrializes the country supplying it (Appendix D). The assumption also removes the East's own knowledge as a state. Part (iii) does two things. Its first half says that with the security motive off, at $\psi = 0$, autarky is the East's best policy against an industrialized customer, so pushing the West down always costs flow payoff. Its second half is a weak sufficient condition that, absent concavity, delivers the shape of the optimal export path. Part (iv) says a large enough import flow pulls the West's knowledge below χ ; the smallest such flow is the benchmark's import fold, written $\bar{m}_\infty$, the counterpart of the import fold $\bar{m}$ of Section 3. Without part (iv) no export path can trap the West. Each object of the benchmark has a counterpart in Section 3. The employment function $\varphi_i(x)$ is $\phi(h; x)$ evaluated on plateau i ; the high autarky stock h_H is $h_H^*(0)$; the fold $\bar{m}_\infty$ is $\bar{m}$; and the threshold χ plays the part of the separatrix h_M^* , which has no counterpart because the step has no interior steady state.

Assumption 3 simplifies both countries' problems. The East starts above the threshold, and the next lemma shows that it stays there whatever it ships, so its productivity is α_H at every date. Employment today therefore adds nothing to the East's productivity tomorrow, and its employment at date t is $n^E(x_t)$, which depends only on that date's shipment. Its payoff at date t from shipping x to a West in regime i is $\Pi_i(x)$. Its reserves do not enter this payoff: the bond's gross return $1/\beta$ is the inverse of the East's discount factor, so the reserve terms cancel from its present value. On the other side, the West's employment and the price it pays depend only on its regime and the shipment, through $\varphi_i(x)$ and $q_i(x)$, and the security

loss takes two values, $g_i = \mathcal{D}(\alpha_i)$, with $g_L < g_H$. Write x_i^c for the shipment that maximizes Π_i , the East's commercial export to a West in regime i . Between two industrialized countries the autarky prices are equal, so $x = 0$ is a critical point of Π_H , and part (iii) of the assumption makes it the global maximum: $x_H^c = 0$. A trapped West pays a higher price, so $x_L^c > 0$.

Lemma 2 (Reduction to a one-state problem). *Under Assumption 3, for every $\psi \geq 0$:*

(i) *an export path $\{x_t \geq 0\}$ is optimal for the East if and only if it solves*

$$\max_{\{x_t \geq 0\}} \sum_{t=0}^{\infty} \beta^t [\Pi_{i_t}(x_t) - \psi g_{i_t}] \quad \text{subject to} \quad h_{t+1}^W = (1 - \delta)h_t^W + \lambda \varphi_{i_t}(x_t), \quad h_0^W = h_H, \quad (36)$$

where $i_t = H$ if $h_t^W > \chi$ and $i_t = L$ otherwise;

(ii) *regime L is absorbing: once $h_t^W \leq \chi$, no export path returns the West above χ ;*

(iii) *among plans that never send the West's knowledge to or below χ , the best ships $x_t = x_H^c = 0$ at every date; and once the West's knowledge is at or below χ , the optimal continuation ships $x_t = x_L^c$ at every later date.*

Proof. (i) With $R_t = 1/\beta$, the reserve terms in the East's budget telescope out of its present value for any bounded reserve path, so its period payoff is consumption net of labor disutility plus the security term, $f(\alpha(h_t^E)n_t^E - x_t) - v(n_t^E) + q_t x_t - \psi g_{i_t}$. Fix any shipment path. Because f is increasing and $\alpha(h) \leq \alpha_H$ for every h , the date- t payoff is at most $\max_n [f(\alpha_H n - x_t) - v(n)] + q_t x_t - \psi g_{i_t}$, and the price q_t depends only on the shipment and on the West. The static employment $n^E(x_t)$ attains this bound at every date provided the East is in regime H throughout, and it is: $dn^E/dx > 0$ by the argument of Lemma 1 applied at $m = -x$, so $n^E(x_t) \geq n^E(0)$, and since the East starts at $h_H = \lambda n^E(0)/\delta$, induction gives $h_t^E \geq h_H > \chi$ at every date. Any other employment path earns strictly less at some date, because $f(\alpha_H n - x) - v(n)$ is strictly concave in n , and the static path is feasible, since $n^E(x)$ leaves the East positive effective capital. Whatever the shipments, the East's payoff at date t is therefore $\Pi_{i_t}(x_t) - \psi g_{i_t}$ on any optimal path, and the only constraint linking periods is the Western knowledge law with φ_{i_t} .

(ii) Let $h_t^W \leq \chi$. Then $h_{t+1}^W = (1 - \delta)h_t^W + \lambda \varphi_L(x_t) \leq (1 - \delta)\chi + \lambda \varphi_L(0) < (1 - \delta)\chi + \delta\chi = \chi$, using φ_L decreasing in x and Assumption 3(i). Induction completes the claim.

(iii) In regime L the regime is permanent by (ii), and the flow payoff $\Pi_L(x_t) - \psi g_L$ involves no state, so it is maximized date by date at x_L^c . In regime H , conditional on never crossing, the flow payoff is $\Pi_H(x_t) - \psi g_H$; its date-by-date maximum is $x_t = 0$ by Assumption 3(iii), and under $x \equiv 0$ the West stays at $h_H > \chi$, so the no-crossing restriction does not bind. $\square$

By the lemma, the East's problem reduces to one decision: whether to move the West's knowledge to or below χ , and if so, at which date. The West's response at each date depends only on that date's state and shipment, and the reduced problem has a single state, so a plan that is optimal at date 0 remains optimal from every state it reaches. The commitment assumed in Section 4.2 is therefore without loss here. If the East never moves the West

below χ , its best policy is to ship nothing, by Lemma 2(iii), and it earns $\Pi_H(0) - \psi g_H$ in every period. If it does move the West below χ , it gives up flow payoff in each period before the crossing, because shipping to an industrialized West pays less than autarky. After the crossing it earns more in every period, for two reasons. First, a trapped West produces few capital goods of its own, so its marginal product of capital is high and it pays a higher price for imports. Second, a trapped West is less of a threat: $g_L < g_H$. Write $K(x) \equiv \Pi_H(0) - \Pi_H(x)$ for the flow payoff the East gives up by shipping x rather than nothing to an industrialized West; it is nonnegative by Assumption 3(iii). An export path $P = \{x_t\}_{t \geq 0}$ is a *crossing plan* if it carries the West's knowledge to or below χ at some finite date. Write T_P for the first such date, and define the cost of the plan as

$$C_P \equiv \sum_{t < T_P} \beta^t K(x_t),$$

the discounted flow payoff the plan gives up before the crossing. The subscript records that T_P and C_P are functions of the path. Both can be computed from the path alone. While the West is in regime H , its employment and the price it pays at date t depend only on that date's shipment: employment is $\varphi_H(x_t)$, and the price is $q_H(x_t) = f'(\alpha_H \varphi_H(x_t) + x_t)$, the West's marginal product of capital, as its import margin (19) requires. The path therefore determines the West's knowledge at every date, and hence T_P . It also determines the East's flow payoff $\Pi_H(x_t)$ at every date, through the East's employment $n^E(x_t)$ and the price $q_H(x_t)$, and hence C_P . Neither computation requires solving the East's problem. Write $\Delta\Pi \equiv \Pi_L(x_L^c) - \Pi_H(0) > 0$ for the commercial gain and $\Delta g \equiv g_H - g_L > 0$ for the security gain.

4.3.1 Predation with free shipments

We first find the critical security weight in a simpler economy, in which the East receives nothing for its shipments. This case isolates the security motive and is a step toward the main result. The West absorbs the shipments as the inflow m of Section 3. At a price of zero the West's import margin has no finite solution, so the quantity is not demanded but delivered: the East chooses it. Write $\Pi^0(x) \equiv \Pi_i(x) - q_i(x)x = f(\alpha_H n^E(x) - x) - v(n^E(x))$ for the East's flow payoff from a gift, which does not depend on the West's regime and falls strictly in x by the envelope theorem; $K^0(x) \equiv \Pi^0(0) - \Pi^0(x)$ for the flow cost of a gift; and $C_P^0 \equiv \sum_{t < T_P} \beta^t K^0(x_t)$ for the discounted cost of a crossing plan's flood. Since Π^0 does not depend on the West's regime, trapping the West brings the East no commercial gain: the gift economy's counterpart of $\Delta\Pi$ is zero.

Proposition 4 (The critical security weight with free shipments). *Let the East receive nothing for its shipments, and let parts (i), (ii), and (iv) of Assumption 3 hold. Then crossing is strictly optimal for $\psi > \psi_0^*$ and strictly suboptimal for $\psi < \psi_0^*$, where*

$$\psi_0^* = \frac{1 - \beta}{\Delta g} \inf_P \beta^{-T_P} C_P^0,$$

the infimum over crossing plans.

Proof. The reduction of Lemma 2 applies, because its proof does not use the price. The one difference is that after a crossing the East's best continuation ships nothing, because Π^0 is decreasing. The value of never crossing is $[\Pi^0(0) - \psi g_H]/(1 - \beta)$, and the value of a crossing plan P is $\sum_{t < T_P} \beta^t [\Pi^0(x_t) - \psi g_H] + \beta^{T_P} [\Pi^0(0) - \psi g_L]/(1 - \beta)$. Subtracting the first from the second, the gain from plan P over never crossing is

$$-C_P^0 + \beta^{T_P} \frac{\psi \Delta g}{1 - \beta}.$$

This gain is affine in ψ with the positive slope $\beta^{T_P} \Delta g/(1 - \beta)$. It is zero at $\psi = z_P^0 \equiv (1 - \beta) \beta^{-T_P} C_P^0 / \Delta g$, positive for $\psi > z_P^0$, and negative for $\psi < z_P^0$. If $\psi > \inf_P z_P^0$, then $\psi > z_P^0$ for some crossing plan, that plan gains, and crossing is strictly optimal. If $\psi < \inf_P z_P^0$, then every crossing plan loses, and crossing is strictly suboptimal. The infimum is taken over a nonempty set: by Assumption 3(iv), shipping a constant x with $\lambda \varphi_H(x) < \delta \chi$ carries the West's knowledge below χ in finite time, so crossing plans exist. $\square$

4.3.2 Predation with paid shipments

With the price restored, each crossing plan's break-even weight is the gift's, adjusted for what the price adds: the revenue on the active shipments, and the commercial gain once the customer is trapped. Write $z_P \equiv [(1 - \beta) \beta^{-T_P} C_P - \Delta \Pi] / \Delta g$ for the weight at which plan P breaks even when the East is paid. Because $K(x) = K^0(x) - q_H(x)x$,

$$z_P = z_P^0 - \frac{(1 - \beta) \beta^{-T_P}}{\Delta g} \sum_{t < T_P} \beta^t q_H(x_t) x_t - \frac{\Delta \Pi}{\Delta g}, \quad (37)$$

where $\Delta \Pi = \Pi_L(x_L^c) - \Pi_H(0)$ is the East's per-period commercial gain from a trapped customer. Both subtracted terms are nonnegative, so $z_P \leq z_P^0$ for every crossing plan. Taking the infimum over plans and truncating at zero gives $\psi^* \leq \psi_0^*$: the security weight at which the East floods is lower when it is paid than when it gives its shipments away. Payment lowers the critical security weight in two ways. The revenue on the shipments offsets part of their cost, and after the crossing the trapped customer pays a higher price. Payment is not necessary, however: by Proposition 4, for every security weight above ψ_0^* crossing is strictly optimal even when the East receives nothing for its shipments.

Proposition 5 (The critical security weight). *Under Assumption 3:*

(i) *There is a unique critical security weight*

$$\psi^* = \max \left\{ 0, \inf_P \frac{(1 - \beta) \beta^{-T_P} C_P - \Delta \Pi}{\Delta g} \right\}, \quad (38)$$

the infimum taken over crossing plans, such that crossing is strictly optimal for $\psi > \psi^$ and strictly suboptimal for $\psi < \psi^*$.*

(ii) *Wherever $\psi^* > 0$, it is strictly decreasing in the security gap Δg .*

Proof. By Lemma 2(iii), the value of never crossing is $V_{\text{stay}}(\psi) = [\Pi_H(0) - \psi g_H]/(1 - \beta)$. A crossing plan P with date T_P and active exports $\{x_t\}_{t < T_P}$ has value

$$V(P; \psi) = \sum_{t < T_P} \beta^t [\Pi_H(x_t) - \psi g_H] + \beta^{T_P} \frac{\Pi_L(x_L^c) - \psi g_L}{1 - \beta},$$

where the continuation uses Lemma 2(ii) and (iii). Subtracting,

$$\Delta(P; \psi) \equiv V(P; \psi) - V_{\text{stay}}(\psi) = -C_P + \beta^{T_P} \frac{\Delta\Pi + \psi \Delta g}{1 - \beta}. \quad (39)$$

Each $\Delta(P; \cdot)$ is affine in ψ with strictly positive slope $\beta^{T_P} \Delta g / (1 - \beta)$, so it has a unique zero, the break-even weight z_P defined before the proposition, and $\Delta(P; \psi) > 0$ exactly when $\psi > z_P$. Crossing plans exist by Assumption 3(iv): a constant export $\bar{x}$ with $\lambda\varphi_H(\bar{x}) < \delta\chi$ drives h^W monotonically toward $\lambda\varphi_H(\bar{x})/\delta < \chi$ and so crosses in finite time. The infimum in (38) is therefore over a nonempty set, and $\psi^* = \max\{0, \inf_P z_P\}$ is finite. If $\psi > \psi^*$, then $\psi > z_P$ for some crossing plan P , so $\Delta(P; \psi) > 0$: some crossing plan beats never crossing, and crossing is strictly optimal. If $\psi < \psi^*$, then $\psi < z_P$ for every crossing plan, so every crossing plan loses to never crossing, whose best version is autarky by Lemma 2(iii): crossing is strictly suboptimal. This proves (i). The comparison is plan by plan, and it has to be: the supremum of $\Delta(P; \psi)$ over crossing plans is never negative, because a plan P preceded by N periods of autarky, which keep the West at h_H , is a crossing plan with gain $\beta^N \Delta(P; \psi)$, and that tends to zero.

For later use, write $\Delta(\psi) \equiv \sup_P \Delta(P; \psi)$ for the East's value gain from predation, the value of its problem less $V_{\text{stay}}(\psi)$. By the argument just given, wherever $\psi^* > 0$ the gain is zero on $[0, \psi^*]$, and it is positive on (ψ^*, ∞) . It is finite, since $\Delta(P; \psi) \leq (\Delta\Pi + \psi \Delta g)/(1 - \beta)$, convex and continuous on $(0, \infty)$ as a finite supremum of affine functions, nondecreasing because each $\Delta(P; \cdot)$ is, and unbounded above, since $\Delta(P; \psi) \rightarrow \infty$ as $\psi \rightarrow \infty$ for any fixed crossing plan P .

For (ii): in (38) the numerator is free of Δg and strictly positive wherever $\psi^* > 0$, so ψ^* falls strictly as Δg rises. $\square$

4.3.3 Predation without a security motive

We now ask whether the East would trap the West with no security motive at all, that is, whether the critical security weight of Proposition 5 can be zero. It can. That happens when some crossing plan is profitable at $\psi = 0$, so that the East traps the West for commercial reasons alone. By (38), the critical security weight is strictly positive if and only if

$$\inf_P \beta^{-T_P} C_P > \frac{\Delta\Pi}{1 - \beta}.$$

The left side is the lowest cost of crossing, with each plan's cost measured at its own crossing date. The right side is the value, at the crossing date, of the commercial gain $\Delta\Pi$ that the

East earns in every period from the crossing on.⁵

Whether the inequality holds depends on how patient the East is. The left side can be written as

$$\beta^{-T_P} C_P = \sum_{t < T_P} \beta^{-(T_P - t)} K(x_t),$$

which carries each period's forgone payoff forward to the crossing date with interest. It falls as β rises, while the right side rises. Nothing else in the comparison depends on β , because the flow payoffs, the West's law of motion, and hence the set of crossing plans do not involve it. The inequality therefore holds when the East is impatient and fails when it is patient enough, and Proposition 6 below locates the cutoff.

The inequality also has a sufficient condition that does not require finding the cheapest plan. Every period of forgone payoff comes at least one period before the crossing, so each term of the sum is at least $K(x_t)/\beta$, and the cost of any crossing plan, measured at its crossing date, is at least C_0/β , where

$$C_0 \equiv \inf \left\{ \sum_{t=0}^{T-1} K(x_t) : h_0 = h_H, h_{t+1} = (1 - \delta)h_t + \lambda \varphi_H(x_t), h_T \leq \chi \right\}. \quad (40)$$

The infimum is taken over export paths of any finite length T . C_0 is the smallest total flow payoff, summed without discounting, that any export path gives up while it carries the West's knowledge from h_H to χ or below. It is computed from the West's law of motion and the flow payoff Π_H alone; neither β , nor ψ , nor the East's problem enters it. Comparing C_0/β with $\Delta\Pi/(1 - \beta)$ gives the condition

$$C_0 > \frac{\beta \Delta\Pi}{1 - \beta}, \quad (D)$$

which is equivalent to $\beta < C_0/(C_0 + \Delta\Pi)$.

Proposition 6 (Commerce and patience). *Under Assumption 3 there is a discount factor $\hat{\beta}$ with $C_0/(C_0 + \Delta\Pi) \leq \hat{\beta} < 1$ such that:*

- (i) *If $\beta > \hat{\beta}$, then $\psi^* = 0$, and at $\psi = 0$ crossing is strictly optimal: the East traps the West for commercial gain alone.*
- (ii) *If $\beta < \hat{\beta}$, then $\psi^* > 0$, and at $\psi = 0$ the optimal path is the commercial equilibrium: the East ships nothing, and the West stays at its high steady state forever.*
- (iii) *Condition (D) is sufficient for case (ii): if it holds, then $\beta < C_0/(C_0 + \Delta\Pi) \leq \hat{\beta}$. The critical security weight then satisfies the explicit bound*

$$\psi^* \geq \frac{(1 - \beta) C_0/\beta - \Delta\Pi}{\Delta g} > 0.$$

⁵The inequality concerns the infimum. If every crossing plan loses money at $\psi = 0$ but the losses, measured at the crossing date, approach zero along a sequence of plans, then any positive security weight makes some plan profitable, and $\psi^* = 0$. This happens only when the two sides of the inequality are equal.

The proof is in [Appendix A](#). Parts (i) and (ii) say that a purely commercial East traps an industrialized customer when it is patient enough and ships nothing when it is not. Part (iii) gives a condition that can be checked without solving the East's problem. The condition is sufficient but not necessary. It bounds the cost of each crossing plan from below by C_0/β , which carries every period's forgone payoff forward by one period only, while the actual cost carries each payoff forward to the crossing date. The cutoff $\hat{\beta}$ can therefore exceed $C_0/(C_0 + \Delta\Pi)$. The difference comes only from compounding within the active phase, and the compounding factors $\beta^{-(T_P-t)}$ approach one as β approaches one. Section 6 computes $\hat{\beta}$ and both bounds.

4.3.4 The two-phase path of predation

Propositions 5 and 6 say when the East crosses, but not how. We now describe the East's optimal export path when $\psi > \psi^*$.

Proposition 7 (Two-phase predation). *Under Assumption 3, let $\psi > \psi^*$. An optimal export path exists, and every optimal export path has exactly two phases.*

- (i) Active phase ($0 \leq t < T^*$, with T^* finite): *the export is positive at every date, is strictly increasing, and reaches its maximum in the period the West's knowledge reaches χ ; the West's knowledge falls strictly throughout.*
- (ii) Passive phase ($t \geq T^*$): *the West's regime is L and remains so, the East ships its commercial export x_L^c .*

T^* is the first date at which $h_t^W \leq \chi$, and at that date $h_{T^*}^W = \chi$.

The proof is in [Appendix A](#). When $\psi^* > 0$, the East's optimal export at $\psi = 0$ is zero at every date, by Proposition 5 and Lemma 2(iii), so every unit shipped in the active phase exceeds the commercial export and is predation.

Two parts of the proposition need explanation: why the crossing date is finite, and why the export rises until the crossing.

The crossing date. Crossing is an investment. The East pays for the flood first, in forgone payoff, and collects the gains afterward: once the West is trapped, it pays a higher price and it is less of a threat, in every period from then on. The East has a reason to take its time. A slower flood is cheaper, because shipping a little to an industrialized West costs the East almost nothing at the margin. It also has a reason to hurry, because the gains begin only at the crossing, and delay discounts them. The second reason eventually wins. Whatever the flood costs, a plan that crosses at date T gains at most β^T times the value of the gains measured at the crossing date, and that bound goes to zero as T grows. When $\psi > \psi^*$, some plan that crosses at a finite date gains a positive amount, so crossing ever later cannot be optimal, and the East crosses at a finite date. [Appendix A](#) shows in addition that the optimal crossing date does not increase with the security weight. When $\psi^* > 0$, it also shows that the crossing date grows without bound as the weight falls to the critical security weight, so just above it the East crosses late, and that no crossing plan breaks even

exactly at the critical security weight, which is the limit of the break-even weights of ever slower plans. Section 6 reports the dates.

The rising export. Once the crossing date is fixed, the East's problem is to bring the West's knowledge down to χ by that date as cheaply as possible. Two things make a late shipment a better buy than an early one. It is paid for later, so it costs less in present value. And more of its effect survives to the crossing date: a shipment lowers the knowledge the West creates in the period it arrives, and knowledge that is not created early would have partly depreciated by the crossing date anyway. The East therefore concentrates the flood near the crossing date. It does not ship everything in the last period, because a shipment's cost rises faster than its effect on Western employment as the shipment grows, which is what Assumption 3(iii) says. Balancing the two, the East ships more in each period than in the one before, and ships the most in the last period before the crossing.

The rising profile is a property of the step. On the plateau nothing but the shipments lowers the West's employment. In the smooth economy the plan has the same two phases, but the peak comes before the crossing rather than at it: as the West's knowledge enters the steep part of the S-curve, its productivity and employment fall on their own, and a smaller shipment is enough to keep its knowledge falling. Section 6 reports both plans at illustrative parameters.

4.3.5 The welfare cost of predation

We now ask whether predation raises or lowers world welfare, that is, whether the East gains more or less than the West loses. Throughout, $\psi^* > 0$, and both countries' welfare is measured relative to the commercial equilibrium, which is then autarky: the East ships nothing and the West stays at h_H .

Write $u_i^W(x) = f(\alpha_i \varphi_i(x) + x) - q_i(x)x - v(\varphi_i(x))$ for the West's welfare in a period in which it is in regime i and imports x : output, less the import bill, less the disutility of work. On an export path $P = \{x_t\}$ with regimes $\{i_t\}$, the West's lifetime loss relative to autarky is

$$L^W(P) = \sum_t \beta^t [u_H^W(0) - u_{i_t}^W(x_t)].$$

For $\psi > \psi^*$, write $\mathcal{P}^*(\psi)$ for the set of the East's optimal plans, which is nonempty by Proposition 7, and write

$$\Delta W^E(\psi) = \sum_t \beta^t [\Pi_{i_t}(x_t) - \psi g_{i_t}] - \frac{\Pi_H(0) - \psi g_H}{1 - \beta} \quad \text{for } \{x_t\} \in \mathcal{P}^*(\psi)$$

for the East's lifetime gain relative to autarky. Every optimal plan gives the East the same value, so its gain does not depend on which optimal plan it follows. The West's loss can: optimal plans with different crossing dates impose different losses. The result below therefore covers every optimal plan.

Near the critical security weight both stakes are small when measured at date zero, for different reasons. The East's gain is small because the critical security weight is the weight at

which crossing just pays. The West's loss is small because just above it the East crosses late, so the trap begins far in the future and is heavily discounted. The informative comparison is therefore made at the crossing date. Measured there, the West's loss $\beta^{-T_P} L^W(P)$ does not shrink. Most of it is the permanent fall in the West's welfare after the crossing, and the value of that fall at the crossing date does not depend on when the crossing happens. The proposition assumes that this loss has a positive lower bound at weights just above the critical security weight.

Proposition 8 (Predation near the critical security weight lowers world welfare). *Under Assumption 3, suppose $\psi^* > 0$. Suppose also that the West's loss, measured at the crossing date, has a positive lower bound just above the critical security weight: there are $\underline{\ell} > 0$ and $\bar{\psi} > \psi^*$ such that*

$$\beta^{-T_P} L^W(P) \geq \underline{\ell} \quad \text{for every } \psi \in (\psi^*, \bar{\psi}) \text{ and every } P \in \mathcal{P}^*(\psi). \quad (41)$$

Then for every weight ψ with $\psi^ < \psi < \min\{\bar{\psi}, \psi^* + (1 - \beta)\underline{\ell}/\Delta g\}$ and every optimal plan $P \in \mathcal{P}^*(\psi)$,*

$$0 < \Delta W^E(\psi) < L^W(P).$$

The East gains from predation, the West loses more, and the sum of the two countries' welfare falls: predation is negative-sum.

The proof is in [Appendix A](#), and its idea is short. On an optimal plan the East's gain is $\beta^{T_P} \Delta g(\psi - z_P)/(1 - \beta)$. No plan breaks even at the critical security weight, so $z_P > \psi^*$, and the gain is less than $\beta^{T_P} \Delta g(\psi - \psi^*)/(1 - \beta)$. For ψ within $(1 - \beta)\underline{\ell}/\Delta g$ of the critical security weight, this is less than $\beta^{T_P} \underline{\ell}$, which by (41) is at most the West's loss. We call $(1 - \beta)\underline{\ell}/\Delta g$ the radius of the proposition. The conclusion covers every optimal plan at every weight in the interval, including weights at which two crossing dates are both optimal, and it also holds for nearby values of Δg , as the proof shows. [Appendix A](#) shows how (41) can be verified at the functional forms of Section 6, and Section 6 reports the bound, the radius, and how far above the critical security weight the conclusion extends when the two stakes are compared directly.

Predation lowers world welfare for two reasons. First, the flood wastes resources. Each unit shipped is worth more to the East at home, where its value is the East's marginal product, than to the West, where its value is the price the West pays. Second, the loss of knowledge is permanent. After the crossing the West's labor produces at α_L instead of α_H in every period. The higher price the trapped West pays is a transfer to the East and does not change the sum of the two countries' welfare; the lost productivity does. Near the critical security weight the East's gain is small, so the fall in world welfare is close to the West's entire loss.

This completes the analysis of predation. Above the critical security weight the East floods the West, traps it, and then sells to it at the higher price a trapped customer pays. Below it, commerce between the two industrialized countries is autarky, and the West keeps its industry. Whether the East floods depends on its security weight and on its patience. The West can observe the East's patience, since the two countries share a discount factor, but not its security weight. The East does not need to be paid: [Appendix E](#) shows that when

its shipments are a gift, the same flood and trap occur above the critical security weight of Proposition 4, and the East then ships nothing at all. Two further points concern the benchmark. First, it makes predation harder, not easier: without the interior of the S-curve, the West's high regime survives much larger inflows, so its import fold is larger than in the smooth economy. Second, the West's side of the model, its employment function and the threshold it must be pushed across, comes from Section 3 and does not depend on why the East ships; the East takes it as given and chooses the flow. Section 6 reports the magnitudes of this subsection. The next section asks what the West can do.

5 The West's Counterpolicy

Section 4 showed what a security-motivated East does to a passive West, one whose households set employment competitively and whose government does nothing: above the critical security weight ψ^* it floods, traps, and returns to commerce with a trapped customer. This section gives the West two instruments, a tariff on the imported capital good and a subsidy to industrial employment, and asks which one stops the flood. We work in the benchmark of Section 4.3 and keep its objects: the regimes $i \in \{H, L\}$, the West's competitive employment $\varphi_i(x)$ and price $q_i(x)$, the East's employment $n^E(x)$ and flow payoff $\Pi_i(x)$ of (34), the step security loss g_i with gap Δg , the high autarky stock h_H , the import fold $\bar{m}_\infty$, and Assumption 3.

The two instruments differ in what the West must know. A tariff deters only if it exceeds a level that depends on the East's motive, which the West does not observe (Section 5.1). The labor subsidy removes the trap at every motive if its rate is not capped, and, capped, deters every motive below a level that rises with the cap; on the equilibrium path it is never paid (Sections 5.2 and 5.3).

5.1 Tariffs

The first instrument is a per-unit tariff $\xi \geq 0$ on imports of the capital good, collected from Western households and rebated to them lump sum. The West's import margin in (19) becomes $f'(k_t^W) = q_t + \xi$, where q_t is the price the East receives: households buy until the marginal product equals the price they pay, tariff included. In Section 4.2 the East chose its price and quantity subject to the West's demand. The same is true here, with one change: the demand ties the quantity to the price the West pays, and the East receives that price less the tariff, so in choosing x the East chooses its after-tax price. At a shipment x in regime i the West's labor margin is unchanged, so its employment is still $\varphi_i(x)$, its marginal product $q_i(x)$ is now the gross price it pays, and the East receives $q_i(x) - \xi$. The tariff shifts the inverse demand the East faces down by ξ at every quantity and changes nothing else at a given shipment. The East's flow payoff from shipping x to a regime- i West is therefore

$$\Pi_i(x) - \xi x, \tag{42}$$

the payoff of (34) less the tariff bill, and the West's flow welfare is $u_i^W(x) + \xi x$, its welfare at that shipment without the tariff plus the receipts. The tariff moves money from the East

to the West. It does not move Western employment.

Three objects of Section 4.3 acquire a tariff argument. The commercial export against a trapped West is now

$$x_L^c(\xi) \in \arg \max_{x \geq 0} [\Pi_L(x) - \xi x].$$

The commercial gain is now

$$\Delta \Pi(\xi) \equiv \max_{x \geq 0} [\Pi_L(x) - \xi x] - \Pi_H(0) \geq 0.$$

It is nonnegative because $\Pi_L(0) = \Pi_H(0)$: the East can always ship nothing. The cost of a crossing plan P is now

$$C_P(\xi) \equiv \sum_{t < T_P} \beta^t [\Pi_H(0) - \Pi_H(x_t) + \xi x_t] = C_P(0) + \xi \sum_{t < T_P} \beta^t x_t,$$

the cost without the tariff plus the discounted tariff bill of the active phase. Crossing plans themselves are unchanged. They are paths of shipments, and the tariff does not change the West's law of motion. The critical security weight under the tariff is

$$\psi^*(\xi) = \max \left\{ 0, \inf_P \frac{(1 - \beta) \beta^{-T_P} C_P(\xi) - \Delta \Pi(\xi)}{\Delta g} \right\}, \quad (43)$$

the formula (38) with the tariffed cost and gain, and $\psi^*(0) = \psi^*$.

Proposition 9 (Tariff absorption). *Under Assumption 3, for every tariff $\xi \geq 0$:*

- (i) *Lemma 2 holds with (42) in place of Π_i : the East's optimal export paths solve the reduced problem (36) with the tariffed payoff, regime L is absorbing, the best never-crossing plan is autarky, and after a crossing the East ships $x_L^c(\xi)$.*
- (ii) *Crossing is strictly optimal for $\psi > \psi^*(\xi)$ and strictly suboptimal for $\psi < \psi^*(\xi)$. The function $\xi \mapsto \psi^*(\xi)$ is continuous and nondecreasing, and wherever it is positive it rises at least at the rate $(1 - \beta)\bar{m}_\infty/\Delta g$:*

$$\psi^*(\xi') \geq \psi^*(\xi) + \frac{(1 - \beta)\bar{m}_\infty}{\Delta g} (\xi' - \xi) \quad \text{whenever } \xi' > \xi \text{ and } \psi^*(\xi) > 0.$$

In particular $\psi^(\xi) \rightarrow \infty$ as $\xi \rightarrow \infty$.*

- (iii) *For $\psi > \psi^*(\xi)$ the East's optimal path crosses: the West's knowledge reaches χ or falls below it at a finite date and stays at or below it, its flow welfare at every date is $u_{i_t}^W(x_t) + \xi x_t$, the welfare it would have at that shipment without the tariff plus the tariff receipts, which are positive at every date with positive shipments. For $\psi < \psi^*(\xi)$ the East ships nothing and the West stays at h_H .*
- (iv) *For every $\psi > \psi^*(0)$ there is a unique $\bar{\xi}(\psi) \in (0, \infty)$ with $\psi^*(\bar{\xi}(\psi)) = \psi$. A tariff below $\bar{\xi}(\psi)$ is absorbed and the West is trapped; a tariff above it deters and the West stays at h_H . The deterrent tariff $\bar{\xi}(\psi)$ is nondecreasing in ψ .*

The proof is in [Appendix A](#).

The proposition states that a tariff changes what predation costs the East and not whether the West is trapped. If the tariff is below $\bar{\xi}(\psi)$, the East pays ξ on every unit it ships and still ships enough to carry the West's knowledge down to χ . The West collects the revenue while its knowledge falls, and it ends at the low steady state, as it would under a tariff of zero. If the tariff is above $\bar{\xi}(\psi)$, the East ships nothing. The deterrent tariff $\bar{\xi}(\psi)$ is finite for every ψ and nondecreasing in ψ , so a large enough tariff deters any given motive. Which case applies depends on ψ and on the damage function behind Δg , and the West observes neither ψ nor $\mathcal{D}$. A West that sets a tariff therefore does not know whether it has set it above or below $\bar{\xi}(\psi)$.

Absorption is a statement about the outcome, not about the path. A tariff raises the cost of every shipped unit, so the East's optimal plan changes, and the crossing date moves with it, though not because the tariff shortens the flood at a given weight. What does not change is where the West ends up. Two further facts bear on how the tariff should be read. First, a prohibitive tariff has no cost here: between two industrialized countries the commercial equilibrium is autarky, so a tariff high enough to stop all trade stops no trade that would have taken place under commerce, and it is equivalent to a quota of zero. Second, the tariffs that deter are small at our parameters, a fraction of the autarky price, as [Section 6](#) reports. A tariff therefore deters at low cost when the East's motive is known, and it gives no protection when the motive exceeds the level the tariff was set to deter.

We do not claim the two-phase shape of [Proposition 7](#) under a tariff. The second half of [Assumption 3\(iii\)](#), which delivers that shape, is a condition on Π_H alone. It need not hold for the tariffed payoff, and the proposition above does not use it.

5.2 The Labor Subsidy

The second instrument is a labor subsidy at rate $\tau_t > -1$. It is paid to Western households and financed by a lump-sum tax, as in [Proposition 3](#). A negative rate is a tax on labor income, rebated lump sum. The household's employment optimality condition, the labor margin in [\(19\)](#), becomes

$$v'(n_t^W) = (1 + \tau_t) \alpha(h_t^W) f'(\alpha(h_t^W) n_t^W + x_t). \quad (44)$$

The rate multiplies the private return to work by $1 + \tau_t$. At $\tau_t = 0$ the condition reduces to that margin. Because the tax is lump sum, the subsidy moves one thing only: Western employment, and through it Western knowledge. It does so indirectly. The West is a decentralized economy, and its government does not choose employment. It sets the rate, and households choose how much to work at that rate. The next lemma says that the distinction does not matter for what follows.

Lemma 3 (Implementability). *Fix the West's knowledge h and the shipment x , and let the rate be set on the two, so that households take it as given. For every rate $\tau > -1$, condition [\(44\)](#) has a unique solution, and every employment level $n > 0$ solves it at exactly one rate,*

$$\tau(n; h, x) = \frac{v'(n)}{\alpha(h) f'(\alpha(h)n + x)} - 1. \quad (45)$$

This rate is strictly increasing in n and zero at the competitive employment $\phi(h; x)$. It is a subsidy, $\tau \geq 0$, if and only if $n \geq \phi(h; x)$, and a tax otherwise.

Proof. At a given rate the left side of (44) rises from zero to infinity in n and the right side falls, as in the proof of Lemma 1 with $(1 + \tau)\alpha(h)$ in place of $\alpha(h)$, so the solution is unique. Solving (44) for the rate gives (45). Its numerator rises in n and its denominator falls, and it is zero at $\phi(h; x)$ by (6). $\square$

The government can therefore induce any employment level, with a subsidy above the competitive level and a tax below it. From here on we describe the government as choosing employment. By the lemma, a choice of employment is the choice of one rate, the rate (45), so nothing is lost by the description. Write the West's flow welfare when it works n , imports x , and has knowledge h as

$$u^W(n; h, x) \equiv f(\alpha(h)n + x) - f'(\alpha(h)n + x)x - v(n). \quad (46)$$

This is output less the import bill less the disutility of work. The bill is priced at the marginal product at which the West's demand absorbs x , as in (28). In the benchmark the same object is $u_i^W(n; x)$, with α_i in place of $\alpha(h)$. The welfare $u_i^W(x)$ of Section 4.3 is its value at the competitive employment $\varphi_i(x)$.

The government chooses employment to maximize the present value of this flow. Its choice differs from the households' for two reasons. The first is the learning externality of Section 3: work today adds to knowledge tomorrow. The second is the price of imports. More work raises the West's capital and lowers the marginal product at which it buys the East's shipment, so it lowers the bill for every unit imported. Households take the price as given and ignore both.

In the benchmark, knowledge matters to the government only through its regime. Productivity is flat on each side of χ , so as long as next period's knowledge stays above χ , it does not matter by how much. Setting knowledge aside, the government's best employment against a shipment x in regime i maximizes $u_i^W(n; x)$. Write $n_i^R(x)$ for it, and call it the government's *static best response*. It solves

$$v'(n) = \alpha_i[f'(k) - xf''(k)], \quad k = \alpha_i n + x. \quad (47)$$

The competitive employment $\varphi_i(x)$, which households choose when the rate is zero, solves the same condition without the term $-xf''(k)$: households take the import price as given, and the government does not. When f is three times differentiable with $f''' \geq 0$, which holds for every f of constant elasticity and in particular for the one of Section 6, u_i^W is strictly concave in n , and $n_i^R(x)$ is unique and nonincreasing in x . The two levels coincide when nothing is shipped, $n_i^R(0) = \varphi_i(0)$, and against every positive shipment the government's level is higher, $n_i^R(x) > \varphi_i(x)$.

The subsidy. The government implements its choice with the rate (45). To see what that rate pays for, write

$$u_n^W(n; h, x) \equiv \alpha(h)[f'(k) - xf''(k)] - v'(n), \quad k = \alpha(h)n + x,$$

for the derivative of the flow (46) in employment. Solving this definition for $v'(n)$ and substituting it into (45) splits the rate into two parts:

$$\tau(n; h, x) = -\frac{x f''(k)}{f'(k)} - \frac{u_n^W(n; h, x)}{\alpha(h) f'(k)}. \quad (48)$$

The first part rewards households for the effect of their work on the import bill, which they ignore because they take the price as given. It is zero when nothing is shipped, and it is below one for every shipment, because $x \leq k$ and $-k f''(k)/f'(k) < 1$ by Assumption 1. The second part is zero at the static best response, where $u_n^W = 0$ by (47), so there the first part is the whole rate. The second part is positive when the government employs more than its static best response, which in the benchmark it does only to keep its knowledge in the high regime; it then pays for the knowledge that the extra work preserves, as the learning subsidy of Proposition 3 does. Neither part depends on why the East ships. The rate responds to the West's knowledge and to the size of the shipment, not to the motive behind them.

The defended margin. Suppose a shipment is large enough that, at the static best response $n_H^R(x)$, next period's knowledge would fall to χ or below. To stay in the high regime, the government must choose a higher employment level, one that places next period's knowledge strictly above χ , because Assumption 3 puts knowledge of exactly χ in the low regime.⁶ Among these levels the government prefers the lowest, because every one of them keeps the high regime and flow welfare falls as employment rises above $n_H^R(x)$. But the set of these levels is open, and it has no lowest element, so the government has no best choice. We restrict its choice to a closed set.

Assumption 4 (Defended margin). *Fix a margin $\varepsilon \in (0, \delta(h_H - \chi))$. The Western government does not induce an employment level that places next period's knowledge in the interval $(\chi, \chi + \varepsilon)$.*

The restriction is needed only because of the step. With a smooth productivity schedule more knowledge is worth more, so the government has a reason to prefer one level of knowledge to another, and no restriction is needed; Appendix F computes that case. The restriction also costs little. The employment levels it rules out lie within ε/λ below the lowest level it allows, so choosing that level instead of a ruled-out one lowers flow welfare by less than ε/λ times the marginal disutility of work at that level. Call $\chi + \varepsilon$ the *defended level*, and write

$$\underline{n}(h) \equiv \frac{\chi + \varepsilon - (1 - \delta)h}{\lambda}$$

⁶Putting χ in the high regime would not help. Both players want to be as close to χ as they can, from opposite sides: the East wants to carry the West's knowledge just into the low regime at the least cost, and the West wants to keep it just inside the high regime. A step must assign χ itself to one side, and whichever side it is, the other player's set is open. With χ in the high regime no export path would be the cheapest way to capture the West, the East would have no optimal plan above the critical security weight, and Proposition 7 would lose its existence claim; we therefore put χ in the low regime and close the West's set with a margin.

for the employment that places next period's knowledge at the defended level from state h . At $h = \chi + \varepsilon$ it equals $\delta(\chi + \varepsilon)/\lambda$, the employment that holds the defended level in place. When $\underline{n}(h) > 0$, the rate that induces it is (45) evaluated at $\underline{n}(h)$,

$$\tau^F(h, x) = \frac{v'(\underline{n}(h))}{\alpha_H f'(\alpha_H \underline{n}(h) + x)} - 1. \quad (49)$$

The bound $\varepsilon < \delta(h_H - \chi)$ ensures that from any state above χ the autarky employment $\varphi_H(0)$ places next period's knowledge at or above the defended level. The margin is held fixed throughout, and we suppress it in the notation.

Lemma 4 (The defended floor). *Suppose Assumptions 3 and 4 hold and f is three times differentiable with $f''' \geq 0$. Let the West be in regime H , at a state $h > \chi$, and let it receive a shipment x .*

- (i) *Among the employment levels that place next period's knowledge at or above the defended level, the one with the highest flow welfare is*

$$N^F(h, x) = \max\{n_H^R(x), \underline{n}(h)\}. \quad (50)$$

It is the static best response when that response keeps knowledge at or above the defended level, and the floor $\underline{n}(h)$ otherwise; in the second case we say that the floor binds. In both cases it is at least the competitive employment $\varphi_H(x)$.

- (ii) *At the defended level, $h = \chi + \varepsilon$, the floor binds if and only if $\lambda n_H^R(x) < \delta(\chi + \varepsilon)$. Where it binds, the rate $\tau^F(\chi + \varepsilon, x)$ is strictly increasing in x .*
- (iii) *For each regime i , every shipment x , and all $n' < n$, $u_i^W(n'; x) - u_i^W(n; x) < v(n) - v(n')$.*

The proof is in [Appendix A](#).

The lemma states three things. First, the government chooses its static best response unless that choice would let its knowledge fall below the defended level; then it chooses the floor. By (48), the rate is then $-xf''(k)/f'(k)$ when the floor does not bind and $\tau^F(h, x)$ when it binds. Second, the floor binds only against large shipments, and against those the rate that holds it rises with the shipment. Third, the floor is cheap. By part (iii), no level below the floor raises the West's flow by more than the floor's disutility $v(\underline{n}(h))$, however large the flood, and that number depends only on the technology and the margin. Section 6 reports the floor, the bound, and the shipments against which the floor binds.

A capped rate. The rate that holds the floor rises with the size of the flood, by part (ii) of the lemma. If the rate cannot exceed a cap $\bar{\tau}$, a large enough flood carries the West's knowledge to χ or below whatever its government does, and an East whose security weight is large enough sends one. A capped subsidy is then an instrument of the tariff's kind: it deters every East whose security weight is below a critical security weight $\psi_{\text{cap}}^*(\bar{\tau})$, which rises with the cap, and no East above it. Section 6 reports it at four caps.

5.3 Deterrence

We now let the East's planner and the West's government interact. The East chooses its shipments knowing that the West holds an instrument. The West responds to each shipment with its subsidy. The state is the West's knowledge $h \geq 0$. Write $i(h)$ for the West's regime at h : $i(h) = H$ if $h > \chi$ and $i(h) = L$ otherwise. The East's knowledge is not a state. Under Assumption 3 the East stays at or above h_H whatever it ships, by the argument of Lemma 2(i), and that argument does not involve how the West responds. Within each period the East ships $x \geq 0$. The West observes x and sets its rate. By Lemma 3 the rate and the employment it induces determine each other, so we describe the West as choosing its employment n , and recover the rate from (45) when it is wanted. The price clears at the West's marginal product. The West's knowledge then updates by (5).

An *export policy* is a function $X : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ from the West's knowledge to the East's shipment: at state h the East ships $x = X(h)$. An *employment policy* is a function $N : \mathbb{R}_+ \times \mathbb{R}_+ \rightarrow \mathbb{R}_+$ from the state and the shipment to the West's employment: at state h , after observing the shipment x , the West works $n = N(h, x)$. By Assumption 4, $N(h, x)$ never places next period's knowledge in the band $(\chi, \chi + \varepsilon)$. Every other positive level is induced by a rate, by Lemma 3.

A pair of policies (X, N) induces the following objects. From any initial state h_0 the path of the game is

$$x_t = X(h_t), \quad n_t = N(h_t, x_t), \quad h_{t+1} = (1 - \delta)h_t + \lambda n_t.$$

The price at date t is

$$q_t = f'(\alpha_{i(h_t)} n_t + x_t), \tag{51}$$

the West's marginal product at that date's employment and shipment. Both flow payoffs are evaluated at that price. The East's flow payoff from shipping x to a West that works n in regime i is

$$\hat{\Pi}_i(x, n) \equiv f(\alpha_H n^E(x) - x) - v(n^E(x)) + f'(\alpha_i n + x) x,$$

the payoff (34) with the price at employment n in place of the competitive price, so that $\hat{\Pi}_i(x, \varphi_i(x)) = \Pi_i(x)$. The hat marks the dependence on the West's employment. The West's flow payoff is $u_i^W(n; x)$ of (46).

Given N , the East's problem at state h is a dynamic program. The state is h , the control is x , the transition is $h' = (1 - \delta)h + \lambda N(h, x)$, and the flow is $\hat{\Pi}_{i(h)}(x, N(h, x)) - \psi g_{i(h)}$. Write $V^E(h; N)$ for its value, the supremum over shipment sequences of the discounted payoff from h . Given X , the West's problem at state h after shipment x is also a dynamic program. Each period the West observes that period's shipment $X(h_t)$ and then chooses its employment. Write $V^W(h, x; X)$ for its value, the supremum over such employment plans of the discounted payoff from (h, x) . Let $\mathcal{N}(h) \subseteq \mathbb{R}_+$ be the set of employment levels the West may choose at state h . By Assumption 4 it is

$$\mathcal{N}(h) = \{n \geq 0 : (1 - \delta)h + \lambda n \leq \chi\} \cup \{n \geq 0 : n \geq \underline{n}(h)\}.$$

The first set contains the levels that let next period's knowledge fall to χ or below; it is empty when $(1 - \delta)h > \chi$. The second contains the levels that keep it at or above the defended level. The two value functions satisfy

$$V^E(h; N) = \sup_{x \geq 0} \left\{ \hat{\Pi}_{i(h)}(x, N(h, x)) - \psi g_{i(h)} + \beta V^E(h'; N) \right\}, \quad h' = (1 - \delta)h + \lambda N(h, x), \quad (52)$$

$$V^W(h, x; X) = \sup_{n \in \mathcal{N}(h)} \left\{ u_{i(h)}^W(n; x) + \beta V^W(h', X(h'); X) \right\}, \quad h' = (1 - \delta)h + \lambda n. \quad (53)$$

We write suprema because nothing guarantees in general that a best shipment or a best employment level exists. Conditions (i) and (ii) of Definition 1 below require the policies to attain these suprema at the states the equilibrium covers, and Proposition 10 shows that they do. The feasible set $\mathcal{N}$ in (53) is a primitive of the game. Appendix F computes the equilibrium defined below with a smooth productivity schedule, where $\mathcal{N}$ is the whole employment grid and no margin is needed. Given a pair of policies, call a set of states $S \subseteq \mathbb{R}_+$ *invariant* if the policies do not leave it: $h \in S$ implies $(1 - \delta)h + \lambda N(h, X(h)) \in S$.

Definition 1 (Markov perfect equilibrium). *A pair of policies (X, N) is a Markov perfect equilibrium on an invariant set S if the following three conditions hold.*

- (i) *The export policy is optimal for the East given the employment policy. For every state $h \in S$, the shipment $X(h)$ attains the supremum in (52): the East's payoff from shipping $X(h)$ at h and following X thereafter, with the West responding by N , equals $V^E(h; N)$, and no shipment sequence from h yields more.*
- (ii) *The employment policy is optimal for the West given the export policy. For every state $h \in S$ and every shipment $x \geq 0$, the employment $N(h, x)$ attains the supremum in (53) over $\mathcal{N}(h)$: the West's payoff from working $N(h, x)$ at (h, x) and following N thereafter, with the East shipping by X , equals $V^W(h, x; X)$, and no employment plan from (h, x) yields more.*
- (iii) *The price clears the West's import margin. At every date the price is (51), the West's marginal product at the realized employment $n_t = N(h_t, x_t)$ and shipment $x_t = X(h_t)$, and the payoffs in (i) and (ii) are evaluated at that price: the East receives q_t per unit shipped, and the West pays q_t per unit received.*

Condition (ii) requires the West's employment to be optimal at every shipment x , not only at the shipment $X(h)$ the East makes in equilibrium. The East's problem in (i) takes as given the West's response $N(h, x)$ to every shipment, including shipments the East does not make. Condition (ii) ensures that each of these responses is one the West would choose if that shipment arrived. In an equilibrium on S , conditions (i) and (ii) are required only at states in S , but the value functions V^E and V^W in those conditions are defined at every state in $\mathbb{R}_+$. This matters because a player that departs from its policy at a state in S can move the West's knowledge outside S , and whether the departure pays then depends on the player's value at that state.

The West moves after the shipment. This is the timing of a subsidy schedule set on the aggregate state, as the schedule of Appendix B is set. The goods arrive, and the rate households face is set knowing what arrived. The timing matters. A subsidy set before the shipment cannot respond to it. Against an East that ships nothing, the best such subsidy is zero, and that leaves the West of Section 4.3 exposed to a flood the moment one is launched.

We now construct a pair of policies and show that it is an equilibrium on the states that play from h_H can reach, with or without a departure by either player. The construction has three parts: an employment policy for the West, an export policy for the East, and a condition that makes the West's policy optimal.

The West's policy is the rule of Lemma 4 above χ and the return to the defended level at or below it. Write N^F for the employment policy with

$$N^F(h, x) = \max\{n_H^R(x), \underline{n}(h)\} \quad \text{for } h > \chi, \quad N^F(h, x) = \underline{n}(h) \quad \text{for } h \leq \chi.$$

Under N^F the West's knowledge never falls to χ once it is above. From $h > \chi$, next period's knowledge is at least $\chi + \varepsilon$, because the West works at least $\underline{n}(h)$. From $h \leq \chi$, it is exactly $\chi + \varepsilon$.

The East's policy is autarky above χ and its flow-optimal shipment at or below χ . At or below χ the West returns to the defended level by working $\underline{n}(h)$, so the East's shipment does not change next period's state. The East's shipment there is then a static choice. Write

$$\tilde{x}(h) \in \arg \max_{x \geq 0} \hat{\Pi}_L(x, \underline{n}(h))$$

for the shipment that maximizes the East's flow payoff against a West that works $\underline{n}(h)$, at the realized price. Write X^c for the export policy with

$$X^c(h) = 0 \quad \text{for } h > \chi, \quad X^c(h) = \tilde{x}(h) \quad \text{for } h \leq \chi.$$

The condition compares two numbers. The first is the most the floor can cost the West in one period. By Lemma 4(iii) that is $v(\underline{n}(h))$, and $\underline{n}(h) \leq (\delta\chi + \varepsilon)/\lambda$ at every $h > \chi$, so it is at most $v((\delta\chi + \varepsilon)/\lambda)$. The second is the least a crossing can cost the West. A West that lets its knowledge cross spends at least one period at or below χ . In that period the East ships $\tilde{x}(h)$ and the West earns at most

$$\bar{u}_L^W \equiv \sup_{h \leq \chi} \max_{n \geq 0} u_L^W(n; \tilde{x}(h)),$$

the best flow a trapped West can obtain against those shipments, whatever it works. After that period the best the West can do is return to the high steady state, whose value under autarky is

$$V_H^W \equiv \frac{u_H^W(0)}{1 - \beta}.$$

So a crossing is worth at most $\bar{u}_L^W + \beta V_H^W$ to the West, against V_H^W for staying above χ . The difference is $u_H^W(0) - \bar{u}_L^W$, and it is incurred one period ahead, so the least a crossing costs is $\beta[u_H^W(0) - \bar{u}_L^W]$. The condition is

$$v\left(\frac{\delta\chi + \varepsilon}{\lambda}\right) < \beta [u_H^W(0) - \bar{u}_L^W]. \quad (\text{C})$$

It states that the most the floor can cost in one period is less than the least a crossing costs. Like Condition (D), it involves only static objects.

Proposition 10 (Deterrence). *Suppose Assumptions 3 and 4 and Condition (C) hold, and f is three times differentiable with $f''' \geq 0$.*

- (i) *Given N^F , for every $\psi \geq 0$ and every $h > \chi$, the East's value is $V^E(h; N^F) = [\Pi_H(0) - \psi g_H]/(1 - \beta)$, and $X^c(h) = 0$ attains the supremum in (52); at every $h \leq \chi$, $X^c(h) = \tilde{x}(h)$ attains it, whatever the weight. Along the path generated by (X^c, N^F) from h_H , the West's knowledge stays at h_H and the subsidy is zero at every date.*
- (ii) *Given X^c , for every $h > \chi$ and every $x \geq 0$, $N^F(h, x)$ attains the supremum in (53). This holds for any employment policy that agrees with N^F at states $h > \chi$, whatever it prescribes at or below χ .*
- (iii) *Suppose in addition that $v(\underline{n}((1 - \delta)\chi)) < \beta[u_H^W(0) - \bar{u}_L^W]$. Then for every $h \in ((1 - \delta)\chi, \chi]$ and every $x \geq 0$, $N^F(h, x) = \underline{n}(h)$ attains the supremum in (53) given X^c . The pair (X^c, N^F) is then a Markov perfect equilibrium on $((1 - \delta)\chi, \infty)$. That set contains h_H , the policies do not leave it, a single departure by the West from any state above χ leaves the state in it, and a departure by the East never takes the state to χ or below.*
- (iv) *For every $\psi > \psi^*$ and every optimal plan P of the East against the competitive West, the West's value under (X^c, N^F) from h_H exceeds its value under the competitive response against P by exactly the loss $L^W(P)$ of Section 4.3.*

Proof of parts (i) and (iv). (i) Fix $h > \chi$. Under N^F the West is in regime H at every date, whatever the East ships. Its employment is at least $\varphi_H(x_t)$, so the price $f'(\alpha_H n_t + x_t)$ is at most $q_H(x_t)$, and the East's flow payoff is $\hat{\Pi}_H(x_t, n_t) \leq \Pi_H(x_t) \leq \Pi_H(0)$, the last by Assumption 3(iii). The security loss is g_H at every date. Summing, every shipment sequence is worth at most $[\Pi_H(0) - \psi g_H]/(1 - \beta)$. Shipping nothing at every date attains it, because $\hat{\Pi}_H(0, n) = \Pi_H(0)$ for every n . Hence $V^E(h; N^F)$ equals that value, and $x = 0$ attains the supremum in (52). At $h \leq \chi$ the West works $\underline{n}(h)$ whatever is shipped, and next period's state is $\chi + \varepsilon$, where the East's value is the one just computed; the East's problem at h is therefore to maximize the flow $\hat{\Pi}_L(x, \underline{n}(h)) - \psi g_L$ over x , and $\tilde{x}(h)$ does so by definition. On the path from h_H : at $x = 0$ the static best response is the competitive employment, $n_H^R(0) = \varphi_H(0)$, which leaves knowledge at h_H , above the defended level because $\varepsilon < h_H - \chi$. So $N^F(h_H, 0) = \varphi_H(0)$, knowledge stays at h_H , and the rate that induces it is zero.

(iv) Under (X^c, N^F) the West's value from h_H is V_H^W by (i). Under the competitive response its value against P is $V_H^W - L^W(P)$, by the definition of L^W in Section 4.3. The proofs of parts (ii) and (iii) are in Appendix A. $\square$

The proposition states three things. First, the floor deters. The security loss is a step, so a West held at the defended level has the same capability as a West at h_H . A flood that does not cross therefore gains the East nothing. Each shipped unit costs the East flow payoff and lowers the price it receives. The East ships nothing, and the West pays no subsidy. Second, the floor is credible. Holding it for one period costs the West at most $v((\delta\chi + \varepsilon)/\lambda)$. Letting

knowledge cross costs at least $\beta[u_H^W(0) - \bar{u}_L^W]$. Condition (C) orders the two. Third, the floor pays. Its value to the West is the loss $L^W(P)$ that predation would have caused. None of this depends on ψ . The West compares two of its own flows. The East compares a flood that gains nothing with autarky. The critical security weight of Section 4.3 is the weight above which the East floods a West whose government does nothing. It does not flood a West that holds the floor, at any ψ .

The West's rule answers a shipment for two reasons. The first is the price. Against any shipment the West works more than its households would, because more work lowers the price of every unit it imports. The employment this motive calls for falls as the flood grows, so by itself it stops only floods below a size that Section 6 reports. The second is knowledge. When a flood would carry knowledge below the defended level, the West chooses the floor, whatever rate that takes. The floor is what makes deterrence hold at every weight. Part (i) needs only two facts about the West: its knowledge never reaches χ , and it works at least its competitive employment. A rule that answers every shipment with the larger of the competitive employment and the floor delivers both, with no price motive at all. Given the floor, the price motive changes how much the West works against a flood, not whether the flood succeeds. Appendix E proves the whole result for that rule, in an economy where the East's shipments are free and there is no price to move.

The result has three limits.

First, the equilibrium covers only the states near the threshold. The pair is an equilibrium on $((1 - \delta)\chi, \infty)$, the states a single departure can reach, and not, as far as we show, on the whole state space. Far below χ , returning to the defended level in one period is costly, and at our parameters the West prefers to climb part of the way first; we do not characterize play there. This does not weaken deterrence from the high steady state. From h_H a single concession cannot bring knowledge to χ , because $(1 - \delta)h_H > \chi$ at our parameters, so deterrence there rests on parts (i) and (ii). Part (ii) holds whatever the West does below χ , because its bound credits a West that crosses with the best flow it could earn there and an immediate return to the high steady state.

Second, we do not characterize equilibria in which the East pays a West to stay trapped. A West that can climb back does so (Appendix C), so an East that wanted to keep it below χ would have to ship enough, every period, that the West's best flow below χ matches its flow at the high steady state. That payment is large, and it pays off only at a security weight well above the critical one (Section 6).

Third, the step in the security loss makes the deterrence exact. A West at the defended level is worth to the East what a West at h_H is worth, so a flood that does not reach χ gains the East nothing. With a smooth security loss every unit of Western knowledge has some security value, and the East might gain from a flood that lowers Western knowledge without trapping the West. Appendix F computes that game and finds no such flood. At every steepness and at every security weight at which it finds an equilibrium, up to a multiple of the benchmark's critical weight, the East ships nothing, the West stays at its high steady state, and the subsidy on the path is the West's domestic correction. At much higher weights it finds no equilibrium. The step is needed for the proof; the computations suggest it is not needed for the result.

Condition (C) holds at our parameters with room to spare, and the condition of part (iii) holds with less room, and only for a small margin. A West that holds its knowledge up against a flood sustained forever is better off than under autarky, because its own employment holds down the price it pays. Section 6 reports these numbers. Appendix E proves the same result when the East's shipments are a gift; there the West's reason to defend itself depends on its patience.

6 A Quantitative Illustration

The results so far are qualitative. They say that a trap can exist, that a large enough flow destroys the high steady state, that a boom outlasting $\bar{T}(m')$ leaves a permanent loss, that above a critical security weight the East floods the West across the threshold, and that a floor on Western employment deters the flood. They say nothing about magnitudes. This section puts numbers on all of it at one parameterization, in the order the paper proved things, and reads durations in years under a period convention. Nothing here is estimated. The parameters that locate the threshold are chosen to make the multiplicity visible, and we report how far the objects move when they change.

6.1 Parameters

We adopt the canonical forms $v(n) = n^{1+\eta}/(1+\eta)$ and $f(k) = k^\theta$ together with the logistic α of (3). Table 1 collects the parameter values, says what disciplines each, and lists the objects of Sections 3 to 5 that they imply.

Two conventions in the table deserve a word. The learning intensity λ is a units normalization: it fixes the scale of the knowledge stock through $h_H^* = (\lambda/\delta) n_H^*$, and rescaling it rescales h , χ , and γ together without changing any observable. And θ is the curvature of f , not a capital income share: because $k = \alpha(h)n + m$ already embeds labor, $f(k) = k^\theta$ exhibits decreasing returns to effective capital, and Assumption 1 requires only $\theta \in (0, 1)$.

The last row of the table and the vertical axis of Figure 4 read a model period as two years. The convention does one thing: it converts the critical duration $\bar{T}$ of Proposition 2 into years. At $\delta = 0.20$ half of the knowledge stock survives six years. Plant-level estimates of organizational forgetting are faster: Benkard (2000) finds that about 61% of a firm's production experience survives a year, a half-life of about 1.4 years. We use the slower rate because the stock here is a sector's collective knowledge rather than one plant's, and nothing in the paper depends on the choice except the clock. The parameters δ , λ , and χ enter the steady states and the fold only through $\delta\chi/\lambda$ and $\gamma\lambda/\delta$: in employment units $\tilde{n} = \delta h/\lambda$ the steady-state condition reads $\tilde{n} = \phi(\lambda\tilde{n}/\delta; m)$, and α depends on $\tilde{n}$ only through those two combinations. A faster forgetting rate with the S-curve rescaled to hold them fixed leaves the steady states in employment units, the fold, and the fold penetration where they are and shortens every duration; at Benkard's rate the twenty years reported below become six.

Parameter	Value	Basis
δ	0.20	Forgetting per period. It sets the clock and nothing else: with χ and γ fixed in employment units the steady states and the fold do not depend on it. Slower than the plant-level evidence; see the text
α_L, α_H	0.30, 3.00	Productivity plateaus. The ratio of ten is a normalization chosen to make the multiplicity visible, not a measured gap; Irwin and Klenow (1994) estimate a learning rate of about 20% per doubling of cumulative output in DRAMs, with no forgetting
γ	4.20	Steepness of the S-curve; not identified, set to yield a visible rather than knife-edge separatrix
χ	4.17	Inflection of the S-curve; not identified, placed inside the operating range
λ	1.00	Units normalization; rescaling changes no observable
θ	0.65	Curvature of f , not a capital income share; conventional in $(0, 1)$
η	1.00	Frisch elasticity of labor supply $1/\eta = 1$
β	0.96	Discount factor; a two-year period at roughly a 2% annual real rate. Enters only the problems of the planners and of the West's government; the households' labor choice is static
<i>Implied objects at the baseline flow $m_0 = 0.05$</i>		
h_L^*, h_M^*, h_H^*	1.849, 4.029, 6.145	Steady states (Proposition 1); slopes $G_h = 0.80, 1.63, 0.80$
$\bar{m}$	3.67	Import fold: above it only the low steady state remains
$\bar{T}(4.50)$	25 periods	Critical duration of a boom of 4.50 (Proposition 2); 125 periods just above the fold, 10 at $m' = 8.07$
h^* at $\zeta = 1$	1.854, 3.043, 6.165	The full steady-state subsidy moves the stable states little and lowers the separatrix from 4.029
$\bar{\zeta}(m_0)$	15.5	Policy fold (Proposition 11); below one at $\gamma = 2.6$, $\delta = 0.15$, or $\lambda = 1.3$
$\underline{x}(0), \underline{x}(1)$	0.285, 0.074	Export folds from autarky (Corollary 1)
Fold in penetration	55%	Import penetration above which the high steady state ceases to exist; independent of the period length
Trap within 20 years	75%	Penetration that traps the West within ten periods, read as two years each

Table 1: Parameters of the numerical example and the objects they imply. The S-curve parameters γ and χ and the plateau ratio α_H/α_L are chosen to make the multiplicity visible and are not measured; λ , θ , and η are normalizations or conventions; δ sets the clock. Every implied object is computed from the model at these values, and the last row reads a period as two years.

6.2 The Mechanism

At these parameters the hypotheses of Proposition 1 hold: the map has three steady states with the stable, unstable, stable pattern, Assumption 2 holds, and the saddle-node of Proposition 1(ii) occurs at $\bar{m} = 3.67$.

Figure 3 measures the hysteresis of Proposition 2. A boom of size $m' = 4.50$, above the fold, has a critical duration of 25 periods: a boom of 24 periods ends 0.035 above the separatrix and the economy climbs back; a boom of 25 ends 0.28 below it and the economy converges to h_L^* . Panel (b) traces $\bar{T}(m')$ across boom sizes: size and duration substitute for one another along a downward-sloping frontier.

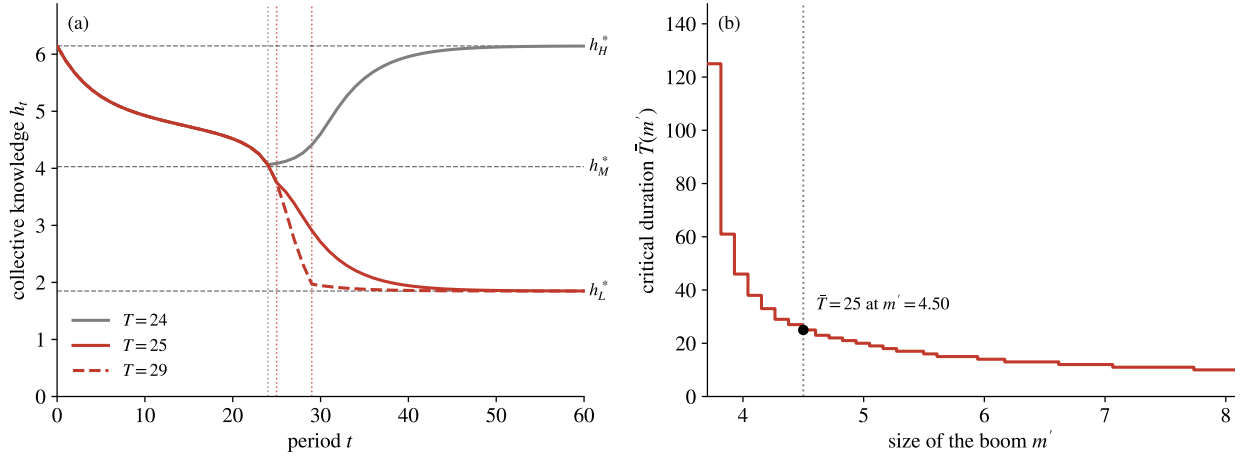


Figure 3: A temporary resource boom with a permanent consequence. Panel (a): knowledge paths when the flow of capital goods rises from $m_0 = 0.05$ to $m' = 4.50$ for T periods and then reverts. A boom of $T = 24$ periods (gray) ends just above the separatrix h_M^* and the economy recovers to h_H^* ; booms of $T = 25$ and $T = 29$ periods (red) end below it and the economy converges to h_L^* . Dotted verticals mark the end of each boom. Panel (b): the critical duration $\bar{T}(m')$ of (13) as a function of the size of the boom, which diverges as m' approaches the fold $\bar{m} = 3.67$ from above.

We measure the size of a surge by the West's *import penetration*, $x/(\alpha(h_H^*)\phi(h_H^*;x) + x)$, imports as a share of total capital-good use, and Figure 4 plots the critical duration of Proposition 2 against it in years. The fold sits at a penetration of about 55%. Below it the high steady state survives a surplus of any duration. Just above it the trap closes over decades, because near the fold the map is close to the diagonal and knowledge falls slowly; a penetration of about 75% or more traps the West within twenty years. Where the fold sits is a property of the S-curve in employment units and of the plateau ratio, neither of which is measured. Moving γ between 2.6 and 16 moves it between 43% and 71%, moving χ between 3.5 and 5.0 moves it between 74% and 20%, and moving α_H/α_L between 6.7 and 13.3 moves it between 12% and 73%. Holding χ and γ fixed in knowledge units instead and moving δ across a half-life of five to ten years moves it between 12% and 91%. What the figure shows is how the theory turns a penetration statistic into a duration, not where any sector stands.

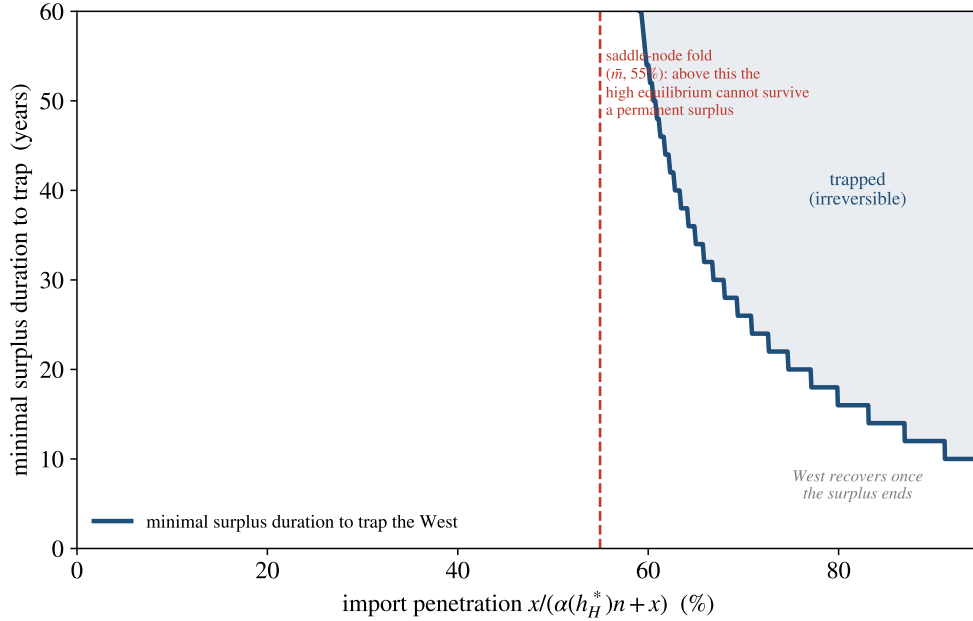


Figure 4: The minimal-trap frontier: for each level of Western import penetration (horizontal), the minimal number of years a surplus must be sustained to push the West below the separatrix and into the low equilibrium, which the West does not leave on its own (vertical); the shaded region is trapped, below the curve the West recovers once the surplus ends. The saddle-node fold (dashed) marks the penetration above which the high equilibrium ceases to exist.

Paying the full steady-state wedge leaves the two stable steady states almost where they are and pulls the separatrix down from 4.029 to 3.043, and the intensity at which the low steady state disappears is $\bar{\zeta}(m_0) = 15.5$, more than fifteen times the Pigouvian wedge, so internalization alone does not close the trap here. A parameter change brings it below one: a steepness of 2.6, a forgetting rate of 0.15, or a learning intensity of 1.3 each suffices. Run in the flow rather than in the policy, the same argument gives the export fold from autarky, $\underline{x}(0) = 0.285$, about eight percent of the import fold, which internalization cuts to $\underline{x}(1) = 0.074$: a country that corrects its own learning externality sits close to its own fold.

6.3 Predation

Proposition 6 decides whether commerce alone would trap a customer. The cheapest undiscounted crossing costs $C_0 = 39.3$, while the perpetuity a trapped customer pays is $\beta\Delta\Pi/(1 - \beta) = 5.07$: Condition (D) holds with a factor of seven to spare, so commercial predation loses money at $\psi = 0$ and the critical security weight is strictly positive. The bound of Proposition 6(iii) puts $\psi^*\Delta g$ above 1.43, close to the computed 1.66. In terms of patience, the condition holds for every discount factor below $C_0/(C_0 + \Delta\Pi) = 0.99465$, and the cutoff of the proposition is $\hat{\beta} = 0.99471$. Reading a period as two years, a purely commercial East

therefore traps the West only if it discounts the future at less than about 0.27% a year. The two cutoffs differ by 5×10^{-5} . Both lie close to one, where the compounding within the active phase that the condition ignores is small.

The critical security weight of Proposition 5 is $\psi^* \Delta g = 1.66$, about one period of the East's autarky flow payoff $\Pi_H(0) = 1.58$. We report it as a numerical approximation, because the infimum in (38) is a limit that no plan attains. The plan that is optimal just above it is worth -16.9 at $\psi = 0$, about eleven periods of that flow, which is the sense in which commerce is far from wanting to predate. The crossing date falls as the weight rises: 22 periods at $\psi \Delta g = 1.67$, 15 at 2.0, and 10 at 4.0. The magnitudes in Table 2 are at $\psi \Delta g = 1.67$, just above the critical security weight, where the active exports rise from 0.4 to 14.3. The benchmark's import fold is $\bar{m}_\infty = 8.85$, against $\bar{m} = 3.67$ in the smooth economy.

Figure 5 shows the optimal plan in the smooth economy, with $\mathcal{D}(z) = z^2/2$, so that the security gap between the two competitive steady states is $\Delta g = 4.45$, and with the weight at $\psi \Delta g = 1.66$. The export rises from 3.4 to a peak of 7.6 over eight periods, while the West's knowledge is still on its upper plateau. It then falls over the next six periods as the West's productivity and employment fall in the steep part of the S-curve. The West crosses the separatrix in period 14, and the export then falls to its commercial level against a trapped West, about 0.9. The benchmark's profile is monotone and peaks at the crossing; the smooth one peaks before it.

6.4 Welfare

The West's lifetime loss is 9.8 at $\psi \Delta g = 1.67$, about six periods of its autarky flow of 1.58. The raw loss is not bounded away from zero as the weight falls to the critical security weight, and it should not be expected to be: the optimal crossing date diverges there, so the trap is pushed into the future and the date-zero stake shrinks. At $\psi \Delta g = 1.6595$ the optimal plan crosses at 29 and costs the West 7.3; at 1.659091, nearer the critical security weight, which is 1.659 to three figures, it crosses at 40 and costs 4.7. Measured at the crossing date the loss barely moves, which is the hypothesis (41) of Proposition 8: $\beta^{-T_P} L^W(P)$ lies between 23.98 and 24.14 at every crossing date we compute, falling monotonically in the date, because the loss is the perpetuity of the West's post-crossing flow gap, 30.3, net of the gain it makes while the flood lasts, and that gain read backward from the crossing hardly depends on the horizon. Three statements about the negative-sum interval rest on different things, and we keep them apart. The first is what the proposition proves. Hypothesis (41) quantifies over crossing dates, including dates beyond any finite computation, so a minimum taken over the dates we compute does not verify it. What does is the pointwise inequality proved in Appendix A: the West's flow gain from a shipment is at most a third of the East's flow cost of making it, at every shipment. That bounds the normalized loss below by $\underline{\ell} = 14.69$ at every crossing date of twenty or more. The shorter dates are finitely many and are checked directly, and they are the only others that matter here: a crossing date of 13 first becomes optimal at $\psi \Delta g = 2.2489$, so at every weight below that the optimal date is 14 or more. Hypothesis (41) therefore holds with $\underline{\ell} = 14.69$ on every optimal plan at every weight within 0.59 of the critical security weight, and Proposition 8 delivers the negative-sum conclusion

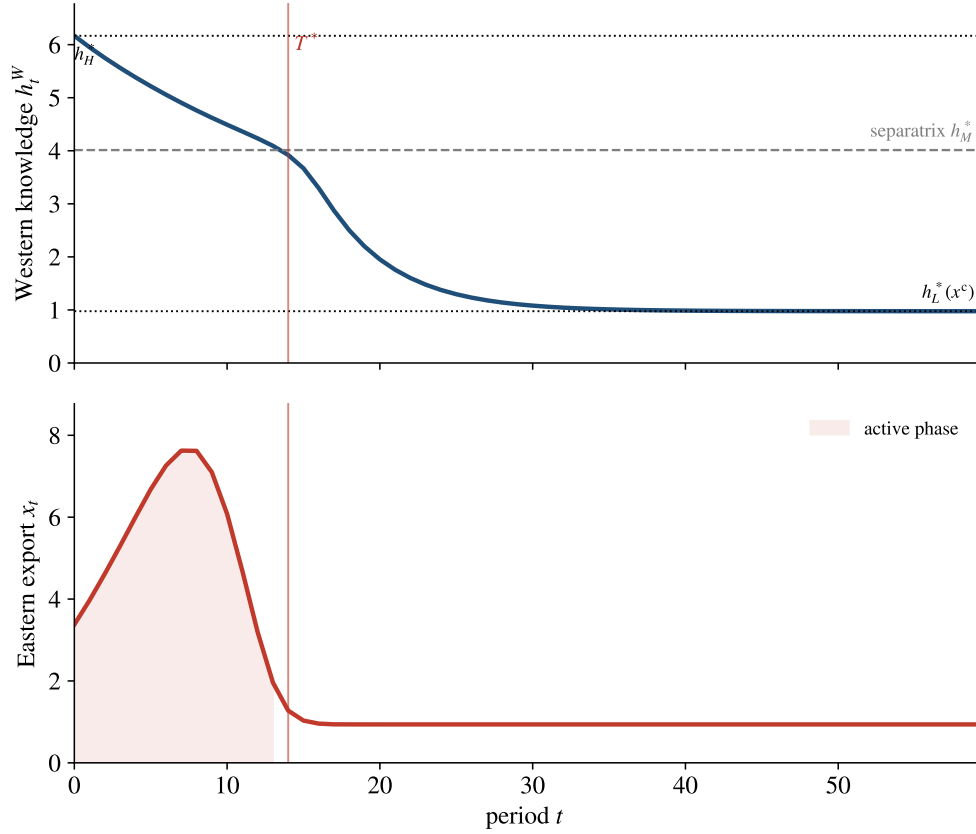


Figure 5: Optimal Eastern predation in the smooth economy, at the parameters of Section 6 with $\mathcal{D}(z) = z^2/2$ and $\psi\Delta g = 1.66$; the East is held on its upper plateau and the West is competitive. Top: Western knowledge h_t^W falls from h_H^* across the separatrix h_M^* at $T^* = 14$ and converges to the low steady state under the East's commercial inflow, $h_L^*(x^c)$. Bottom: the optimal export rises to its peak of 7.6 while the West is on its plateau, falls as the West enters the steep part of the S-curve, and drops to its commercial level after the crossing.

for $\psi\Delta g$ up to 2.25, just inside that date boundary. The second is what the computed plans show: the smallest normalized loss over the dates we compute is 23.98, and the proposition's formula applied to that number gives a radius of 0.96 and an endpoint of 2.62. The third needs no uniform bound at all. Comparing the East's gain with the West's loss on the plan that is optimal at each weight, the gain first exceeds the loss at $\psi\Delta g = 2.72$, and the comparison holds at every weight at which two crossing dates are both optimal, the largest of which is 2.58. So the proposition carries the interval to 2.25, and the direct comparison carries it from there to 2.72. Table 2 collects these magnitudes with the rest of the predation numbers.

Object	Symbol	Value	Where it matters
Benchmark import fold	$\bar{m}_\infty$	8.85	Assumption 3(iv); 3.67 in the smooth economy
Cheapest undiscounted crossing; commercial perpetuity	$C_0; \beta\Delta\Pi/(1-\beta)$	39.3; 5.07	Condition (D), factor seven
Lower bound on the critical security weight	$[(1-\beta)C_0/\beta - \Delta\Pi]/\Delta g$	1.43	Proposition 6(iii)
Discount factor above which commerce alone traps the West	$\hat{\beta}$	0.99471	Proposition 6; lower bound $C_0/(C_0 + \Delta\Pi) = 0.99465$; about 0.27% a year at a two-year period
Value at $\psi = 0$ of the plan optimal at $\psi\Delta g = 1.67$	$\Delta(P; 0)$	-16.9	$\psi^* > 0$; about eleven periods of $\Pi_H(0) = 1.58$
Critical security weight	$\psi^*\Delta g$	1.66	Proposition 5; about one period of $\Pi_H(0)$
Crossing date; active exports	$T^*; x_0 \rightarrow x_{T^*-1}$	22; 0.4 $\rightarrow$ 14.3	Proposition 7, at $\psi\Delta g = 1.67$
West's autarky flow; its loss on the plan	$u_H^W(0); L^W$	1.58; 9.8	Proposition 8, at $\psi\Delta g = 1.67$; about six periods of autarky flow
Loss at the crossing date: proved bound; computed minimum	$\underline{\ell}$	14.69; 23.98	Hypothesis (41); the first on every optimal plan in the proved interval, the second over the dates computed
Upper end of the negative-sum interval: proved; computed	$\psi\Delta g$	2.25; 2.72	Both intervals start at $\psi^*\Delta g$. The first is Proposition 8 at $\underline{\ell} = 14.69$, the second a direct optimal-plan envelope calculation, ties included

Table 2: Magnitudes of the benchmark at the parameters of Table 1: predation. Security gains are in units of the security gap Δg ; objects that depend on the weight are reported at $\psi\Delta g = 1.67$, just above the critical security weight. The magnitudes are illustrative, as the parameters are.

6.5 Defense

The autarky price of the capital good is $q_H(0) = 0.41$, which is the scale against which a tariff should be read. A tariff of 0.1 raises the critical security weight from 1.66 to 2.27; the tariff that deters a security gain of 2.0 per period is 0.054 and the one that deters 2.5 is 0.14, a third of the autarky price. Under a tariff of 1 the plan optimal just above the critical security weight under that tariff crosses after 12 periods rather than 22, which reflects the larger weight a tariff-paying East must have.

With a margin $\varepsilon = 0.01$, the floor at the defended level is 0.84, below the competitive autarky employment of 1.23, and holding it at that level costs the West at most 0.35 per period whatever the flood; the bound 0.36 that Condition (C) uses is the same quantity taken over every state above χ rather than at the defended level alone. Against the floods the East would launch, the price motive does most of the work. The West works 1.01 against a shipment of 10 and 0.93 against the peak shipment of 14.3, well above the floor, and its rate against 14.3 is 0.29, all of it the price motive; at the defended level the floor binds only against shipments above 22.1. By itself the price motive keeps the West out of the trap against any constant shipment below 22.3, where the competitive West's limit is 8.85, and a West that used the subsidy for the price alone would raise the critical security weight from 1.66 to 13.5. Following N^F against a flood of 14.3 forever is worth 61 to the West, against 39.5 at its high steady state under autarky; its knowledge settles at 4.67, above the defended level.

Caps of 0.15, 0.30, 0.50, and 1.00 raise the critical security weight to 5.17, 12.1, 15.7, and 27.9; the cap of 0.15 gives the protection of a tariff of about 0.69 and the cap of 0.50 that of a tariff of about 3.1. These weights are high because the price motive keeps working under the cap. A trapped West cannot climb back under any cap below 1.63, the rate that holds a stock of χ in place in regime L in autarky.

Condition (C) holds by a factor of three, 0.36 against 1.16. The condition in part (iii) of Proposition 10 holds with 0.02 to spare, and it holds for every margin up to $\bar{\varepsilon} \approx 0.02$, which is why the margin is small. Below χ the one-period return is the West's best reply down to about 42 percent of χ , and not below. Paying a West to stay trapped would take a permanent flood of about 10.1 per period, costing the East about 4.4 per period, which pays only if $\psi\Delta g$ exceeds about 4.20, two and a half times the critical security weight.

In the smooth economy, computed in Appendix F, the pair in which the East ships nothing and the West defends its knowledge with the subsidy is an equilibrium at every steepness and at every weight up to $\psi\Delta g = 4$, which is 2.4 times the benchmark's critical security weight, with the subsidy on the path equal to the West's domestic correction of 0.004. Appendix F reports the West's response off that path, which rises steeply as its knowledge falls into the steep region, and the present value the East loses on the floods we computed.

Object	Symbol	Value	Where it matters
<i>Tariffs (Section 5.1)</i>			
Autarky price	$q_H(0)$	0.41	scale of ξ
Critical security weight at $\xi = 0.1$; crossing date at $\xi = 1$	$\psi^*(0.1)\Delta g; T^*$	2.27; 12	Proposition 9(ii)
Tariff that deters $\psi\Delta g = 2.0$; 2.5	$\bar{\xi}(\psi)$	0.054; 0.14	Proposition 9(iv)
<i>The subsidy (Section 5.2)</i>			
Floor at the defended level; competitive autarky employment	$\delta(\chi + \varepsilon)/\lambda; \varphi_H(0)$	0.84; 1.23	floor rule (50)
Employment against 10; 14.3	$n_H^R(x)$	1.01; 0.93	(47)
Rate against 14.3, all price motive	$-xf''/f'$	0.29	(48)
Shipment above which the floor binds at the defended level; the price motive's own limit		22.1; 22.3	Lemma 4(ii)
Critical security weight against a West that uses only the price motive	$\psi\Delta g$	13.5	Section 5.3
Critical security weight against caps 0.15; 0.50	$\psi_{\text{cap}}^*(\bar{\tau})\Delta g$	5.17; 15.7	capped-rate remark; tariff equivalents 0.69; 3.1
<i>Deterrence (Section 5.3)</i>			
Floor bound; continuation loss	$v((\delta\chi + \varepsilon)/\lambda); \beta[u_H^W(0) - \bar{u}_L^W]$	0.36; 1.16	Condition (C), factor three
One-step return; its disutility	$\underline{n}((1 - \delta)\chi); v(\cdot)$	1.51; 1.14	Proposition 10(iii), 0.02 to spare
Value under N^F against 14.3 forever; autarky value	$V^W; V_H^W$	61; 39.5	Section 5.3
Purchase flood; its cost per period; break-even weight	x^b ; cost; $\Pi_H(0) - \Pi_L(x^b)$	10.1; 4.4; 4.20	Section 5.3, second limit

Table 3: Magnitudes of the benchmark at the parameters of Table 1: the counterpolicy. Security gains are in units of Δg , and the margin is $\varepsilon = 0.01$. The magnitudes are illustrative, as the parameters are.

7 Discussion

7.1 Historical Precedents

Treating industrial capability as a strategic, contestable asset is not new. The decline of India’s cotton-textile industry under British rule is the closest historical analogue of the mechanism: India produced about a quarter of world manufacturing output in 1750 and a few percent by 1900 (Bairoch, 1982), as mechanized British cloth displaced Indian production, and the capability did not return when the relative-price shock passed. The decline took place under colonial rule and alongside mechanization, so trade was not the only channel, and we do not claim that it was. The mirror strategy, guarding one’s own technique, also recurs. China held a near-monopoly on silk for roughly two millennia and, by repute, forbade the export of silkworms (Liu, 2010), and Britain prohibited the export of textile machinery and the emigration of skilled artisans for more than half a century (Jeremy, 1977). The model has no instrument of that kind, and we do not analyze export controls.

7.2 Contemporary Industrial Policy

The deterrence result of Section 5.3 bears on the industrial-policy landscape since about 2018, in which the United States, the European Union, and China have run concurrent large-scale subsidies in advanced manufacturing. The model’s equilibrium does not reproduce that picture, and we do not claim that it does: in the benchmark the defending government’s subsidy is a schedule that is in place and never paid, because the rival that faces it does not ship, and in the smooth economy of Appendix F the subsidy on the path is the domestic correction. What the model says is how the two instruments work. A tariff below the prohibitive level is paid, and does not deter, when the rival’s security weight exceeds the critical security weight at that tariff, and the importer does not observe the rival’s weight. In the benchmark, a labor subsidy set as a rule that keeps employment at or above the floor leaves the rival nothing to gain, and what deters is the rule’s response to a flood, not the rate paid on the path. In the smooth economy the West is not trapped at any weight at which an equilibrium is found, and the rival still ships nothing.

7.3 Limitations

Three limitations mark the agenda ahead. The monetary side of the model is spare. A fuller treatment of the bond, its return, and the sustainability of the issuer’s debt would connect the mechanism more tightly to the literature on exorbitant privilege. The strategic analysis is proved only in the sharp benchmark. It bars the Western government from placing its knowledge in a small band just above the threshold, finances the subsidy lump sum, lets it condition on the realized shipment, and leaves its rate uncapped; a cap turns it into an instrument that a large enough motive overwhelms (Section 5.2). It characterizes play only near the threshold, and leaves open the equilibria of a West that has been trapped and is being paid to stay there. The smooth economy is treated numerically in Appendix F, not

analytically, and there the band is not needed. The benchmark also gives two industrialized countries no static gain from trade. A prohibitive tariff then protects as well as the floor does, and it needs neither the credibility condition nor knowledge of where the threshold lies. Ranking the two instruments requires static gains from trade, which we leave open. And the numerical example of Section 6 is an illustration, not an estimate. Estimating the learning, forgetting, and surge parameters against sector time series, and confronting the model with the joint path of Western net imports of semiconductors and Western fabrication capacity, is the natural next step toward a formal empirical test.

8 Conclusion

Industrial knowledge that accumulates through learning by doing and yields a sufficiently steep S-shaped return gives an economy two steady states separated by a threshold. An inflow of capital goods is then not a neutral transfer. It lowers the importer's industrial employment, erodes its knowledge, and, if it is large and persistent enough, carries the importer across the threshold into a loss that outlasts the inflow. In a benchmark in which the return to knowledge is a step, we prove four things. A security-motivated exporter produces such an inflow once its security weight passes a critical value. An exporter with no security motive does the same if it is patient enough, and not otherwise. A tariff protects the importer only if it exceeds a level that depends on a motive the importer cannot observe, although a prohibitive tariff protects against every motive and, in the benchmark, carries no static cost. And a labor subsidy, set as a rule that keeps employment at or above the floor against any shipment, deters the flood without being paid, provided a condition on the primitives makes the rule credible and the rate is uncapped. The mechanism runs both ways: the surplus that deindustrializes the importer can industrialize the exporter, so leadership in the tradable sector is contestable through trade itself.

A Proofs

Proofs for Section 3

Proof of Proposition 1. By Lemma 1, $D_m(h; m) = \lambda \phi_m(h; m) < 0$, so an increase in imports shifts $D(\cdot; m)$ downward pointwise.

For every $m \in [0, m_1]$, $D(0; m) = \lambda \phi(0; m) > 0$. Moreover, since $\alpha_L \leq \alpha(h) \leq \alpha_H$ and f' is decreasing, for $n \geq 1$ we have $\alpha(h)f'(\alpha(h)n + m) \leq \alpha_H f'(\alpha_L)$. Because $v'(n) \rightarrow \infty$, the solution $\phi(h; m)$ remains bounded as $h \rightarrow \infty$. It follows that $D(h; m) = \lambda \phi(h; m) - \delta h \rightarrow -\infty$ as $h \rightarrow \infty$.

Define the values of D at its local minimum and local maximum,

$$L(m) \equiv D(h_1(m); m) \quad \text{and} \quad M(m) \equiv D(h_2(m); m).$$

Since $D_h(h_i(m); m) = 0$, differentiation gives $L'(m) = D_m(h_1(m); m) < 0$ and $M'(m) = D_m(h_2(m); m) < 0$.

Because $L(0) < 0$, we have $L(m) < 0$ for every $m \in [0, m_1]$. Moreover, $M(0) > 0$ and $M(m_1) < 0$. Since M is continuous and strictly decreasing, there exists a unique $\bar{m} \in (0, m_1)$ such that $M(\bar{m}) = 0$.

For $m < \bar{m}$, the local minimum lies below zero and the local maximum lies above zero. Since $D(0; m) > 0$, $D(h; m) \rightarrow -\infty$ as $h \rightarrow \infty$, and Assumption 2 implies that D is decreasing, then increasing, and then decreasing, $D(\cdot; m)$ has exactly one zero in each of the intervals

$$(0, h_1(m)), \quad (h_1(m), h_2(m)), \quad (h_2(m), \infty).$$

Panel (a) of Figure 1 illustrates this configuration.

Hence, the economy has exactly three distinct steady states. At $m = \bar{m}$,

$$D(h_2(\bar{m}); \bar{m}) = 0 \quad \text{and} \quad D_h(h_2(\bar{m}); \bar{m}) = 0.$$

Because

$$D_{hh}(h_2(\bar{m}); \bar{m}) < 0 \quad \text{and} \quad D_m(h_2(\bar{m}); \bar{m}) < 0,$$

the middle and high steady states coalesce in a saddle-node bifurcation. Since $L(\bar{m}) < 0$ and $D(0; \bar{m}) > 0$, there is exactly one additional zero in $(0, h_1(\bar{m}))$. Thus, the economy has exactly two distinct steady states at $m = \bar{m}$.

For $\bar{m} < m \leq m_1$, both the local minimum and the local maximum lie below zero. Since $D(0; m) > 0$, D crosses zero exactly once in $(0, h_1(m))$ and remains negative thereafter. Panel (b) of Figure 1 illustrates this configuration.

Finally, the low and high steady states are downward crossings of $D(\cdot; m)$, so $D_h < 0$ and they are locally asymptotically stable. The middle steady state is an upward crossing, so $D_h > 0$ and it is unstable. For $\bar{m} < m \leq m_1$, the surviving low-level steady state is likewise a downward crossing and is locally asymptotically stable.

It remains to prove part (iv). For $m \neq \bar{m}$ each steady state is a simple zero of $D(\cdot; m)$, so $D_h(h^*(m); m) \neq 0$ and the implicit function theorem applies to the identity $D(h^*(m); m) = 0$, giving

$$\frac{dh^*}{dm} = -\frac{D_m(h^*(m); m)}{D_h(h^*(m); m)}.$$

By Lemma 1, $D_m = \lambda\phi_m < 0$, so the numerator $-D_m$ is positive at every steady state. At the low and high steady states the crossing is downward, $D_h < 0$, and therefore $dh_L^*/dm < 0$ and $dh_H^*/dm < 0$. At the middle steady state the crossing is upward, $D_h > 0$, and therefore $dh_M^*/dm > 0$. The same argument applies to the unique low-level steady state when $\bar{m} < m \leq m_1$. $\square$

Proof of Proposition 2. (i) Since $m' > \bar{m}$, Proposition 1(iii) leaves $h_L^*(m')$ as the unique fixed point of $G(\cdot; m')$, and $D(h; m') < 0$ for every $h > h_L^*(m')$. Because $h_0 = h_H^*(m_0) > h_L^*(m')$, the sequence $h_{t+1} = G(h_t; m')$ satisfies $h_{t+1} < h_t$ as long as the flow is held at m' ; since $G(\cdot; m')$ is increasing, the sequence stays above $h_L^*(m')$. It is therefore strictly decreasing on $t = 0, \dots, T$, and if the flow were never withdrawn the sequence would be monotone and bounded below, hence convergent, with limit the unique fixed point $h_L^*(m')$. For finite T the boom path is the first $T + 1$ terms of that sequence and stops short of the limit.

(ii) By Proposition 1(iv), h_L^* is strictly decreasing in m , so $h_L^*(m') < h_L^*(m_0) < h_M^*(m_0)$. The boom path therefore falls strictly below the separatrix in finite time and $\bar{T}(m')$ is well-defined. Once the dividend stops the law of motion is $G(\cdot; m_0)$, whose basins of attraction are $[0, h_M^*(m_0))$ for $h_L^*(m_0)$ and $(h_M^*(m_0), \infty)$ for $h_H^*(m_0)$. Since h_T is strictly decreasing in T by part (i) and $\bar{T}(m')$ is the first date at which $h_T < h_M^*(m_0)$, we have $h_T \geq h_M^*(m_0)$ for $T < \bar{T}(m')$, with strict inequality for $T \leq \bar{T}(m') - 2$, and $h_T < h_M^*(m_0)$ for $T \geq \bar{T}(m')$. A post-boom path that starts strictly above the separatrix converges to $h_H^*(m_0)$, one that starts strictly below converges to $h_L^*(m_0)$, and one that starts at $h_M^*(m_0)$ stays there, since $G(h_M^*(m_0); m_0) = h_M^*(m_0)$. Equality can hold only at $T = \bar{T}(m') - 1$, because the path is strictly decreasing. For each T the map $m' \mapsto G^T(h_H^*(m_0); m')$ is continuous and strictly decreasing, since $G_m < 0$ and $G_h > 0$, so equality at date T holds for at most one m' ; the set of boom sizes at which the exception arises is countable.

(iii) Let $m'' > m' > \bar{m}$. Then $G(h; m'') \leq G(h; m')$ for every h because $G_m < 0$, and G is increasing in h ; by induction $G^T(h_0; m'') \leq G^T(h_0; m')$ for every T . The first date at which the path falls below $h_M^*(m_0)$ is therefore weakly earlier under m'' , so $\bar{T}(m'') \leq \bar{T}(m')$. $\square$

Proofs for Sections 4 and 5

Throughout, Assumption 3 holds, $h_H = \lambda\varphi_H(0)/\delta$, $\Delta\Pi = \Pi_L(x_L^c) - \Pi_H(0)$, and $\Delta g = g_H - g_L$.

Proof of Proposition 6. The discount factor enters the reduced problem (36) only through the weights β^t . Assumption 3, the flow payoffs Π_i , the cost K , the gains $\Delta\Pi$ and Δg , the set of crossing plans, and C_0 do not depend on it. For a crossing plan P with crossing date $T = T_P$ write

$$a_P(\beta) \equiv (1 - \beta) \beta^{-T} C_P = (1 - \beta) \sum_{t < T} \beta^{-(T-t)} K(x_t), \quad A(\beta) \equiv \inf_P a_P(\beta),$$

the infimum taken over crossing plans. By (38), $\psi^* \Delta g = \max\{0, A(\beta) - \Delta\Pi\}$, and by (39), $\Delta(P; 0) = \beta^T [\Delta\Pi - a_P(\beta)] / (1 - \beta)$. Three facts about A carry the proof.

First, A contracts as β rises. Let $\beta < \beta'$ and $r = (1 - \beta') / (1 - \beta) < 1$. Every term of C_P is nonnegative by Assumption 3(iii), and $\beta^{-(T-t)} \leq \beta'^{-(T-t)}$ for $t < T$, so $a_P(\beta') \leq r a_P(\beta)$ for every crossing plan, and taking infima, $A(\beta') \leq r A(\beta)$.

Second, $A(\beta) < \Delta\Pi$ for β close to one. Fix a crossing plan P ; one exists by Assumption 3(iv), as in the proof of Proposition 5. Its sum $\sum_{t < T} \beta^{-(T-t)} K(x_t)$ has finitely many terms and stays bounded as $\beta \rightarrow 1$, so $A(\beta) \leq a_P(\beta) \rightarrow 0 < \Delta\Pi$.

Third, $A(\beta) \geq (1 - \beta) C_0 / \beta$. For $t \leq T - 1$, $\beta^{-(T-t)} \geq \beta^{-1}$, so

$$\beta^{-T} C_P = \sum_{t < T} \beta^{-(T-t)} [\Pi_H(0) - \Pi_H(x_t)] \geq \beta^{-1} \sum_{t < T} [\Pi_H(0) - \Pi_H(x_t)] \geq \beta^{-1} C_0,$$

where the last inequality holds because the path $\{x_t\}_{t < T}$ carries the West's knowledge from h_H to χ or below, and C_0 in (40) is the infimum of that undiscounted sum over all such paths.

Let $S = \{\beta \in (0, 1) : A(\beta) < \Delta\Pi\}$ and $\hat{\beta} = \inf S$. By the second fact S is nonempty, so $\hat{\beta} < 1$. By the first, if $\beta \in S$ and $\beta' > \beta$, then $A(\beta') \leq rA(\beta) \leq A(\beta) < \Delta\Pi$, so $\beta' \in S$. By the third, Condition (D) gives $A(\beta) \geq (1 - \beta)C_0/\beta > \Delta\Pi$, so no β at which it holds lies in S ; it holds exactly for $\beta < C_0/(C_0 + \Delta\Pi)$, so $\hat{\beta} \geq C_0/(C_0 + \Delta\Pi)$.

(i) Let $\beta > \hat{\beta}$. Some element of S lies below β , so $\beta \in S$, and there is a crossing plan with $a_P(\beta) < \Delta\Pi$. For that plan $\Delta(P; 0) > 0$, so crossing is strictly optimal at $\psi = 0$, and $\psi^* = 0$.

(ii) Let $\beta < \hat{\beta}$, and choose $\beta' \in (\beta, \hat{\beta})$. Then $\beta' \notin S$, so $A(\beta') \geq \Delta\Pi > 0$, and the first fact gives $A(\beta) \geq A(\beta')/r > A(\beta') \geq \Delta\Pi$. Hence $\psi^* > 0$. At $\psi = 0 < \psi^*$ crossing is strictly suboptimal by Proposition 5(i), so the optimal path never crosses; by Lemma 2(iii) it ships $x_H^c = 0$ at every date, and under autarky the West's knowledge stays at h_H .

(iii) Under Condition (D), the third fact gives $\psi^*\Delta g \geq A(\beta) - \Delta\Pi \geq (1 - \beta)C_0/\beta - \Delta\Pi > 0$. $\square$

Proof of Proposition 7. Fix $\psi > \psi^*$ and write $\Delta(\psi) > 0$ for the East's value gain from predation, as in the proof of Proposition 5. Plans with a late crossing date are dominated: $\Delta(P; \psi) \leq \beta^{T_P}(\Delta\Pi + \psi\Delta g)/(1 - \beta)$ vanishes as T_P grows while some plan gains more than $\Delta(\psi)/2$, so attention restricts to $T_P \leq \hat{T}$ for some finite $\hat{T}$. For a date T and active exports $x = (x_0, \dots, x_{T-1})$ write

$$h_T(x) \equiv (1 - \delta)^T h_H + \lambda \sum_{t < T} (1 - \delta)^{T-1-t} \varphi_H(x_t),$$

the West's knowledge at T if it is in regime H at every earlier date; the function is continuous and strictly decreasing in each x_t , because φ_H is. Consider the problem of minimizing $\sum_{t < T} \beta^t K(x_t)$ over $x \geq 0$ subject to $h_T(x) \leq \chi$. Its constraint set is closed, and the objective can be restricted to a compact box, because $K(x) \rightarrow \infty$ as $x \rightarrow \infty$; so whenever the constraint set is nonempty a minimizer $x^{(T)}$ exists, with minimum $C^*(T)$. Every crossing plan with first crossing date T is feasible for this problem, so its value (39) is at most $\Delta_T \equiv -C^*(T) + \beta^T(\Delta\Pi + \psi\Delta g)/(1 - \beta)$. Let T^* maximize Δ_T over the feasible dates $T \leq \hat{T}$, and let $T' \leq T^*$ be the first date at which the exports $x^{(T^*)}$ carry the West's knowledge to χ or below. These exports, followed by x_L^c , form a crossing plan with first crossing date T' . Its cost is at most $C^*(T^*)$, since it drops the nonnegative terms after T' , and $\beta^{T'} \geq \beta^{T^*}$, so its value is at least Δ_{T^*} ; and its value is at most $\Delta_{T'} \leq \Delta_{T^*}$. It is therefore an optimal plan, and an optimal plan exists. Conversely, every optimal plan minimizes cost at its own crossing date T : if some x' with $h_T(x') \leq \chi$ cost less, then x' would cross at a date no later than T at a cost no larger, and its value would exceed the optimum. What follows therefore applies to every optimal plan, with T its crossing date. The passive phase is Lemma 2(ii) and (iii): from T on the regime is L and permanent, and the export is x_L^c .

For the active phase, the constraint binds, $h_T(x^{(T)}) = \chi$. Some x_t is positive, since $h_T(0) = h_H > \chi$; if the constraint were slack, that x_t could be lowered slightly, keeping $h_T \leq \chi$ by continuity and lowering the cost, because $K'(x) > 0$ for $x > 0$ by Assumption 3(iii). Let $\mu \geq 0$ be the multiplier on $h_T(x) \leq \chi$. The Kuhn–Tucker conditions are

$$\beta^t K'(x_t) - \mu \lambda (1 - \delta)^{T-1-t} (-\varphi'_H(x_t)) \geq 0, \quad \text{with equality if } x_t > 0, \quad t < T.$$

If $\mu = 0$ they give $K'(x_t) = 0$, hence $x_t = 0$ for all t , contradicting binding; so $\mu > 0$. At $x_t = 0$ the left side is $-\mu\lambda(1-\delta)^{T-1-t}(-\varphi'_H(0)) < 0$, which violates the condition; so $x_t > 0$ at every active date, and the conditions hold with equality. Dividing successive conditions, the ratio $r(x) \equiv K'(x)/(-\lambda\varphi'_H(x))$ satisfies $r(x_{t+1})/r(x_t) = [\beta(1-\delta)]^{-1} > 1$, and r is strictly increasing by Assumption 3(iii), so x_t is strictly increasing and largest at $t = T - 1$.

Finally, h^W falls strictly. At $t = 0$, $\lambda\varphi_H(x_0) < \lambda\varphi_H(0) = \delta h_H = \delta h_0^W$ because $x_0 > 0$, so $h_1^W < h_0^W$. For the induction step, suppose $\lambda\varphi_H(x_t) < \delta h_t^W$. Then $h_{t+1}^W < h_t^W$, and since $x_{t+1} \geq x_t$,

$$\lambda\varphi_H(x_{t+1}) \leq \lambda\varphi_H(x_t) < \delta[(1-\delta)h_t^W + \lambda\varphi_H(x_t)] = \delta h_{t+1}^W,$$

where the middle inequality is $(1-\delta)\lambda\varphi_H(x_t) < \delta(1-\delta)h_t^W$, the induction hypothesis again. With $h_T^W = \chi$ this gives $h_t^W > \chi$ for $t < T$, consistent with T being the first crossing date, and the largest shipment is made in the period the West's knowledge reaches χ . $\square$

Crossing dates and the critical security weight. Let $C^*(T)$ be the minimum of the fixed-date problem in the proof of Proposition 7, and $z(T) \equiv [(1-\beta)\beta^{-T}C^*(T) - \Delta\Pi]/\Delta g$ the break-even weight of the cheapest plan that crosses at date T .

(a) $z(T+1) < z(T)$ at every date T at which a crossing is feasible. Take the cheapest plan that crosses at T and delay it by one period: the East ships nothing in the first period and then follows the original plan. The delayed copy crosses at $T+1$ and costs $\beta C^*(T)$, so its break-even weight is $z(T)$. Now let the East ship $\epsilon > 0$ in the first period. This lowers the West's knowledge at $T+1$ by an amount proportional to ϵ , so the plan's last shipment, which is positive by the Kuhn–Tucker argument above, can be reduced by an amount proportional to ϵ while the West's knowledge still reaches χ at $T+1$. The first shipment costs $K(\epsilon)$, which is of order ϵ^2 because $K(0) = K'(0) = 0$: the two countries' autarky prices are equal, so $x = 0$ is an interior critical point of Π_H . Reducing the last shipment saves an amount of order ϵ , because $K' > 0$ at a positive shipment. For small ϵ the modified plan costs less than $\beta C^*(T)$. Hence $C^*(T+1) < \beta C^*(T)$, which is equivalent to $z(T+1) < z(T)$.

(b) No crossing plan attains the infimum in (38). A crossing plan P costs at least $C^*(T_P)$, so $z_P \geq z(T_P) > z(T_P+1)$ by (a): some other plan has a lower break-even weight. When $\psi^* > 0$, this gives $z_P > \psi^*$ for every crossing plan, so every crossing plan loses money at $\psi = \psi^*$.

(c) The optimal crossing date does not increase with ψ , and when $\psi^* > 0$ it grows without bound as ψ falls to ψ^* . By the proof above, every optimal plan is the cheapest plan at its own crossing date, and its value is the maximum over T of Δ_T . For $\psi > \psi^*$ the optimal crossing dates are therefore the maximizers over feasible T of $f(T, \psi) \equiv \beta^T[\psi - z(T)]$, a positive multiple of Δ_T . Let $\psi < \psi'$, let T be optimal at ψ , and let T' be optimal at ψ' . Optimality gives $f(T', \psi') \geq f(T, \psi')$ and $f(T, \psi) \geq f(T', \psi)$. Adding the two inequalities and using $f(T, \psi') - f(T, \psi) = (\psi' - \psi)\beta^T$ gives $(\psi' - \psi)\beta^{T'} \geq (\psi' - \psi)\beta^T$, so $T' \leq T$. For the second claim, fix a date T . By (a), $z(s) \geq z(T)$ for every $s \leq T$, and $z(T) > \psi^*$ by (b). For $\psi \in (\psi^*, z(T))$, every plan that crosses by date T therefore loses money, and the optimal crossing date exceeds T .

Proof of Proposition 8. Fix $\psi > \psi^*$ and $P \in \mathcal{P}^*(\psi)$, with crossing date T_P and break-even weight z_P . Because $\Delta(P; \cdot)$ is affine with slope $\beta^{T_P} \Delta g / (1 - \beta)$ and vanishes at z_P ,

$$\Delta W^E(\psi) = \Delta(P; \psi) = \frac{\beta^{T_P} \Delta g}{1 - \beta} (\psi - z_P),$$

which is positive because P gains. No finite crossing plan attains the infimum in (38), by part (b) of the **argument on crossing dates** above, so $z_P > \psi^*$ and

$$\Delta W^E(\psi) < \frac{\beta^{T_P} \Delta g}{1 - \beta} (\psi - \psi^*) \leq \beta^{T_P} \underline{\ell} \leq L^W(P),$$

where the second inequality uses $\psi - \psi^* < (1 - \beta) \underline{\ell} / \Delta g$ and the third is (41), available because $\psi < \bar{\psi}$. World welfare, the sum of the two countries' objectives, changes by $\Delta W^E(\psi) - L^W(P) < 0$ on every optimal plan. The bound holds at every weight in the radius and at every optimal plan, so no separate treatment of the weights at which two crossing dates tie is needed. The conclusion also holds for nearby values of Δg . To see this, note first that the security parameters enter a plan's value (39) only through the product $\psi \Delta g$, so the set of optimal plans, and with it the West's loss on each of them, depends on $(\psi, \Delta g)$ only through that product; a uniform bound at one value of Δg therefore carries to a neighborhood. The primitives Π_i , C_P , and $\Delta \Pi$ do not involve Δg , so $\psi^*(\Delta g)$ from (38) is continuous. The conclusion therefore holds at every pair with $0 < \psi - \psi^*(\Delta g) < \min\{\bar{\psi} - \psi^*(\Delta g), (1 - \beta) \underline{\ell} / \Delta g\}$, and that set is open in $(\psi, \Delta g)$, because all three bounds are continuous in Δg , and nonempty, because it contains points just above $(\Delta g, \psi^*(\Delta g))$. $\square$

A bound on the normalized loss. Hypothesis (41) quantifies over crossing dates, so it cannot be checked by computing finitely many plans. At the functional forms of Section 6 it follows from a pointwise inequality between the two countries' flow terms, which we now prove. Write $q_H(x) = f'(\alpha_H \varphi_H(x) + x)$ for the price a regime- H West pays and $p(x) = f'(\alpha_H n^E(x) - x)$ for the East's own marginal product of capital when it ships x . The West's flow welfare $u_H^W(x)$ is the value of $\max_n f(\alpha_H n + x) - v(n)$ less the bill $q_H(x)x$, and by the envelope theorem that value has derivative $q_H(x)$ in x ; the East's flow payoff $\Pi_H(x)$ is the value of $\max_n f(\alpha_H n - x) - v(n)$, whose derivative is $-p(x)$, plus the revenue $q_H(x)x$. Hence

$$u_H^W(x) - u_H^W(0) = \int_0^x [q_H(s) - q_H(x)] ds, \quad K(x) = \int_0^x [p(s) - q_H(x)] ds. \quad (54)$$

Differentiating the two employment optimality conditions $v'(n) = \alpha_H f'(\alpha_H n \pm x)$ gives

$$q'_H(x) = -Q(k^W(x)), \quad p'(x) = Q(k^E(x)), \quad Q(k) \equiv \frac{-f''(k)}{v'' - \alpha_H^2 f''(k)} > 0,$$

with $k^W(x) = \alpha_H \varphi_H(x) + x$ increasing and $k^E(x) = \alpha_H n^E(x) - x$ decreasing. With f of constant elasticity and v quadratic, $-f''$ is decreasing, so Q is decreasing. Put $D(x) = q_H(0) - q_H(x)$ and $E(x) = p(x) - q_H(0)$, both zero at the origin. Then D is concave and E

is convex, and $k^E(x) \leq k^W(x)$ with Q decreasing gives $E' \geq D'$ and so $E \geq D$. The autarky prices of two industrialized countries are equal, $q_H(0) = p(0)$, so (54) reads

$$u_H^W(x) - u_H^W(0) = xD(x) - \int_0^x D, \quad K(x) = xD(x) + \int_0^x E.$$

The inequality $3[u_H^W(x) - u_H^W(0)] \leq K(x)$ is therefore $2xD(x) \leq 3 \int_0^x D + \int_0^x E$, and $E \geq D$ makes the right side at least $4 \int_0^x D$, while a concave D vanishing at the origin satisfies $D(s) \geq (s/x)D(x)$ on $[0, x]$ and hence $\int_0^x D \geq xD(x)/2$. Both inequalities bind as $x \rightarrow 0$, so the constant one third cannot be improved.

The consequence for (41) is immediate. The West's loss on a crossing plan is the trap perpetuity net of what it gains while the flood lasts,

$$\beta^{-T_P} L^W(P) = \frac{u_H^W(0) - u_L^W(x_L^c)}{1 - \beta} - \beta^{-T_P} \sum_{t < T_P} \beta^t [u_H^W(x_t) - u_H^W(0)],$$

and the subtracted sum is at most $\beta^{-T_P} C_P/3 = [z_P \Delta g + \Delta \Pi]/(3(1 - \beta))$ by the inequality just proved and the definition of z_P . An optimal plan is the cheapest at its own date, so $z_P = z(T_P)$, and z is strictly decreasing by part (a) of the **argument on crossing dates** that follows the proof of Proposition 7. A single evaluation at a date T_0 therefore bounds the normalized loss from below at every date of T_0 or more, leaving the shorter dates finitely many. Since the optimal date falls as the weight rises, only finitely many of them are optimal at weights in a bounded interval above the critical security weight, so checking those completes (41) on that interval, which is all the proposition asks. Section 6 reports the resulting constant, the radius it gives, and how much further the direct comparison of the two stakes reaches.

Proof of Proposition 9. (i) The tariff enters the East's problem only through the revenue term $q_t x_t$, which becomes $(q_t - \xi)x_t$. The East's employment optimality condition (30) does not involve it. The argument of Lemma 2(i) therefore applies unchanged: $\mu^E \equiv 0$ on the plateau, the East's employment is the static $n^E(x)$, and its knowledge stays above χ . Substituting the budget gives the flow payoff $\Pi_{it}(x_t) - \xi x_t - \psi g_{it}$, with the West's law of motion as the only link between dates. The West is competitive, so part (ii) of the lemma holds word for word. For part (iii): in regime H , $\Pi_H(x) - \xi x \leq \Pi_H(0)$ for every $x \geq 0$, by Assumption 3(iii) and $\xi x \geq 0$, so autarky is the best never-crossing plan. In regime L the payoff involves no state. It is maximized date by date at $x_L^c(\xi)$, which exists because $\Pi_L(x) \rightarrow -\infty$ as $x \rightarrow \infty$.

(ii) Fix ξ . Never crossing is worth $[\Pi_H(0) - \psi g_H]/(1 - \beta)$. A crossing plan P is worth

$$\sum_{t < T_P} \beta^t [\Pi_H(x_t) - \xi x_t - \psi g_H] + \beta^{T_P} \frac{\Pi_H(0) + \Delta \Pi(\xi) - \psi g_L}{1 - \beta},$$

using (i) for the continuation. Its advantage over staying is

$$\Delta(P; \psi, \xi) = -C_P(\xi) + \beta^{T_P} \frac{\Delta \Pi(\xi) + \psi \Delta g}{1 - \beta}, \quad (55)$$

which is (39) with the tariffed cost and gain. For fixed ξ the plan-by-plan argument of Proposition 5(i) applies. Each $\Delta(P; \cdot, \xi)$ is affine and strictly increasing in ψ with zero $z_P(\xi)$, the expression inside the infimum in (43); crossing plans exist by Assumption 3(iv); above $\inf_P z_P(\xi)$ some plan gains and below it every plan loses, which is the first claim.

For the dependence on ξ : every crossing plan ships more than $\bar{m}_\infty$ at some active date. Otherwise $\lambda\varphi_H(x_t) \geq \delta\chi$ at every $t < T_P$, and then $h_t^W > \chi$ implies $h_{t+1}^W > (1-\delta)\chi + \delta\chi = \chi$; by induction from $h_0^W = h_H > \chi$, the plan never crosses. So there is $t < T_P$ with $x_t > \bar{m}_\infty$, and

$$\beta^{-T_P} \sum_{s < T_P} \beta^s x_s \geq \beta^{t-T_P} x_t > \bar{m}_\infty.$$

Now $C_P(\xi') - C_P(\xi) = (\xi' - \xi) \sum_{s < T_P} \beta^s x_s$, and $\Delta\Pi$ is nonincreasing, being a maximum of functions that fall in ξ . Hence

$$z_P(\xi') - z_P(\xi) \geq \frac{(1-\beta)\bar{m}_\infty}{\Delta g} (\xi' - \xi) \quad \text{for every } P \text{ and } \xi' > \xi.$$

Taking the infimum over P on both sides gives the same inequality for $\inf_P z_P$. This proves the slope bound wherever $\psi^*(\xi) = \inf_P z_P(\xi) > 0$, the monotonicity of ψ^* , and the divergence. For continuity, $\inf_P z_P(\xi) = \inf_P [a_P + b_P \xi] - \Delta\Pi(\xi)/\Delta g$ with $a_P, b_P \geq 0$. An infimum of affine functions that is finite on $[0, \infty)$ is concave, hence continuous on $(0, \infty)$. At $\xi = 0$, monotonicity gives $\liminf_{\xi \downarrow 0} \geq \inf_P a_P$, and $\inf_P [a_P + b_P \xi] \leq a_P + b_P \xi$ for each P gives $\limsup_{\xi \downarrow 0} \leq \inf_P a_P$. And $\Delta\Pi$ is continuous, as the maximum of a continuous function of (x, ξ) over a set that can be taken compact, since $\Pi_L(x) - \xi x \rightarrow -\infty$ as $x \rightarrow \infty$ uniformly for ξ in a bounded set. The truncation at zero preserves continuity.

(iii) For $\psi > \psi^*(\xi)$, crossing is strictly optimal by (ii). An optimal plan exists by the existence argument in the proof of Proposition 7, which uses only that the capture set $h_T \leq \chi$ is closed, that $\Pi_H(x) - \xi x \rightarrow -\infty$ as $x \rightarrow \infty$, and that late crossings are dominated. On that plan the West is in regime L from the crossing date on, by (i). The West's flow welfare at date t is $u_{it}^W(x_t) + \xi x_t$: households pay the gross price $q_{it}(x_t)$ on x_t units, the treasury returns ξx_t to them, and the East receives the rest. For $\psi < \psi^*(\xi)$, staying is strictly optimal, the best never-crossing plan is autarky by (i), and under autarky the West's knowledge stays at h_H .

(iv) Fix $\psi > \psi^*(0)$ and let $\bar{\xi}(\psi) = \inf\{\xi \geq 0 : \psi^*(\xi) \geq \psi\}$. The set is nonempty by the divergence in (ii) and does not contain 0, so $\bar{\xi}(\psi) > 0$; by continuity $\psi^*(\bar{\xi}(\psi)) = \psi$. For $\xi < \bar{\xi}(\psi)$, $\psi^*(\xi) < \psi$ by the definition of the infimum, and the West is trapped by (iii). For $\xi > \bar{\xi}(\psi)$, $\psi^*(\xi) > \psi$ by the slope bound, since $\psi^*(\bar{\xi}(\psi)) = \psi > 0$, and the West stays at h_H by (iii). Uniqueness follows from the same strict increase. The set in the definition of $\bar{\xi}$ shrinks as ψ rises, so $\bar{\xi}$ is nondecreasing. $\square$

Proof of Lemma 4. (i) The derivative of the flow (46) in n is $\alpha_H f'(k) - v'(n) - \alpha_H x f''(k)$, with $k = \alpha_H n + x$. For $n < \varphi_H(x)$ the first two terms sum to a positive number, because $f(\alpha_H n + x) - v(n)$ is strictly concave with its maximum at $\varphi_H(x)$, and the last term is nonnegative; so the flow is strictly increasing on $[0, \varphi_H(x)]$. Its second derivative is $\alpha_H^2 [f''(k) - x f'''(k)] - v''(n) < 0$ when $f''' \geq 0$, so it is strictly concave, and its unique maximizer $n_H^R(x)$ solves

(47) and is at least $\varphi_H(x)$. Differentiating (47) gives $dn_H^R/dx = \alpha_H x f'''(k) / (\alpha_H^2 [f''(k) - x f'''(k)] - v''(n)) \leq 0$, and at $x = 0$ (47) is the household condition, so $n_H^R(0) = \varphi_H(0)$; the same holds in regime L . The levels that place next period's knowledge at or above $\chi + \varepsilon$ are those with $n \geq \underline{n}(h)$. The maximizer of a strictly concave function over such a set is its unconstrained maximizer if that lies in the set and the endpoint otherwise, which is (50). It is at least $n_H^R(x) \geq \varphi_H(x)$.

(ii) At $h = \chi + \varepsilon$, $\underline{n}(\chi + \varepsilon) = \delta(\chi + \varepsilon)/\lambda$, and by (i) the floor binds exactly when $n_H^R(x) < \underline{n}(\chi + \varepsilon)$. Where it binds, the rate is $v'(\underline{n}) / (\alpha_H f'(\alpha_H \underline{n} + x)) - 1$ with $\underline{n} = \underline{n}(\chi + \varepsilon)$ fixed, which is strictly increasing in x because f' is strictly decreasing.

(iii) Let $n' < n$, and write $k' = \alpha_i n' + x < k = \alpha_i n + x$. Then

$$u_i^W(n'; x) - u_i^W(n; x) = [f(k') - f(k)] - x[f'(k') - f'(k)] + v(n) - v(n').$$

The first bracket is negative because f is strictly increasing, and the second term is nonpositive because f' is decreasing, so the difference is less than $v(n) - v(n')$. $\square$

Proof of Proposition 10, parts (ii) and (iii). (ii) Two bounds on the West's value function under X^c come first. From any $h' > \chi$, the East ships nothing, and any plan earns at most $\max_n u_H^W(n; 0) = u_H^W(0)$ in each period it spends above χ and at most $\bar{u}_L^W$ in each period it spends at or below, since the East ships $\tilde{x}(h)$ there and $\bar{u}_L^W$ bounds the West's flow against every such shipment at every employment. Since $\bar{u}_L^W < u_H^W(0)$ by (C), the best plan stays above χ and works $\varphi_H(0)$ at every date, which is allowed because $(1 - \delta)h' + \lambda\varphi_H(0) > (1 - \delta)\chi + \delta h_H > \chi + \varepsilon$ for $h' > \chi$, by the bound on the margin in Assumption 4; so $V^W(h', 0; X^c) = V_H^W$. Following N^F attains it, because $\underline{n}(h') < (\delta\chi + \varepsilon)/\lambda < \delta h_H/\lambda = \varphi_H(0)$ for $h' > \chi$, so $N^F(h', 0) = \varphi_H(0)$. From any $h' \leq \chi$, the same bounds show that the value is at most $\bar{u}_L^W + \beta V_H^W$: at least one period is spent at or below χ , and the bound is largest when exactly one is.

Now fix $h > \chi$, a shipment x , and an employment policy that agrees with N^F above χ . The levels in $\mathcal{N}(h)$ split in two. Those that place next period's knowledge at or above $\chi + \varepsilon$ all leave the continuation at V_H^W by the first bound, so among them $N^F(h, x)$ is best, by Lemma 4(i). Those that place it at or below χ leave the continuation at most at $\bar{u}_L^W + \beta V_H^W$, by the second bound, so each loses at least $\beta V_H^W - \beta[\bar{u}_L^W + \beta V_H^W] = \beta[u_H^W(0) - \bar{u}_L^W]$ in continuation. Take such a level n' . If $N^F(h, x) = n_H^R(x)$, the unconstrained maximizer of the flow, then n' has no higher flow either, and it does not gain. Otherwise $N^F(h, x) = \underline{n}(h) > n'$, and by Lemma 4(iii) the gain of n' in flow is less than $v(\underline{n}(h)) \leq v((\delta\chi + \varepsilon)/\lambda)$, because $\underline{n}(h) \leq (\delta\chi + \varepsilon)/\lambda$ for $h > \chi$. By (C) the loss exceeds the gain.

(iii) Fix $h \in ((1 - \delta)\chi, \chi]$ and a shipment x . The levels that return the West to the high regime are those with $n \geq \underline{n}(h)$, and they all leave the continuation at V_H^W by the first bound in (ii). Among them the flow $u_L^W(\cdot; x)$ is highest at $\underline{n}(h)$, because it is strictly concave with maximizer $n_L^R(x) \leq n_L^R(0) = \varphi_L(0) < \delta\chi/\lambda < \underline{n}(h)$, by the proof of Lemma 4(i) and Assumption 3(i). Returning to the defended level means working $\underline{n}(h) < \underline{n}((1 - \delta)\chi)$. The other levels in $\mathcal{N}(h)$ leave the West at or below χ for another period, so their continuation is at most $\bar{u}_L^W + \beta V_H^W$ by the second bound in (ii), and each lies below $\underline{n}(h)$. By Lemma 4(iii), the gain in flow of each over $\underline{n}(h)$ is less than $v(\underline{n}(h))$. Its loss in continuation is at least

$\beta[u_H^W(0) - \bar{u}_L^W]$. The stated condition makes the loss exceed the gain for every such h . Neither bound involves x , so the return is optimal whatever the East ships. Finally, from h_H the policies visit only h_H . From any $h \in ((1 - \delta)\chi, \chi]$ the policies move the state to $\chi + \varepsilon$, and from $h > \chi$ they keep it above χ , so the policies do not leave $((1 - \delta)\chi, \infty)$. A single departure by the West from a state $h > \chi$ reaches $h' \geq (1 - \delta)h > (1 - \delta)\chi$, and a single departure by the East is answered by N^F and leaves the state above χ . On $((1 - \delta)\chi, \infty)$ condition (i) of Definition 1 holds by (i), condition (ii) by (ii) and the argument just given, and condition (iii) because both payoffs are defined at the price (51). $\square$

B Industrial Policy and the Steady-State Set

The multiplicity of Proposition 1 is a property of the competitive equilibrium, in which no one is paid for the knowledge their work creates. This appendix asks how much of it survives when policy pays for that knowledge. We index the correction by its intensity, from laissez-faire to the full Pigouvian subsidy of Proposition 3, and follow the steady-state set as the intensity rises. Two results come out. Under a condition on the shape of the steady-state residual, enough policy destroys the low steady state, so the trap is a property of the allocation mechanism, not of the technology. And the same argument, run in the flow m rather than in the policy, delivers an export fold: a country that ships enough capital goods abroad loses its low steady state too.

Section 3.3 gives the closed form of the corrective subsidy at a steady state. Consider the family of subsidy rates

$$\tau(h, n; \zeta) = \zeta \cdot \frac{\beta \lambda \alpha'(h) n}{\alpha(h) [1 - \beta(1 - \delta)]}, \quad \zeta \geq 0, \quad (56)$$

where the government sets the rate each period on the aggregate state (h_t, n_t) and households, being atomistic, take it as given. The intensity ζ measures how much of the steady-state Pigouvian wedge the policy pays: $\zeta = 0$ is laissez-faire, and at $\zeta = 1$ the steady states of the subsidized economy coincide with the steady states of the planner's first-order conditions, by construction. Along a transition the $\zeta = 1$ schedule is not the optimal subsidy, because the multiplier equals its steady-state value only at a steady state, and away from a steady state it is not even a mild approximation to one: the rate (56) is proportional to employment, so the private return to work rises with employment wherever α' is large, and near the inflection the schedule induces employment orders of magnitude above the laissez-faire level. The two stable steady states barely move, because α' is almost zero at both, and what the schedule does move is what the appendix reports: it pulls the separatrix down and shrinks the trough of the residual, which are the comparative statics the fold analysis needs. What the behavior near the inflection shows is that the schedule is a device for tracing the steady-state set as the correction strengthens, and that it should not be read as a transition policy. With that understood, the subsidized economy is a competitive economy with a static labor margin at every ζ , so the apparatus of Section 3.2 applies to it unchanged.

Equilibrium employment solves $v'(n) = (1 + \tau(h, n; \zeta)) \alpha(h) f'(\alpha(h)n + m)$. Write the solution as $\phi(h; m, \zeta)$, extending the employment function (10) with the policy as a third ar-

gument, $\phi(h; m, 0) = \phi(h; m)$, and write $D(h; m, \zeta) \equiv \lambda\phi(h; m, \zeta) - \delta h$ for the corresponding steady-state residual. We now also allow the flow to be negative: $m < 0$ means the country ships capital goods abroad, and the solution keeps effective capital $k = \alpha(h)n + m$ positive: exports are produced with domestic labor. As in Section 3.2, the S-shape of α does not by itself guarantee the required shape of D , so we impose it directly.

Assumption 5 (Shape restriction under policy). *There exist $x_1 > 0$, $\zeta_1 > 1$, and m_1 as in Assumption 2 such that for every $(m, \zeta) \in [-x_1, m_1] \times [0, \zeta_1]$, the equilibrium employment condition has a unique solution $\phi(h; m, \zeta) > 0$, continuously differentiable, at which the condition crosses in the usual direction, $\Psi_n > 0$ for the Ψ of the proof of Lemma 5, and $D(\cdot; m, \zeta)$ satisfies the shape restriction of Assumption 2: D_h has exactly two interior zeros $h_1(m, \zeta) < h_2(m, \zeta)$, negative below h_1 , positive between, negative above, with $D_{hh}(h_1) > 0$, $D_{hh}(h_2) < 0$, and h_1, h_2 continuously differentiable.*

Lemma 5 (The subsidy lifts the residual where the S-curve is steep). *Under Assumptions 1 and 5,*

$$\frac{\partial \phi}{\partial \zeta}(h; m, \zeta) > 0 \quad \text{and} \quad \frac{\partial \phi}{\partial m}(h; m, \zeta) < 0,$$

with $\partial \phi / \partial \zeta = \alpha'(h) \Lambda(h; m, \zeta)$ for a strictly positive continuous Λ . Hence D is strictly increasing in ζ and strictly decreasing in m , pointwise in h , and the lift in ζ carries the factor $\alpha'(h)$: it vanishes where $\alpha' = 0$ and is concentrated where α' is large.

The proofs are at the end of this appendix.

The factor $\alpha'(h)$ locates the lift. On the two plateaus of the S-curve α' is near zero, so the subsidy pays almost nothing and moves the residual very little. In the steep interior α' is large, and the lift is concentrated there. The interior is also where the trough of D sits, the local minimum h_1 between the low and middle steady states. The subsidy therefore lifts the residual most where its trough is.

Proposition 11 (Industrial policy and the steady-state set). *Suppose Assumptions 1 and 5 hold and the economy has three steady states $h_L^* < h_M^* < h_H^*$ at (m, ζ) with $m \in [0, m_1]$. Then:*

(i) *On the three-steady-state region,*

$$\frac{dh_L^*}{d\zeta} > 0, \quad \frac{dh_M^*}{d\zeta} < 0, \quad \frac{dh_H^*}{d\zeta} > 0 :$$

the subsidy raises both stable steady states and lowers the separatrix.

(ii) *If $D(h_1(m, \zeta_1); m, \zeta_1) > 0$, there is a unique intensity $\bar{\zeta}(m) \in (0, \zeta_1)$ at which the low and middle steady states coalesce in a saddle-node,*

$$D(h_1; m, \bar{\zeta}) = 0 \quad \text{and} \quad D_h(h_1; m, \bar{\zeta}) = 0,$$

and for $\zeta > \bar{\zeta}(m)$ the economy's unique steady state is the high one.

(iii) $\bar{\zeta}$ is strictly increasing in m where it is defined: a larger inflow of capital goods requires more policy to close the trap.

The proposition states that the trap is an equilibrium object, not a technological one. Proposition 1 shows that a large enough inflow destroys the *high* steady state; this proposition shows that a strong enough policy destroys the *low* one. The same technology that gives a laissez-faire economy two stable steady states gives an economy with enough industrial policy a unique, industrialized one. Whether internalization alone suffices is the question whether $\bar{\zeta}(m)$ falls below one. That is a quantitative question, and the answer depends on how deep the trough is relative to the Pigouvian lift.

The same argument, run in the flow rather than in the policy, gives the exporter's version of Proposition 1.

Corollary 1 (The export fold). *Fix $\zeta \in [0, \zeta_1]$ and suppose the economy has three steady states at $m = 0$. Write $x = -m \geq 0$ for net exports of capital goods. Then on the three-steady-state region,*

$$\frac{dh_L^*}{dx} > 0, \quad \frac{dh_M^*}{dx} < 0, \quad \frac{dh_H^*}{dx} > 0,$$

and if $D(h_1(-x_1, \zeta); -x_1, \zeta) > 0$ there is a unique fold $\underline{x}(\zeta) \in (0, x_1)$ at which the low and middle steady states coalesce in a saddle-node; for a sustained surplus $x > \underline{x}(\zeta)$ the economy's unique steady state is the high one. The fold is strictly decreasing in ζ .

The economics of the corollary run through the same margin as everything else in this appendix. An export surplus reduces domestic absorption: at every level of employment, shipping x abroad lowers effective capital, raises its marginal product, and raises the return to industrial work. Employment rises, learning accumulates, and the map lifts. A surplus sustained above $\underline{x}(\zeta)$ removes the low steady state altogether, so the exporter has no trap left to fall back into when the surplus ends. The fold falls with ζ because policy and surplus do the same work at the same margin: a country that pays part of its learning externality needs a smaller surplus to free itself. Section 6 reports $\bar{\zeta}$ and $\underline{x}$.

Proof of Lemma 5. Write the employment condition as

$$\Psi(n; h, m, \zeta) \equiv v'(n) - \alpha(h) f'(\alpha(h)n + m) [1 + \zeta w(h)n] = 0,$$

where

$$w(h) \equiv \frac{\beta \lambda \alpha'(h)}{\alpha(h) [1 - \beta(1 - \delta)]} > 0.$$

Then

$$\Psi_\zeta = -\alpha f'(k) w(h) n < 0, \quad \Psi_m = -\alpha f''(k) [1 + \zeta w(h)n] > 0,$$

the first because $\alpha' > 0$ under Assumption 1, the second because $f'' < 0$. Assumption 5 imposes $\Psi_n > 0$ at the root, so the implicit function theorem gives $\partial \phi / \partial \zeta = -\Psi_\zeta / \Psi_n > 0$ and $\partial \phi / \partial m = -\Psi_m / \Psi_n < 0$. Both statements about $D = \lambda \phi - \delta h$ follow, and $\partial \phi / \partial \zeta = \alpha'(h) \Lambda$ with $\Lambda \equiv \beta \lambda f'(k) n / ([1 - \beta(1 - \delta)] \Psi_n) > 0$. $\square$

Proof of Proposition 11. (i) Each steady state is a simple zero of $D(\cdot; m, \zeta)$, so the implicit function theorem gives $dh^*/d\zeta = -D_\zeta/D_h$ with $D_\zeta > 0$ by Lemma 5. The outer steady states are downward crossings ($D_h < 0$) and the middle one an upward crossing ($D_h > 0$), which signs the three derivatives.

(ii) Let $L(\zeta) \equiv D(h_1(m, \zeta); m, \zeta)$ be the value of D at its interior local minimum. Since $D_h(h_1; m, \zeta) = 0$, differentiating gives $L'(\zeta) = D_\zeta(h_1; m, \zeta) > 0$. Three steady states at ζ require $L(\zeta) < 0$, and $L(\zeta_1) > 0$ by hypothesis, so continuity and strict monotonicity deliver a unique $\bar{\zeta} \in (\zeta, \zeta_1)$ with $L(\bar{\zeta}) = 0$. There $D = D_h = 0$ with $D_{hh}(h_1) > 0$: a saddle-node annihilating the low and middle steady states. Let $M(\zeta) \equiv D(h_2(m, \zeta); m, \zeta)$ be the value of D at its interior local maximum. As for L , $M'(\zeta) = D_\zeta(h_2; m, \zeta) > 0$. Three steady states at the given ζ require $M(\zeta) > 0$, since the middle and high steady states straddle h_2 , so $M > 0$ on $[\zeta, \zeta_1]$. For $\zeta' > \bar{\zeta}$ the local minimum lies above zero, so $D(\cdot; m, \zeta')$ is positive on $(0, h_2]$: it starts at $D(0; m, \zeta') = \lambda\phi(0; m, \zeta') > 0$, falls to $L(\zeta') > 0$ at h_1 , and rises to $M(\zeta') > 0$ at h_2 . Beyond h_2 it is decreasing and tends to $-\infty$, because $\phi(\cdot; m, \zeta')$ is bounded as in the proof of Proposition 1, so it has exactly one zero there, a downward crossing: the high steady state.

(iii) The fold condition $L(m, \bar{\zeta}(m)) = 0$, with L now written as a function of both arguments, has $\partial L/\partial m = D_m(h_1; \cdot) < 0$ and $\partial L/\partial \zeta = D_\zeta(h_1; \cdot) > 0$ by Lemma 5 and $D_h(h_1; \cdot) = 0$. The implicit function theorem gives $d\bar{\zeta}/dm = -L_m/L_\zeta > 0$. $\square$

Proof of Corollary 1. By Lemma 5, $D(\cdot; m, \zeta)$ rises pointwise as m falls, so the argument of Proposition 11 applies along m at fixed ζ , with $-D_m > 0$ in place of $D_\zeta > 0$ and with the local maximum M now increasing in x from its positive value at $x = 0$. For the last claim, the fold condition $L(m, \zeta) = 0$ defines $\underline{m}(\zeta) = -\underline{x}(\zeta)$ with $d\underline{m}/d\zeta = -L_\zeta/L_m > 0$, since $L_\zeta > 0$ and $L_m < 0$; hence $\underline{x}$ is strictly decreasing. $\square$

At the parameters of Section 6, policy moves the trap in the way Lemma 5 predicts. Paying the full steady-state wedge, the $\zeta = 1$ schedule of (56), moves the two stable steady states barely at all but pulls the separatrix down by a full unit and cuts the depth of the trough of D nearly in half, because the lift carries the factor α' . Internalization alone does not close the trap there, and the export fold from autarky is a small fraction of the import fold, which internalization cuts further: a country that corrects its own learning externality sits close to its own fold. Section 6 reports the policy fold, the two export folds, and the parameter changes that bring the policy fold below one.

C Efficient Basins of Attraction

The multiplicity of Section 3 is a property of the competitive equilibrium. This appendix asks when it is also a property of the efficient allocation. The setting is a single closed economy in the sharp-threshold benchmark of Section 4.3, run by a planner:

$$\max_{\{n_t \geq 0\}} \sum_{t=0}^{\infty} \beta^t [f(\alpha(h_t) n_t) - v(n_t)] \quad \text{subject to} \quad h_{t+1} = (1 - \delta)h_t + \lambda n_t, \quad h_0 \text{ given,} \quad (57)$$

with $\alpha(h) = \alpha_L$ for $h \leq \chi$ and $\alpha(h) = \alpha_H$ for $h > \chi$, the convention of Assumption 3. The problem is not concave, so we argue from plans, not from first-order conditions.

Write $u_i(n) \equiv f(\alpha_i n) - v(n)$ for the regime- i flow, n_i for its maximizer, the solution of $v'(n) = \alpha_i f'(\alpha_i n)$, and $u_i^* \equiv u_i(n_i)$. Two facts organize the analysis. The flow depends on the knowledge stock only through the regime. And the flow maximizer sustains its own steady state: n_i is also the employment that holds h constant at $h_i = \lambda n_i / \delta$. We impose the autarky part of Assumption 3(i), $\lambda n_L < \delta \chi < \lambda n_H$, so that $h_L < \chi < h_H$; note that $n_i = \varphi_i(0)$.

Three numbers govern the results. The gain from crossing is the flow gap $\Delta u \equiv u_H^* - u_L^*$, positive because $u_H(n) > u_L(n)$ for every $n > 0$. The floor is

$$s^* \equiv u_L^* - u_L(\delta \chi / \lambda) > 0 :$$

a plan that crosses χ must, in its crossing period, carry knowledge from at most χ to above it, so it employs more than $\delta \chi / \lambda$, which exceeds n_L by assumption, and u_L is strictly concave, so the crossing period sacrifices at least s^* of flow. The climb cost is

$$C_0^{\text{up}}(h_0) \equiv \inf \left\{ \sum_{t=0}^{T-1} [u_L^* - u_L(n_t)] : h_{t+1} = (1 - \delta)h_t + \lambda n_t, h_T > \chi \right\},$$

the undiscounted counterpart of (40), run uphill; it is nonincreasing in h_0 and at least s^* .

Proposition 12 (Basins of the efficient allocation). *Let $\lambda n_L < \delta \chi < \lambda n_H$.*

- (i) *From every $h_0 > \chi$ the unique optimal plan sets $n_t = n_H$ at every date, and $h_t \rightarrow h_H$.*
- (ii) *If $\beta \Delta u < (1 - \beta) s^*$, then from every $h_0 \leq \chi$ the unique optimal plan sets $n_t = n_L$ at every date, and $h_t \rightarrow h_L$. The efficient allocation then has two basins of attraction, separated at χ , and two efficient steady states.*
- (iii) *If $(1 - \beta) s^* < \beta \Delta u < (1 - \beta) C_0^{\text{up}}(h_L)$, the efficient allocation still has two basins: there is a threshold $\hat{h} \in (h_L, \chi)$ such that from every $h_0 < \hat{h}$ the plan $n_t = n_L$ at every date is optimal and knowledge converges to h_L , while from every $h_0 \in (\hat{h}, \chi]$ every plan that never crosses χ is strictly suboptimal.*

Proof. (i) From $h_0 > \chi$, the plan $n_t = n_H$ keeps $h_{t+1} > (1 - \delta)\chi + \lambda n_H > \chi$, so its flow is u_H^* at every date and its value is $u_H^* / (1 - \beta)$. No plan does better, since every flow is at most $\max\{u_L^*, u_H^*\} = u_H^*$. A plan attaining the bound must earn u_H^* at every date, which forces regime H and $n_t = n_H$ throughout, by strict concavity of u_H . The law of motion then converges to h_H .

(ii) From $h_0 \leq \chi$, the plan $n_t = n_L$ stays in regime L , because $h_{t+1} \leq (1 - \delta)\chi + \lambda n_L < \chi$; its value is $u_L^* / (1 - \beta)$ and it converges to h_L . Every plan that never crosses has flows at most u_L^* , so none does better, and one that ties must set $n_t = n_L$ throughout, by strict concavity of u_L . Now take a plan that first crosses at date T . Its pre-crossing flows are $u_L^* - s_t$ with $s_t \geq 0$ and $s_{T-1} \geq s^*$, and its flows from T on are at most $u_L^* + \Delta u$. Its value therefore exceeds $u_L^* / (1 - \beta)$ by at most

$$-\beta^{T-1} s^* + \beta^T \frac{\Delta u}{1 - \beta} = \beta^{T-1} \left[\frac{\beta \Delta u}{1 - \beta} - s^* \right] < 0.$$

(iii) Three steps. First, the gain from crossing is monotone in the initial stock. Let $V^x(h_0)$ be the supremum of values over crossing plans; the best never-crossing value is $u_L^*/(1 - \beta)$ from every $h_0 \leq \chi$, as in (ii). For $h'_0 < h''_0 \leq \chi$, any crossing plan from h'_0 , run from h''_0 with the same employment path, has a pointwise higher knowledge path, crosses no later, and earns weakly higher flows at every date, so V^x is nondecreasing, and the set of stocks from which crossing dominates is an interval reaching to χ . Let $\hat{h}$ be its infimum. Second, crossing dominates near χ . From $h_0 \leq \chi$ the plan with $n_0 = (\chi' - (1 - \delta)h_0)/\lambda$ for some $\chi' > \chi$ and $n_t = n_H$ thereafter crosses at once; its gain over staying is $\beta\Delta u/(1 - \beta) - [u_L^* - u_L(n_0)]$, and as $h_0 \uparrow \chi$ and $\chi' \downarrow \chi$ the bracket falls to s^* , which lies below $\beta\Delta u/(1 - \beta)$ by the left inequality. Third, staying dominates near h_L . The bound in (ii), run with the full climb cost in place of the single-period floor, caps the gain of any plan crossing at T from h_0 by $\beta^{T-1}[\beta\Delta u/(1 - \beta) - C_0^{\text{up}}(h_0)]$, which is negative at $h_0 = h_L$ by the right inequality. And C_0^{up} is continuous at h_L from above: it is nonincreasing, and concatenating the one-period step from h_L to h_0 , whose sacrifice $u_L^* - u_L((h_0 - (1 - \delta)h_L)/\lambda)$ vanishes as $h_0 \downarrow h_L$, with the cheapest climb from h_0 gives $C_0^{\text{up}}(h_L) \leq u_L^* - u_L((h_0 - (1 - \delta)h_L)/\lambda) + C_0^{\text{up}}(h_0)$. So staying dominates on an interval above h_L , and the threshold $\hat{h}$ lies strictly between h_L and χ . $\square$

This is the history dependence of [Skiba \(1978\)](#) and [Dechert and Nishimura \(1983\)](#): in a planner problem with a convex-concave technology, optimal paths converge to different steady states depending on the initial stock, and $\hat{h}$ plays the part of the Skiba point: it separates the two basins. We have not shown that the planner is indifferent at $\hat{h}$ between staying and an attained crossing plan, which is how the point is usually defined; that would require the crossing value to be continuous and attained at $\hat{h}$, and part (iii) does not need it. Under the convention that places χ itself in the low regime, the set of plans that cross is open, so from $h_0 \in (\hat{h}, \chi]$ the supremum over crossing plans need not be attained; the three parts of the proposition are proved by bounds and do not need it to be. The benchmark adds explicit boundaries. At the parameters of [Section 6](#), $u_L^* = 0.17$, $u_H^* = 1.58$, $s^* = 0.11$, and $C_0^{\text{up}}(h_L) = 1.01$, so part (ii) applies for $\beta < 0.07$ and part (iii) for β between 0.07 and 0.42. At $\beta = 0.96$ neither applies, and no trap survives: the climb from h_L gains 31.2 in present value over staying, the climb from $h_0 = 0$ gains 28.4, and by the monotonicity in the proof of (iii) the planner's low basin is empty. The climb is a planner's: a government whose subsidy rate is capped below the rate the floor employment requires in regime L , which [Section 6](#) reports, cannot make it. The competitive economy of [Section 3](#) has two basins at every discount factor, because no private margin prices the knowledge externality. The planner's two-basin region requires heavy discounting, and at the discount factor of the numerical example it is empty, so the market's low basin is pure inefficiency there. Closing it is not a matter of the steady-state margin alone: the steady-state schedule of [Appendix B](#) at $\zeta = 1$ leaves the low steady state in place at these parameters, while the efficient allocation escapes it.

D Predation from a Pre-industrial Position

Assumption 3(ii) starts both countries industrialized. This appendix examines what that leaves out: a West that begins in regime H facing an East that begins at or below χ . With the East's own regime free to change, its knowledge is a second state, the reduction of Lemma 2 fails, and we do not solve the problem. One structural fact can still be proved: no plan traps the West while the East remains pre-industrial. Trapping requires a flood, the flood requires labor, and at low productivity that labor industrializes the country that supplies it. The condition is

$$\lambda \bar{m}_\infty \geq \alpha_L \chi, \quad (\text{P})$$

which says that producing the fold flow at low productivity, employment of $\bar{m}_\infty/\alpha_L$, generates at least the threshold stock of knowledge in a single period. At the parameters of Section 6 the left side is 8.85 against $\alpha_L \chi = 1.25$: the condition holds by a factor of seven.

Proposition 13 (Predation from below industrializes the exporter). *Let the West begin in regime H , and let part (iv) of Assumption 3 and Condition (P) hold. Consider any export plan that carries the West's knowledge to χ or below, and let T be the first date at which it is there. Then the East is in regime H at date $T - 1$ or at date T . In particular, no plan traps the West while the East remains pre-industrial throughout.*

Proof. The West is in regime H at every date before T , so $h_{T-1}^W > \chi$ and

$$\lambda \varphi_H(x_{T-1}) = h_T^W - (1 - \delta)h_{T-1}^W \leq \chi - (1 - \delta)h_{T-1}^W < \chi - (1 - \delta)\chi = \delta\chi.$$

Since φ_H is decreasing and $\lambda \varphi_H(\bar{m}_\infty) = \delta\chi$, the crossing shipment satisfies $x_{T-1} > \bar{m}_\infty$. If the East is in regime H at date $T - 1$, the claim holds at that date. If it is in regime L , feasibility $x_{T-1} \leq \alpha_L n_{T-1}$ gives $n_{T-1} > \bar{m}_\infty/\alpha_L$, so

$$h_T^E = (1 - \delta) h_{T-1}^E + \lambda n_{T-1} > \lambda \bar{m}_\infty/\alpha_L \geq \chi$$

by Condition (P), so the East is in regime H at date T . $\square$

The proposition sorts predation from below into two cases, and both end at the configuration the text assumes. If the East industrializes before the final flood, then from that date it is an industrialized country facing a still-industrialized West, deciding whether to finish the crossing: the position of Assumption 3(ii). If instead it is still pre-industrial in the crossing period, its own flood completes its industrialization in the same period the West falls, and the two countries exchange positions at once. This exchange is the role reversal announced in the introduction. Nothing ties the exporter to the high regime. A country at or below χ that ships a sustained surplus raises its own employment and knowledge while depressing its customer's, and by the logic of Corollary 1 a surplus above $\underline{x}(\zeta)$ carries it across its own threshold. An East that begins at the low steady state and runs a large enough surplus for long enough ends at the high one, and the West it ships to ends at the low one. Whether such a surplus is commerce is a different question, and we answer only part of it. The next proposition, which holds in the smooth model and in the benchmark alike, says that while the East remains capital-scarce, shipping has no commercial value.

Proposition 14 (A capital-scarce East sells nothing). *Set $\psi = 0$. Call a plan of the East capital-scarce if*

$$\alpha(h_t^E) n_t^E \leq \alpha(h_t^W) n_t^W + x_t \quad \text{for all } t :$$

keeping every shipped unit at home, the East would still have less effective capital than the West holds with them. Removing every shipment from a capital-scarce plan weakly raises its value, and strictly so if $x_t > 0$ at some date. At $\psi = 0$, therefore, no optimal plan is both capital-scarce at every date and exports a positive amount at some date: an optimal plan that keeps the East capital-scarce throughout ships nothing.

Proof. Fix a capital-scarce plan $\{n_t, x_t\}$ and let the alternative plan keep the same employment and ship nothing. The East's knowledge path depends only on employment, so it is unchanged, and so is the disutility of labor. The value difference between the alternative and the original is

$$\sum_{t=0}^{\infty} \beta^t \left[f(\alpha(h_t^E) n_t) - f(\alpha(h_t^E) n_t - x_t) - q_t x_t \right].$$

By strict concavity of f , $f(\alpha n_t) - f(\alpha n_t - x_t) \geq x_t f'(\alpha n_t)$, strictly if $x_t > 0$. By capital scarcity and $f'' < 0$, $f'(\alpha(h_t^E) n_t) \geq f'(\alpha(h_t^W) n_t^W + x_t) = q_t$. Every term is therefore nonnegative, and positive at any date with $x_t > 0$: removing a shipment loses the revenue $q_t x_t$ but returns goods worth more at home. $\square$

At the natural configuration of a lagging East, its low steady state against the West's high one, the East's output of capital goods is a fraction of the West's effective capital, 0.12 against 3.7 at the parameters of Section 6, and a shipment only widens the gap, since it raises the West's effective capital (Lemma 1) and leaves the East's output unchanged. A plan that keeps employment near its steady-state level is therefore capital-scarce at every date, and the proposition says that such an East gains nothing from shipping: whatever revenue a shipment earns, the goods are worth more at home. The class does not contain every plan a lagging East can run. A plan that raises employment enough to industrialize the East leaves it once the East's productivity is high, and from that date the East faces the export decision of Section 4, including the commercial gain $\Delta\Pi$ from a trapped customer. A role-reversal surplus is therefore not covered by the proposition, and whether a pre-industrial East would choose one is a value comparison, bundling its own industrialization gain, the subject of Appendix C, with the gain from trapping the West, that we do not characterize. What the two propositions establish is what any such plan does to the East itself: predation cannot be completed from below, executing it manufactures exactly the industrial capacity that Assumption 3(ii) assumes, and while the East remains capital-scarce it has no commercial reason to ship at all.

E Predation with Gift

Section 4.3 lets the East sell its shipments at the West's marginal product. This appendix removes the payment. The East ships capital goods and receives nothing for them, so every

unit is a gift, and the question is whether the security motive alone produces the flood, the trap, and the return to autarky of Proposition 7. It does, above the critical security weight of Proposition 4, which Proposition 15 shows is at least ψ^* , and the plan ends differently from the paid one: once the West is trapped the East ships nothing at all.

Environment. Everything is as in Section 4.3 except that the East receives nothing for its shipments. At a price of zero the West's import margin (19) has no finite solution, so the quantity is not demanded but delivered: the East chooses it, and the West's households absorb it as the inflow of Section 3 with $m_t = x_t$. Absorbing it is individually rational, since a free unit of capital raises a household's output, and only the government could refuse or destroy it; we take the delivery as given and return to refusal below. The West's labor margin is unchanged, its employment in regime i is $\varphi_i(x)$, and its flow welfare is $u_i^{W,0}(x) \equiv f(\alpha_i \varphi_i(x) + x) - v(\varphi_i(x))$, which exceeds the autarky flow $u_i^W(0)$ for every $x > 0$. The East's budget is $c_t^E = f(\alpha_H n_t^E - x_t)$ and its objective is (22). Its employment at a given shipment is the same $n^E(x)$ as before, because the price never entered the East's employment optimality condition (30), and its flow payoff from shipping x is

$$\Pi^0(x) \equiv f(\alpha_H n^E(x) - x) - v(n^E(x)), \quad (58)$$

the gift payoff of Section 4.3, the payoff (34) without its revenue term. It does not depend on the West's regime. By the envelope theorem, $\Pi^{0'}(x) = -f'(\alpha_H n^E(x) - x) < 0$: every unit shipped costs the East its marginal product at home, so the commercial export is zero in both regimes. The flow cost K^0 and the plan cost C_P^0 are as defined there.

Assumption 6 (Gift benchmark). *The step (35) and parts (i), (ii), and (iv) of Assumption 3 hold, and the ratio $-\Pi^{0'}(x)/(-\lambda\varphi'_H(x))$ is strictly increasing on $[0, \infty)$.*

The ratio condition replaces part (iii) of Assumption 3. The first half of that part is not needed, since Π^0 is decreasing and so is maximized at zero. The second half is the same shape restriction as before, on the cost of a gift rather than of a sale. It holds at the parameters of Section 6.

Lemma 2 holds without change, because its proof does not use the price: the East stays industrialized, regime L is absorbing, the best plan that never crosses ships nothing, and the best continuation after a crossing ships nothing, since Π^0 is decreasing and g_L is fixed. The reduced problem is (36) with Π^0 in place of Π_{it} , and Proposition 4 gives the critical security weight ψ_0^* and, through (37), its ordering against ψ^* . Two more facts hold.

Proposition 15 (Predation with gift). *Under Assumption 6:*

(i) *The critical security weight with gifts is strictly positive,*

$$\psi_0^* \geq \frac{1 - \beta}{\beta} \frac{r_0 (h_H - \chi)}{\Delta g}, \quad r_0 \equiv \frac{f'(\alpha_H n^E(0))}{-\lambda\varphi'_H(0)} > 0,$$

and if Assumption 3(iii) also holds, then $\psi_0^ \geq \psi^*$: a gift requires at least the weight a sale requires.*

(ii) For $\psi > \psi_0^*$ an optimal plan exists, and every optimal plan has two phases. In the active phase, $0 \leq t < T^*$ with T^* finite, the shipment is strictly positive, strictly increasing, and largest in the period the West's knowledge reaches χ , and the West's knowledge falls strictly. In the passive phase, $t \geq T^*$, the West's regime is L and permanent, and the East ships nothing.

Proof. (i) For the bound, write $K^0(x) = \int_0^x r(y)(-\lambda\varphi'_H(y)) dy$ with r the ratio of Assumption 6, so that $K^0(x) \geq r_0\lambda[\varphi_H(0) - \varphi_H(x)]$, because $r \geq r(0) = r_0$. A plan that crosses at T_P has $h_{T_P}^W \leq \chi$, and since $h_H = (1 - \delta)^{T_P} h_H + \lambda \sum_{t < T_P} (1 - \delta)^{T_P-1-t} \varphi_H(0)$,

$$\lambda \sum_{t < T_P} [\varphi_H(0) - \varphi_H(x_t)] \geq \lambda \sum_{t < T_P} (1 - \delta)^{T_P-1-t} [\varphi_H(0) - \varphi_H(x_t)] = h_H - h_{T_P}^W \geq h_H - \chi.$$

Hence $C_P^0 \geq \beta^{T_P-1} \sum_{t < T_P} K^0(x_t) \geq \beta^{T_P-1} r_0(h_H - \chi)$, so $\beta^{-T_P} C_P^0 \geq r_0(h_H - \chi)/\beta$ for every crossing plan, and the infimum obeys the same bound. For the ordering, both bracketed terms in (37) are nonnegative, the revenue because $q_H(x)x \geq 0$ and $\Delta\Pi$ because $\Pi_L(x_L^c) \geq \Pi_L(0) = \Pi_H(0)$ under Assumption 3(iii); so each plan's zero is at least as large under the gift as under the sale, and so is the infimum in (38).

(ii) The proof of Proposition 7 applies with K^0 in place of K and $\Delta\Pi = 0$. Existence uses the closed capture set and $\Pi^0(x) \leq \Pi_H(x) \rightarrow -\infty$ as $x \rightarrow \infty$. The constraint binds because $K^{0'} > 0$, the multiplier is positive, the Kuhn–Tucker conditions hold with equality at every active date, the ratio argument runs under Assumption 6, and the induction for h^W is unchanged. The passive phase is Lemma 2(ii) together with the fact that, once the regime is L , the flow $\Pi^0(x) - \psi g_L$ is maximized at $x = 0$ at every date. $\square$

What the gift changes. Relative to Section 4.3 the gift removes two things, adds one, and changes a third. It removes the commercial motive, so there is no Condition (D) to check and no trade with the trapped customer afterward. It removes the revenue that pays part of the flood, so the critical security weight is higher. And it adds a transfer to the West. Because the goods are free, every active period raises the West's flow welfare above its autarky level, so the West's lifetime loss from predation is the trap net of the gift, and it can have either sign: an impatient West can be better off accepting the flood, a patient one is not, because the trap is permanent and the gift is not. This is the qualification under which the deterrence results of Section 5 carry over. The gift also changes the border instrument. A tariff acts on the West's import margin, which a delivered quantity bypasses, so under a gift the border instruments are refusal and a ban. Both are costless here, as the prohibitive tariff of Section 5.1 is costless in the paid benchmark because the commercial export between industrialized countries is zero, and neither is formalized in this paper.

Deterrence with gift. The game of Section 5.3 simplifies in the same way. The East's flow payoff from a shipment is $\Pi^0(x)$ whatever the West works, so the West's employment reaches the East only through the state, and the price condition of Definition 1 is empty. The West's flow welfare when it works n and receives x in regime i is $u_i^{W,0}(n; x) = f(\alpha_i n + x) - v(n)$, which is strictly concave in n with maximizer $\varphi_i(x)$. There is no price to move, so the

West's static best response is its competitive employment, and Assumption 4, with the same defended level $\chi + \varepsilon$, is the only restriction on its choice. Export and employment policies, the rule N^F , and the Bellman equations are as before with these payoffs; above χ the rule is $N^F(h, x) = \max\{\varphi_H(x), \underline{n}(h)\}$. The East's export policy X^c ships nothing at every state: above χ as in Section 5.3, and at or below χ because the maximizer of Π^0 is zero. The best flow a trapped West can earn is therefore its autarky flow $u_L^W(0)$, and Condition (C) becomes a comparison of autarky objects:

$$v\left(\frac{\delta\chi + \varepsilon}{\lambda}\right) < \beta [u_H^W(0) - u_L^W(0)]. \quad (C^0)$$

Its right side rises with β , so once it holds it holds for every more patient West.

Proposition 16 (Deterrence with gift). *Suppose Assumptions 4 and 6 and Condition (C⁰) hold.*

- (i) *Given N^F , for every $\psi \geq 0$ and every $h \geq 0$, shipping nothing is the unique maximizer in the East's Bellman equation, and at every $h > \chi$, $V^E(h; N^F) = [\Pi^0(0) - \psi g_H]/(1 - \beta)$.*
- (ii) *Given X^c , for every $h > \chi$ and every $x \geq 0$, $N^F(h, x)$ attains the supremum in the West's Bellman equation, whatever the policy prescribes at or below χ .*
- (iii) *If in addition $v(\underline{n}((1 - \delta)\chi)) < \beta[u_H^W(0) - u_L^W(0)]$, then (X^c, N^F) is a Markov perfect equilibrium on $((1 - \delta)\chi, \infty)$.*
- (iv) *Let P be a finite flood: a plan that ships x_t for $t < T_P$, carries a West using the competitive response to χ or below at T_P , and ships nothing afterward, as the optimal plan of Proposition 15(ii) does. The West's value under (X^c, N^F) from h_H exceeds its value under the competitive response against P by*

$$\sum_{t < T_P} \beta^t [u_H^W(0) - u_H^{W,0}(x_t)] + \beta^{T_P} \frac{u_H^W(0) - u_L^W(0)}{1 - \beta}.$$

The first term is the gift; it is negative and bounded. The second is the trap; it is positive and grows without bound as $\beta \rightarrow 1$. For every such plan there is therefore a discount factor below one above which the West prefers the floor to accepting the gift.

Proof. (i) From $h > \chi$, under N^F the West's regime is H at every date whatever is shipped, so a shipment sequence is worth $\sum_t \beta^t [\Pi^0(x_t) - \psi g_H]$. Since Π^0 is strictly decreasing, this is at most $[\Pi^0(0) - \psi g_H]/(1 - \beta)$, with equality only if every shipment is zero. From $h \leq \chi$ the West returns to the defended level whatever is shipped, so the East's continuation is fixed and it maximizes the flow $\Pi^0(x) - \psi g_L$, shipping nothing.

(ii) Fix $h > \chi$ and x . The levels that keep knowledge at or above $\chi + \varepsilon$ all leave the continuation at V_H^W , by the first bound in the proof of Proposition 10(ii), and among them $N^F(h, x)$ has the highest flow, by strict concavity. Take a level n' that sends knowledge to χ or below. If $N^F(h, x) = \varphi_H(x)$, the unconstrained maximizer, n' has no higher flow. Otherwise $N^F(h, x) = \underline{n}(h) > n'$, and the gain in flow of n' is $u_H^{W,0}(n'; x) - u_H^{W,0}(\underline{n}(h); x) <$

$v(\underline{n}(h)) \leq v((\delta\chi + \varepsilon)/\lambda)$, because $f(\alpha_H \underline{n}(h) + x) > f(\alpha_H n' + x)$ and $\underline{n}(h) \leq (\delta\chi + \varepsilon)/\lambda$ for $h > \chi$. At or below χ the East ships nothing, so the West's flow there is at most $\max_n u_L^{W,0}(n; 0) = u_L^W(0)$, and the best it can do afterward is return to $V_H^W = u_H^W(0)/(1 - \beta)$; its loss in continuation is at least $\beta[u_H^W(0) - u_L^W(0)]$, which exceeds the gain by (C⁰).

(iii) For the West at $h \in ((1 - \delta)\chi, \chi]$, the best return is to the defended level, because $\varphi_L(x) \leq \varphi_L(0) < \delta\chi/\lambda < \underline{n}(h)$. Relative to any level that stays at or below χ , it costs less than $v(\underline{n}((1 - \delta)\chi))$ in flow, by the argument in (ii), and it secures V_H^W , while staying at or below χ costs at least $\beta[u_H^W(0) - u_L^W(0)]$ by the bound in (ii); the stated condition orders the two, whatever the East ships. The policies do not leave $((1 - \delta)\chi, \infty)$, from h_H they visit only h_H , and a single departure by either player leaves the state in that set, as in the proof of Proposition 10(iii).

(iv) Under (X^c, N^F) the West's value from h_H is V_H^W by (i). Under the competitive response against P , its knowledge crosses at T_P and it receives nothing afterward, so its value is $\sum_{t < T_P} \beta^t u_H^{W,0}(x_t) + \beta^{T_P} u_L^W(0)/(1 - \beta)$. Subtracting gives the display. The first sum has finitely many terms, each between $u_H^W(0) - u_H^{W,0}(x_t)$ and zero, so it is bounded uniformly in β ; the second term grows without bound as $\beta \rightarrow 1$ because $u_H^W(0) > u_L^W(0)$, which holds since $\alpha_H > \alpha_L$ and f is increasing. $\square$

The gift isolates the two forces of Section 5.3. The floor deters because a West that pays for its knowledge leaves the East nothing to gain from a flood, and the floor is credible because holding it for a period costs a bounded disutility while a crossing costs a permanent fall in the autarky flow. What the gift adds is the one reason a West might not want to be defended: it is being paid in goods for the capability it loses, and whether that payment covers the trap is a question of patience. The ratio condition of Assumption 6, the ordering of the two critical security weights, and the sign of the West's lifetime loss are verified numerically at the parameters of Section 6.

F The Counterpolicy in the Smooth Economy

Section 5 proves its results in the sharp benchmark. This appendix computes the same game with the logistic productivity schedule (3), at the parameters of Section 6 and at larger values of the steepness γ , to see which of the results depend on the step. The West that pays for its knowledge is not trapped at any weight at which we find an equilibrium, the East ships nothing on the equilibrium path, and the East's flood against it is unprofitable. Two objects change form: the subsidy on the path, zero in the benchmark, becomes the West's domestic correction, and the floor at χ becomes a level inside the steep region.

The game. The state is the West's knowledge h . Productivity is $\alpha(h)$, and the security loss is $g(h) = \mathcal{D}(\alpha(h))$ with $\mathcal{D}(z) = z^2/2$. The gap between the two competitive steady states is then $\Delta g = g(h_H^*) - g(h_L^*) = 4.45$, and we report the security weight as $\psi \Delta g$, as in Section 4.3. We hold the East on its upper plateau, where $\alpha'(h_H^*) = 0.003$, so its employment is the static $n^E(x)$ of Section 4.3 and its knowledge is not a state. Timing and payoffs are

those of Definition 1 with $\alpha(h)$ in place of α_i : each period the East ships $x = X(h)$, the West observes x and works $n = N(h, x)$, and the price is $f'(\alpha(h)n + x)$. The West chooses its employment at realized welfare (46), which is the criterion in its Bellman equation (53), and it chooses from the whole employment grid. The equilibrium concept is Definition 1 with the feasible set $\mathcal{N}(h)$ equal to that grid. This is the game of Section 5.3 with $\alpha(h)$ in place of α_i , and it needs no margin: productivity rises with knowledge, so the West has a reason to prefer one level of knowledge to another. We call it the *unrestricted* smooth game. As in Section 5.2, the West’s choice answers two motives, its knowledge and the price it pays for imports. On the equilibrium path nothing is shipped, so the price motive is absent and the subsidy there is the domestic correction and nothing else. Off the path both are at work, and we separate them when we report the West’s response to a flood. What the learning motive alone gives is reported separately below.

Method. We solve each player’s problem by value-function iteration on grids: 240 points of knowledge on $[0.3, 8]$; shipments in steps of 0.01 on $[0, 1]$ and 0.2 beyond, to 16, or to 30 in the runs with large weights; employment in steps of 0.0125 on $[0, 3]$. Continuation values are interpolated linearly between knowledge points. The object we compute is therefore an equilibrium of the discretized game, in which each player’s actions are restricted to the grids, and not of the continuous game. We iterate best responses. Given the East’s export policy, the West’s employment policy solves its dynamic problem at every knowledge point and every shipment on the grid; given the West’s policy, the East’s export policy solves its problem at every knowledge point. The iteration stops when neither policy changes at any grid point, and we run it from two starting policies for the West, the competitive response and a uniformly high employment. A pair at which the iteration stops is then checked directly. We evaluate each player’s value from following its policy against the other’s and compute the largest gain from a one-stage deviation over the whole grid: for the East over every shipment at every knowledge point, for the West over every employment at every pair of knowledge and shipment, both at realized welfare, as condition (ii) of Definition 1 requires. A policy that no one-stage deviation improves is optimal in a discounted dynamic program, so a gain of zero at every grid state rules out every deviation plan on the grids. We call the pair an equilibrium only when neither gain exceeds 10^{-8} ; in every cell of Table 4 the East’s largest gain is below 10^{-9} and the West’s is zero. Where the iteration does not stop we say so, describe the iterates, and claim no equilibrium. No policy or best response sits at the top of an action grid, and no path reaches the top of the knowledge grid. The policy arrays and the deviation gains behind every entry of Table 4 are archived with the replication code. A no-defense benchmark fixes the West at its competitive response and lets the East optimize against it.

Results. Table 4 reports the outcomes. Against the passive West, the East ships at most 0.03 at $\psi\Delta g = 0.3$ and floods at 0.6, so the critical security weight in the smooth economy lies between the two, below the benchmark’s 1.66. The flood that traps the passive West peaks at 8.0 at $\gamma = 4.2$ and at 13.4 at $\gamma = 32$, toward the benchmark’s 14.3. These peaks are on this appendix’s shipment grid, whose step is 0.2 above one unit; on the finer grid of Figure 5 the peak at $\gamma = 4.2$ is 7.6. Against the West that uses the corrective subsidy the answer is

Passive West		Unrestricted smooth-game equilibrium					Price-taking West		
$\psi\Delta g = 1.66$		all three weights					deviation gain at $\psi\Delta g =$		
γ	peak x	h	x	τ	h		1.66	2.5	4
4.2	8.0	0.9	0	0.004	6.19		0.036	0.042	none found
8	9.6	0.9	0	0.004	6.19		0.041	0.040	0.044
16	11.8	0.9	0	0.004	6.19		0.044	0.044	0.047
32	13.4	0.9	0	0.004	6.19		0.046	0.046	0.051

Table 4: The counterpolicy game with a smooth productivity schedule, at the parameters of Section 6 and larger steepness. Left: the East’s optimal plan against a West that does nothing. Middle: the equilibrium of the unrestricted smooth game, in which the West chooses employment from the whole grid at realized welfare, which is the same at all three security weights, $\psi\Delta g \in \{1.66, 2.5, 4\}$; x is the largest shipment on the path, τ is the West’s subsidy rate on the path, and h is knowledge after 300 periods. The rate equals the domestic correction of Section 3.3 at the high steady state, 0.004 at $\gamma = 4.2$ and below 10^{-4} beyond, to within one step of the employment grid, which is 0.0125 in employment and about 0.01 in the rate. In every cell the best-response iteration stops from both starting policies at the same pair, and neither player has a profitable one-stage deviation at any grid state. Right: the largest gain in realized welfare the West could obtain by changing its employment for one period, at the pair reached when it is restricted to choose its employment taking the import price as given; this is what the restriction costs it, and it is why that pair is not an equilibrium of Definition 1. At $\gamma = 4.2$ and $\psi\Delta g = 4$ the restricted iteration does not stop and no such pair was found. The separatrix is 4.01 at $\gamma = 4.2$.

the same in every cell of the table. At every steepness and at every weight up to $\psi\Delta g = 4$, which is 2.4 times the benchmark’s critical security weight, the iteration stops from both starting policies at the same pair, and the pair is an equilibrium of the discretized game: neither player has a profitable one-stage deviation at any grid state, the East’s largest gain being below 10^{-9} and the West’s zero. In it the East ships nothing, the West’s knowledge sits at the planner’s high steady state, 6.19, and the subsidy on the path equals the domestic correction of Section 3.3 at that state to the precision of the employment grid: the rate is 0.004 at every steepness, and the correction is 0.004 at $\gamma = 4.2$ and below 10^{-4} beyond, the two agreeing within one step of that grid. The West is never trapped. The exact autarky of the benchmark therefore survives the smooth schedule, and for the benchmark’s reason: no flood against a West that pays for its knowledge is worth its cost. On a grid with twice the knowledge points and half the action steps the passive West’s peaks move at the first decimal, to 7.5 at $\gamma = 4.2$ and 13.5 at $\gamma = 32$, the trapped West’s knowledge settles at 0.97 rather than 0.94, the East ships nothing at $\psi\Delta g = 0.3$, and the bracket on the critical security weight is unchanged. The equilibrium at $\psi\Delta g = 4$ is found again on that grid, at $\gamma = 4.2, 8$, and 32, from the competitive start, and at $\gamma = 4.2$ also from the base-grid policy mapped onto it: the East ships nothing, knowledge settles at 6.19 at $\gamma = 4.2$ and 6.16 at the two larger steepnesses, and the subsidy on the path is within one grid step of the domestic correction.

The learning motive alone. The unrestricted game lets the West act on the price it pays as well as on its own knowledge, so it answers whether the no-trap outcome survives a smooth schedule, not whether the learning correction alone delivers it. To isolate the second question we restrict the government to choose employment taking the import price as given, which removes the price motive. That choice is not optimal at realized welfare, so the object is no longer an equilibrium of Definition 1 with $\mathcal{N}$ the grid; we call it a price-taking policy fixed point and report what it looks like. The iteration stops at eleven of the twelve cells, at pairs with the same deterrence outcome: the East ships nothing, except 0.07 every fifth period at $\gamma = 4.2$ and $\psi\Delta g = 2.5$, and the West's knowledge stays at the high steady state with the same subsidy on the path. So in the cells where the restricted fixed point exists, the price channel is not what deters; the learning correction alone is enough to deter. Those pairs are not equilibria of Definition 1: at each of them the West could raise its value by between 0.036 and 0.051 by changing its employment for one period at some state, against a value of about 39 at the high steady state. The restriction also costs the computation its stability. At $\gamma = 4.2$ and $\psi\Delta g = 4$ the restricted iteration does not stop from either starting policy, nor at the intermediate weights from 2.75 to 3.75 when each is started from the last pair found, and on the finer grid it does not stop at $\gamma = 4.2$ at the weight we ran there, while the unrestricted iteration stops at every cell we ran. A government that chooses under one flow and is evaluated under another does not solve a contraction, and its inner iteration need not converge; a government that both chooses and is evaluated at realized welfare does.

Off the equilibrium path the West's response to a flood is what deters, and Figure 6 shows it, from the equilibrium at $\gamma = 4.2$ and $\psi\Delta g = 4$. Against a shipment of 8, the West's wedge is 0.27 at the high steady state, 0.40 at $h = 5.0$, 1.7 at the inflection $h = 4.2$, 3.0 at the separatrix, and 11.0 at $h = 3.5$. Two motives are at work in these numbers, and they separate at the two ends. At the high steady state the shadow value of knowledge is nil, so the 0.27 is entirely the price term: a West facing a shipment of 8 works more than its competitive level to lower what it pays for it. Deep in the steep region the price term is a small part of a much larger wedge, and what drives the response is the shadow value of knowledge, which grows steeply below the separatrix, where without the subsidy knowledge would slide to the low steady state. The response is not proportional to $\alpha'(h)$ at the current state, which is smaller at $h = 3.5$ than at the separatrix. On the equilibrium path, where nothing is shipped, the price term is zero and the wedge is the domestic correction alone. Under a permanent flood of 8 the West's knowledge falls from 6.17 to 5.44 and stays there, well above the separatrix, while the competitive West falls below it and settles at 0.5. Every flood we computed loses money. A shipment of 16 sustained for 40 periods costs the East 240 in present value and never carries the West's knowledge below 5.1. A shipment of 30 sustained for 60 periods costs 1,042 and never carries it below 4.95. The logic of Proposition 10 carries over: the West's cost of holding a level is bounded, the East's cost of pushing is not, and a push that does not reach the steep region gains the East little, because g is nearly flat on the plateau.

At weights far above the critical security weight, $\psi\Delta g = 10$ and 20, the iteration does not stop from either starting policy, with or without the price-taking restriction, and no equilibrium was found; we draw no conclusion at those weights.

Tariffs behave as in Proposition 9. Against the passive West at $\gamma = 4.2$, a tariff of 0.1 is

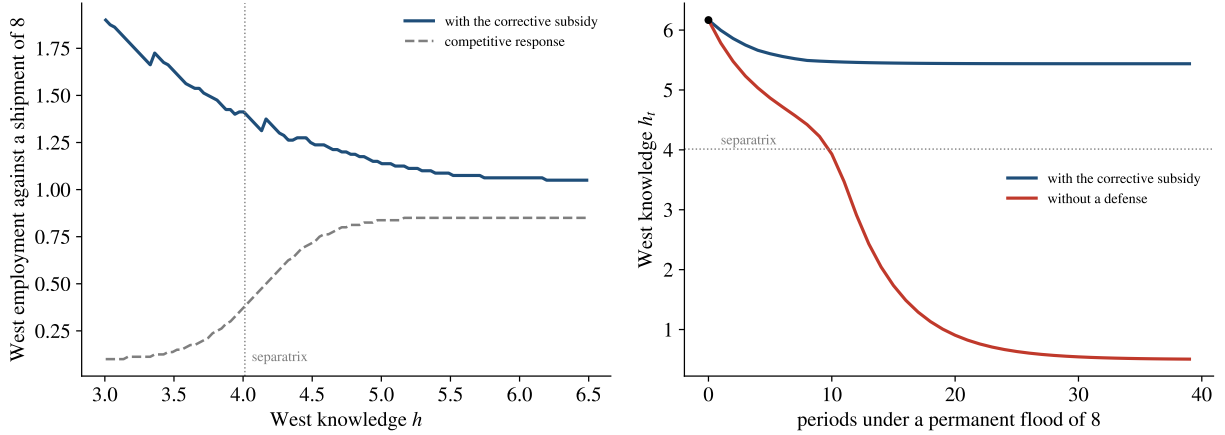


Figure 6: The West’s defense in the smooth economy, at the parameters of Section 6 and $\psi\Delta g = 4$, from the equilibrium of Table 4. Left: the West’s employment against a shipment of 8 as a function of its knowledge, with the corrective subsidy (solid) and under the competitive response (dashed); the gap widens as knowledge falls into the steep region of the S-curve and below the separatrix. Right: the West’s knowledge under a permanent flood of 8, with the corrective subsidy (blue) and without a defense (red); the dotted line is the separatrix.

absorbed at every weight tested, and a tariff of 0.3 deters at $\psi\Delta g = 1.66$ and is absorbed at 2.5 and 4.

What the step changes. Two things change. On the equilibrium path the subsidy is the domestic correction rather than zero, and that correction vanishes as the plateau flattens, falling below 10^{-4} at $\gamma \geq 8$, which the employment grid cannot resolve. The defended level becomes a level between the separatrix and the high steady state, 5.4 against a flood of 8 here, that the West holds by an employment response that rises with its shadow value of knowledge. What the step does not change is the content of Section 5: the West that pays for its knowledge is not trapped, the East ships nothing on the equilibrium path, the East’s flood against it is unprofitable, and a tariff below a level that depends on the East’s motive is absorbed.

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